

Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

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Chapter 1

Low Dimensional Asymptotic Normality

1.1 Basic Setup

Given a "high dimensional" data set $\{\mathbf{x}_i\}_{i=1}^n$ where $\mathbf{x}_i \in \mathbb{R}^p$ for all $i = 1, \dots, n$, the statistician fixes a positive integer $q < p$ and wishes to find the q principal components. Assuming that the data is centred (that is, with mean 0), the probabilistic principal component analysis model is a statistical model that treats $\mathbf{x}_i$'s as realizations of independent random variables $\mathbf{X}_i = \mathbf{W}\mathbf{Z}_i + \boldsymbol{\epsilon}_i$, where $\mathbf{W}$ is an unknown $p \times q$ loading matrix of full rank, $\mathbf{Z}_i \sim N_q(\mathbf{0}, \mathbf{I}_q)$ is the latent normal variable, and $\boldsymbol{\epsilon}_i \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I}_p)$ is the Gaussian noise with unknown standard deviation $\sigma > 0$, and $(\mathbf{Z}_1, \dots, \mathbf{Z}_n, \boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_n)$ are all independent (for now). Put $\theta = (\mathbf{W}_\theta, \sigma_\theta^2)$ for the unknown parameter to estimate and Θ for all possible choices of θ . Θ is thus the collection of all tuples $(\mathbf{W}, \sigma^2)$ such that $\mathbf{W}$ is a real $p \times q$ matrix with $\text{rank}(\mathbf{W}) = q$ and $\sigma^2 \in (0, \infty)$. Given that the collection of all real $p \times q$ matrix with rank q is an open subset of the Euclidean space $\mathbb{R}^{pq}$, we consider Θ as an open subset of the Euclidean space $\mathbb{R}^{pq+1}$ and our statistical model is

$$\mathcal{P} = \{N_p(\mathbf{0}, \mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 \mathbf{I}_p)\}_{\theta \in \Theta}. \quad (1.1)$$

To resolve the non-identifiability of $\mathcal{P}$, we introduce the quotient manifold structure. For $(\mathbf{A}, s) \in \Theta$ and $(\mathbf{B}, t) \in \Theta$, we define $(\mathbf{A}, s) \sim (\mathbf{B}, t)$ to mean that they give rise to the same covariance matrix, namely when $\mathbf{A}\mathbf{A}^T + s\mathbf{I}_p = \mathbf{B}\mathbf{B}^T + t\mathbf{I}_p$. The equivalence classes are denoted by $[(\mathbf{A}, s)]$ or more briefly $[\mathbf{A}, s]$. Unless otherwise mentioned, $(\mathbf{A}, s) \in \Theta$ is the default representative we pick when we write an equivalence class as $[\mathbf{A}, s]$. Writing $O(q)$ for the collection of all $q \times q$ real orthogonal matrices, it was shown in Datta's thesis that $[\mathbf{A}, s] = \{(\mathbf{A}\mathbf{Q}, s) \mid \mathbf{Q} \in O(q)\}$. Therefore, $M := \Theta / \sim$ can be identified as the orbits of the right action of $O(q)$ on Θ and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see [Lee(2013)], Theorem 21.10) shows that there is a unique smooth structure making M a manifold of dimension $pq + 1 - \frac{q(q-1)}{2}$ and the map $\pi : \Theta \rightarrow M$ defined by

$$\pi((\mathbf{A}, s)) = [\mathbf{A}, s] \quad (1.2)$$

is a smooth submersion. Moreover, this action is verticle, transitive on fibers and is isomertic with respect to the Euclidean metric. Thus, there exists a unique metric g on M making (M, g) a Riemannian manifold and π a Riemannian submersion (see [Lee(2018)], Theorem 2.28). In more concrete terms, if g^Θ is the standard Euclidean metric given by

$$g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)) = \text{tr}(\mathbf{V}\mathbf{U}^T) + cb, \quad (1.3)$$

whenever $(\mathbf{A}, s) \in \Theta$, $(\mathbf{V}, c), (\mathbf{U}, b) \in T_{(\mathbf{A}, s)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$, then

$$g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{h}) = g_{[\mathbf{A}, s]}(d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c)), d\pi_{(\mathbf{A}, s)}((\mathbf{U}, b))) = g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)), \quad (1.4)$$

whenever $\mathbf{v}, \mathbf{h} \in T_{[\mathbf{A}, s]}M$ lifting to $(\mathbf{V}, c)$ and $(\mathbf{U}, b)$ respectively. For later convenience, let us define two functions $\Phi : \Theta \rightarrow \mathbb{R}^{p \times p}$ and $\Sigma : M \rightarrow \mathbb{R}^{p \times p}$ by setting, for every $[\mathbf{A}, s] \in M$:

$$\Sigma([\mathbf{A}, s]) := \Phi((\mathbf{A}, s)) := \mathbf{A}\mathbf{A}^T + s\mathbf{I}_p. \quad (1.5)$$

In other words, $\Sigma \circ \pi = \Phi$ gives the covariance matrix. (1.1) is then reparametrized as

$$\mathcal{P} = \{N_p(\mathbf{0}, \Sigma([\mathbf{A}, s]))\}_{[\mathbf{A}, s] \in M} \quad (1.6)$$

and this is the identifiable model we shall work with.

Finally, throughout the entire document, $\|\mathbf{A}\|$ will denote the spectral norm of a matrix $\mathbf{A}$, which is defined as the largest singular value: the largest eigenvalue of $(\mathbf{A}^T \mathbf{A})^{\frac{1}{2}}$. On the other hand, the smallest singular value of $\mathbf{A}$ will be denoted by $\sigma_{\min}(\mathbf{A})$.

1.2 Basic Facts about (M, g)

Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and verticle tangent spaces.

Proposition 1.

1. The map Σ defined in (1.5) is a smooth injective immersion.
2. At every $(\mathbf{A}, s) \in \Theta$, the verticle tangent space

$$V_{(\mathbf{A}, s)} := \ker(d\pi_{(\mathbf{A}, s)}) = \ker(d\Phi_{(\mathbf{A}, s)}) \quad (1.7)$$

equals to the collection of all tangent vectors of the form $(\mathbf{A}\mathbf{\Omega}, 0) \in T_{(\mathbf{A}, s)}\Theta$, with $\mathbf{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

3. At every $(\mathbf{A}, s) \in \Theta$, the horizontal tangent space $H_{(\mathbf{A}, s)} := V_{(\mathbf{A}, s)}^\perp$ equals to the collection of all tangent vectors of the form $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$ ranged over all $c \in \mathbb{R}$ and all $p \times q$ matrices $\mathbf{V}$ satisfying $\mathbf{V}^T \mathbf{A} = \mathbf{A}^T \mathbf{V}$.

Proof. Fix a point $(\mathbf{A}, s) \in \Theta$ throughout this proof. Before we start the proof, let us first study the set $\ker(d\Phi_{(\mathbf{A}, s)})$, where Φ is given in (1.5). For $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$, define a curve $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$ by setting $\gamma(t) = (\mathbf{A} + t\mathbf{V}, s + tc)$. Then:

$$d\Phi_{(\mathbf{A}, s)}((\mathbf{V}, c)) = \left. \frac{d}{dt} \right|_{t=0} \Phi(\gamma(t)) = \mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p.$$

If $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A}, s)})$, then

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p = \mathbf{0}. \quad (1.8)$$

If $\mathbf{u} \in \mathbb{R}^p$ is any nonzero vector orthogonal to all column vectors of $\mathbf{A}$, then $\mathbf{A}^T \mathbf{u} = \mathbf{0}$ and

$$0 = \mathbf{u}^T (\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p) \mathbf{u} = c \mathbf{u}^T \mathbf{u}.$$

Given that $\mathbf{u} \neq \mathbf{0}$, we must have $c = 0$ and (1.8) reduces to

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T = \mathbf{0}. \quad (1.9)$$

Apply $\mathbf{u}$ to the right to get:

$$\mathbf{A}\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Given that $\mathbf{A}$ has full rank, rank nullity theorem forces

$$\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Therefore, we have proven that every vector $\mathbf{u}$ orthogonal to columns vectors of $\mathbf{A}$ will also be orthogonal to column vectors of $\mathbf{V}$. This means $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ for some $q \times q$ matrix $\mathbf{\Omega}$. Substitute this back to (1.9) gives

$$\mathbf{A}(\mathbf{\Omega}^T + \mathbf{\Omega})\mathbf{A}^T = \mathbf{0}.$$

Applying $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ to the left and $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$ to the right yields $\mathbf{\Omega}^T + \mathbf{\Omega} = \mathbf{0}$. Therefore, $\mathbf{\Omega}$ is skew-symmetric. Conversely, (1.9) is satisfied by $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ with any skew-symmetric matrix $\mathbf{\Omega}$. We have proven that $\ker(d\Phi_{(\mathbf{A},s)})$ equals to the collection of all tangent vectors of the form $(\mathbf{A}\mathbf{\Omega}, 0) \in T_{(\mathbf{A},s)}\Theta$, with $\mathbf{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

We now move back to prove 1 and 2 at the same time by showing (1.7). Clearly, Σ is well-defined and is injective. The smoothness of Σ is guaranteed by relation $\Sigma \circ \pi = \Phi$ (see for example [Lee(2013)] Theorem 4.29). By the chain rule, we have

$$d\Phi_{(\mathbf{A},s)} = d\Sigma_{[\mathbf{A},s]} \circ d\pi_{(\mathbf{A},s)}. \quad (1.10)$$

It follows immediately that $\ker(d\pi_{(\mathbf{A},s)}) \subseteq \ker(d\Phi_{(\mathbf{A},s)})$. The other direction will follow if we can prove that $d\Sigma_{[\mathbf{A},s]}$ is injective.

Suppose that $\mathbf{v} \in T_{[\mathbf{A},s]}M$ is such that $d\Sigma_{[\mathbf{A},s]}(\mathbf{v}) = 0$. We prove that $\mathbf{v} = \mathbf{0}$. Let $(\mathbf{V}, c) \in T_{(\mathbf{A},s)}\Theta$ be the horizontal lift of $\mathbf{v}$ at $(\mathbf{A}, s)$, meaning that $(\mathbf{V}, c)$ is the unique horizontal tangent vector at $(\mathbf{A}, s)$ such that $\mathbf{v} = d\pi_{(\mathbf{A},s)}((\mathbf{V}, c))$. (1.10) then implies that $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A},s)})$ and the proceeding discussion shows that $c = 0$ and $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ for some skew-symmetric $\mathbf{\Omega}$. To proceed, we look at the curve $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$ defined by $\gamma(t) = (\mathbf{A}e^{t\mathbf{\Omega}}, s)$. Clearly $\gamma(0) = (\mathbf{A}, s)$ and $\gamma'(0) = (\mathbf{A}\mathbf{\Omega}, 0)$. Moreover, the matrix $e^{t\mathbf{\Omega}}$ is always orthogonal for any t and any skew-symmetric $\mathbf{\Omega}$ so that $\pi(\gamma(t)) = [\mathbf{A}, s]$ for all t . We have

$$\mathbf{v} = d\pi_{(\mathbf{A},s)}((\mathbf{A}\mathbf{\Omega}, 0)) = d\pi_{(\mathbf{A},s)}(\gamma'(0)) = \left. \frac{d}{dt} \right|_{t=0} \pi(\gamma(t)) = \mathbf{0} \in T_{[\mathbf{A},s]}M.$$

Therefore, $d\Sigma_{[\mathbf{A},s]}$ is injective and the verticle tangent space is the same as $\ker(d\Phi_{(\mathbf{A},s)})$. Since $[\mathbf{A}, s]$ is arbitrary, Σ is an immersion.

To compute the horizontal tangent space, any $(\mathbf{V}, c) \in V_{(\mathbf{A},s)}^\perp$ will satisfy

$$0 = g_{(\mathbf{A},s)}^\Theta((\mathbf{V}, c), (\mathbf{A}\mathbf{\Omega}, 0)) = \text{tr}(\mathbf{A}^T \mathbf{V} \mathbf{\Omega}) \quad (1.11)$$

for all skew-symmetric $\mathbf{\Omega}$. Decomposing $\mathbf{A}^T \mathbf{V} = \mathbf{S} + \mathbf{K}$, where $\mathbf{S}$ is the symmetric part and $\mathbf{K}$ is the skew-symmetric part, we find out that

$$0 = \text{tr}(\mathbf{S}\mathbf{\Omega}) + \text{tr}(\mathbf{K}\mathbf{\Omega}) = 0 + \text{tr}(\mathbf{K}\mathbf{\Omega}) = \text{tr}(\mathbf{K}\mathbf{\Omega}).$$

Setting $\mathbf{\Omega} = \mathbf{K}^T$ will yield that $\mathbf{K} = \mathbf{0}$. Therefore, $\mathbf{A}^T \mathbf{V}$ is a symmetric matrix. Conversely, any $\mathbf{V}$ such that $\mathbf{A}^T \mathbf{V}$ is symmetric will satisfy (1.11). The proof is complete. $\square$

Proposition 2. Given $[\mathbf{A}, s] \in M$ and $\mathbf{v} \in T_{[\mathbf{A},s]}M$, geodesics in M starting at $[\mathbf{A}, s]$ with initial velocity $\mathbf{v}$ takes the format of curves $\gamma : I \rightarrow M$, where I is an open interval containing 0, defined by $\gamma(t) = [\mathbf{A} + t\mathbf{V}, s + tc]$, where $(\mathbf{V}, c) \in H_{(\mathbf{A},s)}$ is the horizontal lift of $\mathbf{v}$ at $(\mathbf{A}, s)$. Consequently, the exponential map is computed as $\text{Exp}_{[\mathbf{A},s]}(\mathbf{v}) = [\mathbf{A} + \mathbf{V}, s + c]$, at least well-defined over an open ball $\mathcal{E}_{[\mathbf{A},s]}$ of $\mathbf{0} \in T_{[\mathbf{A},s]}M \cong \mathbb{R}^{pq+1-\frac{q(q-1)}{2}}$ of radius $\min(\sigma_{\min}(\mathbf{A}), s)$.

Proof. First of all, let us quickly check that the curve γ is well defined. Indeed, if $(\mathbf{A}\mathbf{Q}, s)$, where $\mathbf{Q} \in O(q)$, is another representative of $[\mathbf{A}, s]$, let $(\mathbf{V}', c')$ be the horizontal lift of $\mathbf{v}$ to $(\mathbf{A}\mathbf{Q}, s)$. With the function $R_{\mathbf{Q}} : \Theta \rightarrow \Theta$ defined by $R_{\mathbf{Q}}((\mathbf{A}, s)) = (\mathbf{A}\mathbf{Q}, s)$, we have

$$\mathbf{v} = d\pi_{(\mathbf{A},s)}(\mathbf{V}, c) = d(\pi \circ R_{\mathbf{Q}})_{(\mathbf{A},s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q},s)}(dR_{\mathbf{Q}})_{(\mathbf{A},s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q},s)}(\mathbf{V}\mathbf{Q}, c).$$

Thus, $\mathbf{V}' = \mathbf{V}\mathbf{Q}$ and $c' = c$. Substitute this into the definition of γ will yield the exact same curve.

A well-known property of geodesics in case of Riemannian submersion (see for example [O'Neill(1967)] Section 2 Corollary 2 or [Valencia(2021)] Theorem 3.6) is that any geodesic c in Θ with initial velocity being horizontal will always have horizontal velocities and $\gamma = \pi \circ c$ will remain a geodesic in M . Therefore, $\text{Exp}_{[\mathbf{A},s]}(\mathbf{v}) = [\mathbf{A} + \mathbf{V}, s + c]$ is well defined for all vectors $\mathbf{v} \in T_{[\mathbf{A},s]}M$ of small enough norm making sure that $(\mathbf{A} + \mathbf{V}, s + c) \in \Theta$. To have a quantified upper bound of the norm, we cite Weyl's perturbation inequality (Chapter 7.3 Problem 16 equation (7.3.15) of [Horn and Johnson(2013)]) to get

$$\sigma_{\min}(\mathbf{A} + \mathbf{V}) \geq \sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\| \geq \sigma_{\min}(\mathbf{A}) - |\mathbf{v}| > 0,$$

whenever $|\mathbf{v}| < \min(\sigma_{\min}(\mathbf{A}), s)$. The last inequality is because the spectral norm of $\mathbf{V}$ is always smaller than the Euclidean norm (Frobenius norm) of $\mathbf{V}$ and hence norm of $\mathbf{v}$. Thus, for all $\mathbf{v} \in \mathcal{E}_{[\mathbf{A},s]}$, $\text{rank}(\mathbf{A} + \mathbf{V}) = q$ and the exponential map is well-defined. $\square$

To compute the logarithmic map, we need first preparatory lemmas that work out an orthogonal matrix based on given points in M .

Lemma 3. If $\mathbf{L}$ is an invertible $q \times q$ matrix and $\mathbf{Q} \in O(q)$ are such that $\mathbf{L}\mathbf{Q}$ is symmetric, then there exists matrices $\mathbf{R} \in O(q)$, $\mathbf{P} \in O(q)$, a $q \times q$ diagonal matrix $\mathbf{D}$ whose diagonal entries are 1 or -1 , and a $q \times q$ diagonal matrix $\mathbf{S}$ containing the singular values of $\mathbf{L}$ in increasing order such that $\mathbf{L} = \mathbf{R}\mathbf{S}\mathbf{P}^T$ and $\mathbf{Q} = \mathbf{P}\mathbf{D}\mathbf{R}^T$.

Proof. Let us start by applying the singular value decomposition of $\mathbf{L}$ to get matrices $\mathbf{R}' \in O(q)$, $\mathbf{P}' \in O(q)$ and $\mathbf{S}$ such that $\mathbf{L} = \mathbf{R}'\mathbf{S}(\mathbf{P}')^T$. Substituting the given condition that $\mathbf{L}\mathbf{Q}$ is symmetric will yield that $\mathbf{S}\mathbf{J} = \mathbf{J}^T\mathbf{S}$, where $\mathbf{J} := (\mathbf{P}')^T\mathbf{Q}\mathbf{R}'$.

Clearly $\mathbf{J} \in O(q)$. The equation

$$\mathbf{S}^2 = \mathbf{S}\mathbf{J}\mathbf{J}^T\mathbf{S} = (\mathbf{S}\mathbf{J})(\mathbf{S}\mathbf{J})^T = (\mathbf{J}^T\mathbf{S})(\mathbf{J}^T\mathbf{S})^T = \mathbf{J}^T\mathbf{S}^2\mathbf{J}$$

shows that

$$\mathbf{J}\mathbf{S}^2 = \mathbf{S}^2\mathbf{J}.$$

Looking at each entry of both size, we end up with

$$\mathbf{J}_{ij}(\mathbf{S}_{jj}^2 - \mathbf{S}_{ii}^2) = 0$$

for $i, j = 1, \dots, q$. Given that $\mathbf{L}$ is invertible, its singular values are all strictly positive. Therefore, in fact we have

$$\mathbf{J}_{ij}(\mathbf{S}_{jj} - \mathbf{S}_{ii}) = 0 \tag{1.12}$$

for all i and all j so that $\mathbf{J}\mathbf{S} = \mathbf{S}\mathbf{J} = \mathbf{J}^T\mathbf{S}$. Invert $\mathbf{S}$ yields that $\mathbf{J}$ is actually symmetric. Being a symmetric orthogonal matrix, $\mathbf{J}^2 = \mathbf{I}_q$ and the eigenvalues of $\mathbf{J}$ must be 1 or -1 . Another look at (1.12) shows that $\mathbf{J}$ is a block-diagonal matrix, diagonal if and only if all diagonal entries of $\mathbf{S}$ are distinct. Furthermore, each block of $\mathbf{J}$ must still be orthogonal and symmetric.

We now diagonalize $\mathbf{J}$ based on the above information. Suppose that $\mathbf{S} = \bigoplus_{i=1}^k \mathbf{s}_i \mathbf{I}_{m_i}$, where $\mathbf{s}_1 < \dots < \mathbf{s}_k$ are the unique diagonal entries of $\mathbf{S}$ and each repeats for m_i times. Thus, $\mathbf{J}$ has k blocks. Write the i th block by $\mathbf{J}_i$ so that $\mathbf{J} = \bigoplus_{i=1}^k \mathbf{J}_i$. The spectral theorem then allows us to orthogonally diagonalize $\mathbf{J}_i = \mathbf{K}_i \mathbf{D}_i \mathbf{K}_i^T$, where $\mathbf{K}_i \in O(m_i)$ and $\mathbf{D}_i$ is an $m_i \times m_i$ diagonal matrix with 1 or -1 diagonal entry. Put $\mathbf{K} = \bigoplus_{i=1}^k \mathbf{K}_i$ and $\mathbf{D} = \bigoplus_{i=1}^k \mathbf{D}_i$. Then $\mathbf{K} \in O(q)$, $\mathbf{D}$ is a $q \times q$ diagonal matrix with 1 or -1 diagonal entry, and $\mathbf{J} = \mathbf{K}\mathbf{D}\mathbf{K}^T$. This involved construction leads us to the outcome that $\mathbf{K}$ in fact commutes with $\mathbf{S}$:

$$\mathbf{K}\mathbf{S} = \bigoplus_{i=1}^k \mathbf{s}_i \mathbf{K}_i = \mathbf{S}\mathbf{K}.$$

Put $P = P'K$, $R = R'K$. We have

$$Q = P'J(R')^T = P'KDK^T(R')^T = PDR^T$$

and

$$RSP^T = R'KSK^T(P')^T = R'SKK^T(P')^T = R'S(P')^T = L.$$

This completes the proof. □

Lemma 4. If L is an invertible $q \times q$ matrix with singular value decomposition $L = RSP^T$, then

$$PR^T = (L^T L)^{-\frac{1}{2}} L^T.$$

This means, although neither P nor R is uniquely determined, PR^T is uniquely determined.

Proof. Given that

$$L^T L = PS^2 P^T,$$

we have

$$(L^T L)^{\frac{1}{2}} = PSP^T.$$

Therefore:

$$L = RP^T PSP^T = RP^T (L^T L)^{\frac{1}{2}}.$$

The result follows from inverting $(L^T L)^{\frac{1}{2}}$ and take transpose. □

Proposition 5. For every $[A, s] \in M$ there exists an open neighborhood $U \subset M$ containing $[A, s]$ such that

1. For every $[B, t] \in U$, $A^T B$ is invertible.
2. The logarithmic map $\text{Exp}_{[A, s]}^{-1} : U \rightarrow T_{[A, s]}M$ is well-defined and is smooth in U . In fact:

$$\text{Exp}_{[A, s]}^{-1}([B, t]) = d\pi_{(A, s)}((BQ(A, B) - A, t - s)), \quad (1.13)$$

where

$$Q(A, B) = (B^T A A^T B)^{-\frac{1}{2}} B^T A \in O(q). \quad (1.14)$$

Proof. First of all, if $R \in O(q)$ and $P \in O(q)$, then $Q(AP, BR) = R^T Q(A, B)P$ and the same computation as in the proof of Proposition 2 shows that (1.13) is indeed well-defined.

It is a well-known fact that $\text{Exp}_{[A, s]}$ is a local diffeomorphism and thus there exists an open neighborhood containing $[A, s]$ in which the logarithmic map $\text{Exp}_{[A, s]}^{-1}$ is well-defined and is smooth. Given that $\text{Exp}_{[A, s]}^{-1}([A, s]) = 0$, we shall only take an open subset U_1 of this open neighborhood such that, whenever $[B, t] \in U_1$ and $v = \text{Exp}_{[A, s]}^{-1}([B, t])$, we always have $|v| < \epsilon$, where

$$\epsilon \in \left(0, \min \left(\sigma_{\min}(A), 2 \left[\frac{2}{\sigma_{\min}(A)} + 1 \right]^{-1} \right) \right)$$

is a fixed constant.

Next, let $U^\Theta \subset \Theta$ be the collection of all $(B, t) \in \Theta$ such that $\|A - B\| < \epsilon$. Then U^Θ is an open neighborhood of (A, s) and any $(B, t) \in U^\Theta$ has the property that $A^T B$ is invertible thanks to the condition that $\epsilon < \sigma_{\min}(A)$. To see this, suppose for the sake of contradiction that there exists a unit vector $x \in \mathbb{R}^q$ with $A^T Bx = 0$. Then

$$|Ax|^2 = x^T A^T A x = -x^T A^T (B - A)x \leq |Ax| \|B - A\| \cdot 1$$

and $\|B - A\| \geq |Ax| \geq \sigma_{\min}(A)$, a contradiction. The fact that $A^T B$ is invertible guarantees that the matrix $Q(A, B)$ given at (1.14) is well defined for any $(B, t) \in U_\Theta$ and, by continuity of $Q(A, B)$ and the fact that $Q(A, A) = I_q$, we can shrink U^Θ to also guarantee that $\|Q(A, B) - I_q\| < \epsilon$ whenever $(B, t) \in U_\Theta$. We define $U_2 := \pi(U^\Theta)$. Given that π is an open map, U_2 is an open neighborhood of $[A, s]$. If $[B, t] \in U_2$, then *every* representative (B, t) of $[B, t]$ will have $A^T B$ being invertible because $\det(A^T B R) = \det(A^T B) \neq 0$ for every $R \in O(q)$, although only *some* representatives (B, t) will satisfy both $\|A - B\| < \epsilon$ and $\|Q(A, B) - I_q\| < \epsilon$. Given that our results do not depend on representatives, we shall always choose a representative (B, t) such that this is the case.

We now compute $\text{Exp}_{[A, s]}^{-1}$ for $[B, t] \in U := U_1 \cap U_2$. By Proposition 2, $[B, t] = [A + V, s + c]$, where (V, c) is the horizontal lift of $v = \text{Exp}_{[A, s]}^{-1}([B, t])$ to (A, s) . It follows that $c = t - s$ and that there exists a $Q \in O(q)$ such that $V = BQ - A$. Or equivalently:

$$\text{Exp}_{[A, s]}^{-1}([B, t]) = d\pi_{(A, s)}((BQ - A, t - s)). \quad (1.15)$$

Given that (V, c) is a horizontal tangent vector, Proposition 1 shows that

$$A^T(BQ - A) = (BQ - A)^T A$$

and consequently $A^T BQ$ is symmetric. Lemma 3 with $L = A^T B$ shows the existence of $R, P \in O(q)$, a diagonal matrix S with the singular values of $A^T B$ in increasing order, and a $q \times q$ diagonal matrix D with diagonal entries 1 or -1 such that $A^T B = RSP^T$ and $Q = PDR^T$. (1.15) now becomes

$$\text{Exp}_{[A, s]}^{-1}([B, t]) = d\pi_{(A, s)}((BPDR^T - A, t - s)). \quad (1.16)$$

Now, given that Lemma 4 shows that $Q(A, B) = PR^T$, the only thing remaining is to rule out all $D \neq I_q$. We achieve this by comparing the two matrices directly:

$$\|Q - Q(A, B)\| = \|QQ(A, B)^T - I_q\| = \|PDP^T - I_q\| = \|D - I_q\|.$$

Thus, $\|Q - Q(A, B)\|$ equals to either 0 or 2 and takes no other values. It suffices to bound it away from 2. We have

$$\begin{aligned} \|QQ(A, B)^T - I_q\| &\leq \|(Q - I_q)Q(A, B)^T + (Q(A, B)^T - I_q)\| \leq \|Q(A, B)^T - I_q\| + \|Q - I_q\| \\ &< \epsilon + \frac{1}{\sigma_{\min}(A)} \|AQ - A\| \leq \epsilon + \frac{1}{\sigma_{\min}(A)} \|AQ - BQ + V\| \leq \epsilon + \frac{1}{\sigma_{\min}(A)} (\|A - B\| + \|V\|) \\ &\leq \epsilon + \frac{2\epsilon}{\sigma_{\min}(A)} < 2. \end{aligned}$$

This completes the proof. $\square$

Remark 6. Unfortunately, the logarithmic map is not globally defined in the standard way even over all $[B, t]$ such that $A^T B$ is invertible. For a counterexample, take $p = 2, q = 1, A = [1, 0]^T, B = [1, 1]^T$ and $s = t = 1$. In this case, $O(1) = \{1, -1\}$. Then, although $A^T B = 1$ is invertible, we still have

$$\text{Exp}_{[A, s]}(d\pi_{(A, s)}((B - A, 0))) = [B, 1] = [-B, 1] = \text{Exp}_{[A, s]}(d\pi_{(A, s)}((-B - A, 0))).$$

The reason is that, even when we assume $A^T B$ is invertible, deducing $D = I_q$ relies on the local smoothness of the logarithmic map and the procedure of letting $[B, t] \rightarrow [A, s]$. If $[B, t]$ is far away from $[A, s]$, then $Q = PDR^T$ with some $D \neq I_q$ can still produce a valid horizontal vector whose image under $\text{Exp}_{[A, s]}$ is $[B, t]$, so that $\text{Exp}_{[A, s]}$ fails to be injective and admits no globally defined inverse.

Nevertheless, the choice $D = I_q$ has an interesting property of selecting the geodesic of minimal speed and length when $[B, t]$ is any point such that $A^T B$ is invertible. To see this, consider a general candidate $Q = PDR^T$

and suppose that $\mathbf{v} = d\pi_{(\mathbf{A}, s)}((\mathbf{BQ} - \mathbf{A}, t - s))$ satisfies $\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}) = [\mathbf{B}, t]$. Given that so that $(\mathbf{BQ} - \mathbf{A}, t - s)$ is the horizontal lift of $\mathbf{v}$ at $(\mathbf{A}, s)$, we can compute the squared norm of $\mathbf{v}$:

$$g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{v}) = g_{(\mathbf{A}, s)}^\Theta((\mathbf{BQ} - \mathbf{A}, t - s), (\mathbf{BQ} - \mathbf{A}, t - s)) \\ = \text{tr}((\mathbf{BQ} - \mathbf{A})(\mathbf{BQ} - \mathbf{A})^T) + (t - s)^2 = \text{tr}(\mathbf{B}\mathbf{B}^T) + \text{tr}(\mathbf{A}\mathbf{A}^T) - 2\text{tr}(\mathbf{S}\mathbf{D}) + (t - s)^2,$$

where the last equality substituted $\mathbf{A}^T \mathbf{BQ} = \mathbf{RSP}^T \mathbf{PDR}^T = \mathbf{RSDR}^T$ and used the cyclic property of the trace. The fact that the diagonal entries of $\mathbf{S}$ are strictly positive show that $g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{v})$ is minimized precisely when $\mathbf{D} = \mathbf{I}_q$. Given that the geodesic starting at $[\mathbf{A}, s]$ with initial velocity $\mathbf{v}$ reaches $[\mathbf{B}, t]$ at time 1 with constant speed $\sqrt{g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{v})}$, its length is $\sqrt{g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{v})}$. The choice $\mathbf{D} = \mathbf{I}_q$ gives the slowest and hence shortest geodesic. To illustrate, let us go back to the counterexample in the previous paragraph. Put $\mathbf{v}_\pm := d\pi_{(\mathbf{A}, s)}((\pm \mathbf{B} - \mathbf{A}, 0))$. Direct computation shows $g_{[\mathbf{A}, s]}(\mathbf{v}_+, \mathbf{v}_+) = 1$ and $g_{[\mathbf{A}, s]}(\mathbf{v}_-, \mathbf{v}_-) = 5$ so that the option corresponding to $\mathbf{D} = \mathbf{I}_1 = 1$ gives the slowest speed. Therefore, it is actually possible to define $\text{Exp}_{[\mathbf{A}, s]}^{-1}$ over all $[\mathbf{B}, t]$ such that $\mathbf{A}^T \mathbf{B}$ is invertible by manually asking it to output the minimal speed candidate and this definition will yield the same $Q(\mathbf{A}, \mathbf{B})$ formula.

However, the situation totally breaks down in case $\mathbf{A}^T \mathbf{B}$ is not invertible. Consider again $p = 2, q = 1$, $\mathbf{A} = [1, 0]^T$, $s = t = 1$, but this time $\mathbf{B} = [0, 1]^T$. Then $g_{[\mathbf{A}, s]}(\mathbf{v}_+, \mathbf{v}_+) = g_{[\mathbf{A}, s]}(\mathbf{v}_-, \mathbf{v}_-) = 2$ and there are two distinct geodesics of the same speed so $\text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t])$ cannot be well-defined.

(Picture: there is a hole at $[0, 0]^T$. Starting $\mathbf{A} = [1, 0]^T$, as $\mathbf{B} = [0, 1]^T$ is glued to $[0, -1]^2$, $\mathbf{v}_+$ points from $[1, 0]^T$ to $[0, 1]^T$, $\mathbf{v}_-$ points from $[1, 0]^T$ to $[0, -1]^T$)

Given the above situation and (the expectation that) our work should not require a global definition, we stick to the standard definition of logarithmic map only defined locally around $[\mathbf{A}, s]$.

1.3 Properties of the Probabilistic Principal Component Analysis Model

We now show that the statistical model $\mathcal{P}$ given by (1.6) is *differentiable in quadratic mean* and has *positive definite Fisher information operator* in the sense developed in [Sun et al.(2026)Sun, Lin, and Liu], definition III.1 and III.2.

Let us introduce a few notations. For each $[\mathbf{A}, s] \in M$, let

$$p(\mathbf{x}; [\mathbf{A}, s]) = (2\pi)^{-\frac{p}{2}} \det^{-\frac{1}{2}}(\Sigma([\mathbf{A}, s])) \exp\left(-\frac{1}{2} \mathbf{x}^T (\Sigma([\mathbf{A}, s]))^{-1} \mathbf{x}\right) \quad (1.17)$$

be the density of $N_p(\mathbf{0}, \Sigma([\mathbf{A}, s]))$. For $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]} \subset T_{[\mathbf{A}, s]}M \cong \mathbb{R}^{pq+1-\frac{q(q-1)}{2}}$, write $l_{[\mathbf{A}, s]}(\mathbf{v})$ for the log-likelihood function of the local experiment

$$l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) = \log\left(p(\mathbf{x}; \text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}))\right). \quad (1.18)$$

Define the square root likelihood function $r_{[\mathbf{A}, s]} : \mathcal{E}_{[\mathbf{A}, s]} \rightarrow L^2(\mathbb{R}^p)$ by setting, for all $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$, that

$$r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) = \sqrt{p(\mathbf{x}; \text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}))}. \quad (1.19)$$

Theorem 7. $r_{[\mathbf{A}, s]}$ is Fréchet differentiable at any $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$. In particular, $\mathcal{P}$ is differentiable in quadratic mean with score

$$s(\mathbf{x}; [\mathbf{A}, s])(\mathbf{h}) = \frac{1}{2} \text{tr} \left\{ (\Sigma([\mathbf{A}, s]))^{-1} [\mathbf{x}\mathbf{x}^T - (\Sigma([\mathbf{A}, s]))] (\Sigma([\mathbf{A}, s]))^{-1} (D_{\mathbf{h}} \Sigma([\mathbf{A}, s])) \right\}, \quad (1.20)$$

where

$$D_{\mathbf{h}} \Sigma([\mathbf{A}, s]) := \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(t\mathbf{h})) = \mathbf{U}\mathbf{A}^T + \mathbf{A}\mathbf{U}^T + b\mathbf{I}_p, \quad (1.21)$$

if $\mathbf{h} \in T_{[\mathbf{A}, s]}M$ lifts to $(\mathbf{U}, b)$ at $(\mathbf{A}, s)$. Furthermore, $\mathcal{P}$ is a regular model in the sense that the Fisher information operator

$$G_{[\mathbf{A}, s]}(\mathbf{h}_1, \mathbf{h}_2) = \frac{1}{2} \text{tr} \{ (\Sigma([\mathbf{A}, s]))^{-1} D_{\mathbf{h}_1} \Sigma([\mathbf{A}, s]) (\Sigma([\mathbf{A}, s]))^{-1} D_{\mathbf{h}_2} \Sigma([\mathbf{A}, s]) \} \quad (1.22)$$

is positive definite.

Proof. The proof for the Fréchet differentiability is long but routine. We compute the Gâteaux derivative of $r_{[\mathbf{A}, s]}$ at $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ in a given direction $\mathbf{h} \in T_{[\mathbf{A}, s]}M$ and show that the Gâteaux derivative is the Fréchet derivative by a direct analysis of the remainder term. At the very end, we specialize our computations to $\mathbf{v} = \mathbf{0}$ to prove (1.20) and (1.22).

Given $\mathbf{v}$ and $\mathbf{h}$, there exists an $\epsilon > 0$ such that $\mathbf{v} + t\mathbf{h} \in \mathcal{E}_{[\mathbf{A}, s]}$ whenever $t \in (-\epsilon, \epsilon)$. If they respectively lift to $(\mathbf{V}, c)$ and $(\mathbf{U}, b)$ at $(\mathbf{A}, s)$, then $\mathbf{v} + t\mathbf{h}$ lifts to $(\mathbf{V} + t\mathbf{U}, c + tb)$ at $(\mathbf{A}, s)$. First of all:

$$\begin{aligned} D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) &:= \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})) \\ &= \left. \frac{d}{dt} \right|_{t=0} \{ (\mathbf{A} + \mathbf{V} + t\mathbf{U})(\mathbf{A} + \mathbf{V} + t\mathbf{U})^T + (s + c + tb)\mathbf{I}_p \} \\ &= \mathbf{U}\mathbf{V}^T + \mathbf{V}\mathbf{U}^T + \mathbf{U}\mathbf{A}^T + \mathbf{A}\mathbf{U}^T + b\mathbf{I}_p, \end{aligned} \quad (1.23)$$

proving (1.21) with special case $\mathbf{v} = \mathbf{0}$. Secondly, for all $\mathbf{x} \in \mathbb{R}^p$:

$$\begin{aligned} D_{\mathbf{h}} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) &:= \left. \frac{d}{dt} \right|_{t=0} l_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) \\ &= \left. \frac{d}{dt} \right|_{t=0} \left\{ -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})) \right) - \frac{1}{2} \mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})) \right)^{-1} \mathbf{x} \right\} \\ &= -\frac{1}{2} \text{tr} \left[\left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(\left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})) \right) \right] \\ &\quad + \frac{1}{2} \mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(\left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \mathbf{x} \\ &= -\frac{1}{2} \text{tr} \left[\left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \right] \\ &\quad + \frac{1}{2} \mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \mathbf{x} \\ &= \frac{1}{2} \text{tr} \left\{ \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left[\mathbf{x} \mathbf{x}^T - \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \right] \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \right\}. \end{aligned} \quad (1.24)$$

Finally:

$$\begin{aligned} D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) &= D_{\mathbf{h}} \exp \left(\frac{1}{2} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) \right) = \frac{1}{2} \exp \left(\frac{1}{2} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) \right) D_{\mathbf{h}} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) \\ &= \frac{1}{2} r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) D_{\mathbf{h}} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}). \end{aligned} \quad (1.25)$$

To prove that (1.25) indeed defines a valid Gâteaux derivative, we must prove that $D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v}) \in L^2(\mathbb{R}^p)$. Luckily, the expectation and covariance of normal quadratic forms have already been very well studied in [Gupta and Nagar(2000)]. For $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma)$, Theorem 7.7.1(i) states that

$$\mathbb{E} [\mathbf{X}^T \mathbf{Z} \mathbf{X}] = \text{tr} (\mathbf{Z} \Sigma)$$

for any $p \times p$ matrix $\mathbf{Z}$ and Theorem 7.7.2 states that

$$\text{Cov}(\mathbf{X}^T \mathbf{Z} \mathbf{X}, \mathbf{X}^T \mathbf{Y} \mathbf{X}) = \text{tr}(\mathbf{Z} \Sigma \mathbf{Y}^T \Sigma) + \text{tr}(\mathbf{Z}^T \Sigma \mathbf{Y}^T \Sigma)$$

for any two $p \times p$ matrices $\mathbf{Y}$ and $\mathbf{Z}$. Therefore, we can confirm without effort that the expectation of derivative of log density is zero:

$$\begin{aligned} & \int_{\mathbb{R}^p} D_{\mathbf{h}} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}))) d\mathbf{x} \\ &= -\frac{1}{2} \text{tr} \left[\left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \right] \\ &+ \frac{1}{2} \int_{\mathbb{R}^p} \left[\mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \mathbf{x} \right] p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}))) d\mathbf{x} = 0 \end{aligned}$$

and, whenever $\mathbf{h}_1$ and $\mathbf{h}_2 \in T_{[\mathbf{A}, s]}M$:

$$\begin{aligned} & \int_{\mathbb{R}^p} D_{\mathbf{h}_1} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) D_{\mathbf{h}_2} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}))) d\mathbf{x} \\ &= \frac{1}{4} \text{Cov}(\mathbf{X}^T \mathbf{Z}_1 \mathbf{X}, \mathbf{X}^T \mathbf{Z}_2 \mathbf{X}) = \frac{1}{4} \cdot 2 \text{tr} \left(\mathbf{Z}_1 \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \mathbf{Z}_2 \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \\ &= \frac{1}{2} \text{tr} \left\{ \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}_1} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}_2} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right\}, \end{aligned} \quad (1.26)$$

where $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})))$ and

$$\mathbf{Z}_j := \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}_j} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \quad (1.27)$$

is symmetric for $j = 1, 2$, so that the two terms in Theorem 7.7.2 of [Gupta and Nagar(2000)] coincide. Thus, in the particular case $\mathbf{h} = \mathbf{h}_1 = \mathbf{h}_2$:

$$\begin{aligned} & \|D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v}); L^2(\mathbb{R}^p)\|^2 = \int_{\mathbb{R}^p} [D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x})]^2 d\mathbf{x} \\ &= \frac{1}{4} \int_{\mathbb{R}^p} (D_{\mathbf{h}} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}))^2 p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}))) d\mathbf{x} = \frac{1}{8} \text{tr} \left\{ \left[\left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right]^2 \right\}. \end{aligned} \quad (1.28)$$

Therefore, $D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v}) \in L^2(\mathbb{R}^p)$ and (1.25) is indeed a valid Gâteaux derivative at $\mathbf{v}$ in the direction $\mathbf{h}$.

To prove that this is indeed the Fréchet derivative, let us look at the remainder term:

$$R(\mathbf{v}, \mathbf{h})(\mathbf{x}) := r_{[\mathbf{A}, s]}(\mathbf{v} + \mathbf{h})(\mathbf{x}) - r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) - D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x})$$

and prove that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{1}{|\mathbf{h}|^2} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\|^2 = 0.$$

To start, the fundamental theorem of calculus shows that

$$\begin{aligned} & r_{[\mathbf{A}, s]}(\mathbf{v} + \mathbf{h})(\mathbf{x}) - r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) = \int_0^1 \frac{d}{dt} r_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) dt \\ &= \int_0^1 \lim_{s \rightarrow 0} \frac{r_{[\mathbf{A}, s]}(\mathbf{v} + (t+s)\mathbf{h})(\mathbf{x}) - r_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})(\mathbf{x})}{s} dt = \int_0^1 D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) dt. \end{aligned}$$

Therefore, by Minkowski's integral inequality and the linearity of $D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})$ in $\mathbf{h}$, we have

$$\begin{aligned} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\| &= \left\| \int_0^1 (D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) - D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})(\mathbf{x})) dt; L^2(\mathbb{R}^p) \right\| \\ &\leq \int_0^1 \|D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt = |\mathbf{h}| \int_0^1 \|D_{\frac{\mathbf{h}}{|\mathbf{h}|}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\frac{\mathbf{h}}{|\mathbf{h}|}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \end{aligned}$$

and

$$\begin{aligned} \frac{1}{|\mathbf{h}|^2} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\|^2 &\leq \left\{ \int_0^1 \|D_{\frac{\mathbf{h}}{|\mathbf{h}|}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\frac{\mathbf{h}}{|\mathbf{h}|}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \right\}^2 \\ &\leq \left\{ \int_0^1 \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \right\}^2. \end{aligned}$$

The proof will be complete once we can show that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \int_0^1 \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt = 0. \quad (1.29)$$

We must therefore examine the continuity of $D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})$, which is easy to see if we revisit the formulas. (1.23) shows that $D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{A},s]}(\mathbf{v}))$ is smooth in both $\mathbf{h}$ and $\mathbf{v}$. Thus, (1.25) shows that, for each fixed $\mathbf{x} \in \mathbb{R}^p$, $D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})(\mathbf{x})$ is smooth in both $\mathbf{h}$ and $\mathbf{v}$ and (1.28) shows that $\|D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\|$ is smooth in both $\mathbf{h}$ and $\mathbf{v}$. Scheffé's lemma shows that $D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})$, as a map to $L^2(\mathbb{R}^p)$, is continuous in $\mathbf{h}$ and in $\mathbf{v}$. The integrand inside (1.29) is thus bounded uniformly over all $t \in [0, 1]$ and all $\mathbf{h}$ in a compact neighborhood of $\mathbf{0}$ and it suffices to show that, for every $t \in [0, 1]$, that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| = 0, \quad (1.30)$$

thanks to the bounded convergence theorem. To prove (1.30), let $\mathbf{e}_1, \dots, \mathbf{e}_d$ be a $g_{[\mathbf{A},s]}$ orthonormal basis where $d = pq + 1 - \frac{q(q-1)}{2}$. Every unit vector $\mathbf{u}$ can be expressed as $\mathbf{u} = \sum_{i=1}^d u_i \mathbf{e}_i$ with $\sum_{i=1}^d u_i^2 = 1$. By linearity of $D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})$ in $\mathbf{h}$, we have

$$\begin{aligned} \|D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| &\leq \sum_{i=1}^d |u_i| \|D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| \\ &\leq \sqrt{d} \max_{i=1, \dots, d} \|D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\|. \end{aligned}$$

Taking supremum yields

$$\sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| \leq \sqrt{d} \max_{i=1, \dots, d} \|D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\|.$$

Given that $D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) \rightarrow D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v})$ in $L^2(\mathbb{R}^p)$ as $\mathbf{h} \rightarrow \mathbf{0}$ for every i , every t , and every $\mathbf{v}$, the proof of Fréchet differentiability is complete.

Finally, the formulas (1.23), (1.24), and (1.25) simplifies greatly when $\mathbf{v} = \mathbf{0}$:

$$\begin{aligned} D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{0})(\mathbf{x}) &= \frac{1}{2}r_{[\mathbf{A},s]}(\mathbf{0})(\mathbf{x})D_{\mathbf{h}}l_{[\mathbf{A},s]}(\mathbf{0})(\mathbf{x}) \\ &= \frac{1}{4}\sqrt{p(\mathbf{x}; [\mathbf{A}, s])} \text{tr} \left\{ (\Sigma([\mathbf{A}, s]))^{-1} [\mathbf{x}\mathbf{x}^T - (\Sigma([\mathbf{A}, s]))] (\Sigma([\mathbf{A}, s]))^{-1} (D_{\mathbf{h}}\Sigma([\mathbf{A}, s])) \right\}. \end{aligned}$$

The score $s(\mathbf{x}; [\mathbf{A}, s])(\mathbf{h})$ therefore equals to $D_{\mathbf{h}}l_{[\mathbf{A},s]}(\mathbf{0})(\mathbf{x})$ and is indeed given by (1.20). The Fisher information operator (1.22) is obtained by taking $\mathbf{v} = \mathbf{0}$ in (1.26). To show that $G_{[\mathbf{A},s]}$ is positive definite, let $\mathbf{h} \in T_{[\mathbf{A},s]}M$ be

given. We show that $G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}) \geq 0$ and equality holds only when $\mathbf{h} = \mathbf{0}$. The non-negative part is clear from (1.22) (1.22):

$$G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \text{tr} \left\{ [(\Sigma([\mathbf{A}, s]))^{-1} D_{\mathbf{h}} \Sigma([\mathbf{A}, s])]^2 \right\} \geq 0.$$

Assume from now on that the equality holds. A slight annoyance here is that the product of two symmetric matrices is not necessarily symmetric and we cannot immediately conclude that $D_{\mathbf{h}} \Sigma([\mathbf{A}, s]) = \mathbf{0}$. We easily overcome this difficulty by using the cyclic property of the trace to get a symmetric matrix

$$\mathbf{N} := (\Sigma([\mathbf{A}, s]))^{-\frac{1}{2}} D_{\mathbf{h}} \Sigma([\mathbf{A}, s]) (\Sigma([\mathbf{A}, s]))^{-\frac{1}{2}}$$

such that

$$0 = G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \text{tr}(\mathbf{N}^2).$$

Therefore, $\mathbf{N} = \mathbf{0}$ and $D_{\mathbf{h}} \Sigma([\mathbf{A}, s]) = \mathbf{0}$. Now (1.21) reads the same as (1.8) in the proof of Proposition 1. We therefore get $(\mathbf{U}, b) \in \ker(d\Phi_{(\mathbf{A}, s)}) = V_{(\mathbf{A}, s)}$. However, being lifted from $\mathbf{h}$, $(\mathbf{U}, b)$ must also be a horizontal vector. Therefore, $\mathbf{h} = \mathbf{0}$. We have proven that $G_{[\mathbf{A}, s]}$ is positive definite. $\square$

Proposition 8. Suppose that $[\mathbf{A}, s] \in M$ is the true parameter so that each $\mathbf{X}_i$ has density $p(\cdot; [\mathbf{A}, s])$ as in (1.17). We have the local asymptotic normality expansion

$$\sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \text{Exp}_{[\mathbf{A}, s]}\left(\frac{\mathbf{h}}{\sqrt{n}}\right)\right)}{p(\mathbf{X}_i; [\mathbf{A}, s])} = \sum_{i=1}^n s(\mathbf{X}_i; [\mathbf{A}, s]) \left(\frac{\mathbf{h}}{\sqrt{n}}\right) - \frac{1}{2} G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}) + o_{\mathbb{P}}(1) \quad (1.31)$$

whenever n large enough that $\frac{\mathbf{h}}{\sqrt{n}} \in \mathcal{E}_{[\mathbf{A}, s]}$. In particular,

$$\sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \text{Exp}_{[\mathbf{A}, s]}\left(\frac{\mathbf{h}}{\sqrt{n}}\right)\right)}{p(\mathbf{X}_i; [\mathbf{A}, s])} \rightarrow N\left(-\frac{1}{2} G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}), G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h})\right) \quad (1.32)$$

in distribution as $n \rightarrow \infty$.

Proof. Given that $\mathcal{P}$ is differentiable in quadratic mean and has positive definite Fisher operator, this follows from Proposition III.1 in [Sun et al.(2026)Sun, Lin, and Liu]. $\square$

1.4 Properties of the Maximum Likelihood Estimator of $\mathcal{P}$

We now move to the core of the subject.

Chapter 2

Draft

From here on we change the names of the coordinates on Θ and write a point of Θ as $(\mathbf{W}, \sigma^2)$, with $\mathbf{W}$ the loading matrix and σ^2 the noise variance of (1.1), and a point of M as $[\mathbf{W}, \sigma^2]$, following the notation of [Datta and Chakrabarty(2023)] and [Datta(2026)]. This is nothing but a change of names — the generic representative called $(\mathbf{A}, s)$ in the previous sections is now called $(\mathbf{W}, \sigma^2)$, so that $\Sigma([\mathbf{W}, \sigma^2]) = \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}_p$ by (1.5), and every statement proved so far applies verbatim — but it is the notation in which the statistical results of this section are usually quoted, and it frees the letter s for the score (1.20). Orthonormal eigenvector matrices, which in this section are extracted from the sample covariance matrix rather than from the parameter, are correspondingly denoted by $\mathbf{K}$ and $\mathbf{K}_q$ below. Singular values keep the notation $\sigma_j(\cdot)$ and $\sigma_{\min}(\cdot)$ of Section 1.1; they always carry an explicit matrix argument, so they are never confused with the noise variance σ^2 .

Throughout this section we fix a *true parameter* $[\mathbf{W}_0, \sigma_0^2] \in M$, we assume that $\mathbf{X}_1, \mathbf{X}_2, \dots$ are independent and identically distributed random vectors with common density $p(\cdot; [\mathbf{W}_0, \sigma_0^2])$ as in (1.17), and we abbreviate

$$\Sigma_0 := \Sigma([\mathbf{W}_0, \sigma_0^2]) = \mathbf{W}_0\mathbf{W}_0^T + \sigma_0^2\mathbf{I}_p, \quad \mathbf{S}_n := \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i\mathbf{X}_i^T, \quad d := pq + 1 - \frac{q(q-1)}{2} = \dim M. \quad (2.1)$$

We write $\rightsquigarrow$ for convergence in distribution and, following (1.4), we abbreviate $|\mathbf{v}| := \sqrt{g_{[\mathbf{W}, \sigma^2]}(\mathbf{v}, \mathbf{v})}$ for $\mathbf{v} \in T_{[\mathbf{W}, \sigma^2]}M$.

The whole point of the construction of Section 1.1 is that the naive statement “ $\sqrt{n}(\hat{\mathbf{W}}_n - \mathbf{W}_0)$ is asymptotically normal” is not merely false but meaningless: by non-identifiability the estimator $\hat{\mathbf{W}}_n$ produced by any algorithm is only determined up to a right multiplication by an element of $O(q)$, so that $\hat{\mathbf{W}}_n$ need not converge to $\mathbf{W}_0$ at all, and only its orbit can. The correct object to look at is the intrinsic estimation error $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1}([\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]) \in T_{[\mathbf{W}_0, \sigma_0^2]}M$ of [Sun et al.(2026)Sun, Lin, and Liu], which by Proposition 5 is the ordinary error *after optimal rotational alignment*. The goal of this section is to prove that this tangent vector is asymptotically linear in the sense of [Sun et al.(2026)Sun, Lin, and Liu], Lemma III.3, with influence operator $G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}s(\cdot; [\mathbf{W}_0, \sigma_0^2])$, and consequently asymptotically normal with covariance operator $G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}$.

It will be convenient to name the horizontal lift. By Proposition 1 and the definition (1.4) of g , the restriction of $d\pi_{(\mathbf{W}, \sigma^2)}$ to $H_{(\mathbf{W}, \sigma^2)}$ is a linear isometry onto $T_{[\mathbf{W}, \sigma^2]}M$; we denote its inverse by

$$\iota_{(\mathbf{W}, \sigma^2)} := \left(d\pi_{(\mathbf{W}, \sigma^2)}|_{H_{(\mathbf{W}, \sigma^2)}} \right)^{-1} : T_{[\mathbf{W}, \sigma^2]}M \longrightarrow H_{(\mathbf{W}, \sigma^2)}, \quad \iota_{(\mathbf{W}, \sigma^2)}(\mathbf{h}) =: (\mathbf{U}_{\mathbf{h}}, b_{\mathbf{h}}), \quad (2.2)$$

so that $\iota_{(\mathbf{W}, \sigma^2)}$ is a *linear* isometry and $|\mathbf{h}|^2 = \text{tr}(\mathbf{U}_{\mathbf{h}}\mathbf{U}_{\mathbf{h}}^T) + b_{\mathbf{h}}^2$. Finally, since $G_{[\mathbf{W}, \sigma^2]}$ is a positive definite bilinear form on $T_{[\mathbf{W}, \sigma^2]}M$ by Theorem 7, the map $\mathbf{h} \mapsto G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \cdot)$ is a linear isomorphism from $T_{[\mathbf{W}, \sigma^2]}M$ onto its dual $T_{[\mathbf{W}, \sigma^2]}^*M$; we write $G_{[\mathbf{W}, \sigma^2]}^{-1}$ for its inverse, characterized by

$$G_{[\mathbf{W}, \sigma^2]} \left(G_{[\mathbf{W}, \sigma^2]}^{-1}(\varphi), \mathbf{h} \right) = \varphi(\mathbf{h}) \quad \text{for all } \varphi \in T_{[\mathbf{W}, \sigma^2]}^*M \text{ and all } \mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M. \quad (2.3)$$

Since the score $s(\mathbf{x}; [\mathbf{W}, \sigma^2])$ of (1.20) is a linear functional on $T_{[\mathbf{W}, \sigma^2]}M$, the composition $G_{[\mathbf{W}, \sigma^2]}^{-1}s(\mathbf{x}; [\mathbf{W}, \sigma^2])$ is a genuine tangent vector.

Preliminaries on the sample covariance matrix

Before anything else we collect the elementary properties of the sample covariance matrix $\mathbf{S}_n$ of (2.1) that will be used over and over below. The reason for gathering them here is that the sample enters the problem exclusively through $\mathbf{S}_n$ — the log-likelihood (2.16) is a function of $\mathbf{S}_n$ alone — and that the maximum likelihood estimator produced by Proposition 15 is a function of nothing but the eigenvalues of $\mathbf{S}_n$ and the eigenspace attached to the q largest of them. Everything in this subsection is classical and the proofs are short; they are given so that the section remains self-contained and so that the statements are available in the notation of this document.

We write $\text{Sym}(p)$ for the vector space of real symmetric $p \times p$ matrices, which we equip with the Frobenius inner product $\langle \mathbf{B}, \mathbf{C} \rangle := \text{tr}(\mathbf{B}\mathbf{C})$ and the associated norm $\|\cdot\|_F$, so that $\text{Sym}(p)$ becomes a Euclidean space of dimension $\frac{p(p+1)}{2}$; as in Section 1.1, $\|\cdot\|$ continues to denote the spectral norm. For $\mathbf{B} \in \text{Sym}(p)$ we write $\lambda_1(\mathbf{B}) \geq \dots \geq \lambda_p(\mathbf{B})$ for the eigenvalues of $\mathbf{B}$, repeated according to multiplicity and arranged in non-increasing order, and we write $\mathbf{B} \succeq \mathbf{0}$, respectively $\mathbf{B} \succ \mathbf{0}$, to say that $\mathbf{B}$ is positive semi-definite, respectively positive definite. The first lemma is pure linear algebra and applies to any symmetric matrix.

Lemma 9. Let $\mathbf{S} \in \text{Sym}(p)$ and write $\lambda_j := \lambda_j(\mathbf{S})$.

1. (Spectral decomposition and its ambiguity.) There is $\mathbf{K} = (\mathbf{k}_1, \dots, \mathbf{k}_p) \in O(p)$ with

$$\mathbf{S} = \mathbf{K} \text{diag}(\lambda_1, \dots, \lambda_p) \mathbf{K}^T = \sum_{j=1}^p \lambda_j \mathbf{k}_j \mathbf{k}_j^T. \quad (2.4)$$

The eigenvalues are uniquely determined by $\mathbf{S}$, and so are the orthogonal projectors $\Pi_1, \dots, \Pi_r$ onto the eigenspaces $E_1, \dots, E_r$ of the distinct eigenvalues $\mu_1 > \dots > \mu_r$ of $\mathbf{S}$, since

$$\Pi_k = \prod_{l \neq k} \frac{\mathbf{S} - \mu_l \mathbf{I}_p}{\mu_k - \mu_l}, \quad k = 1, \dots, r. \quad (2.5)$$

The matrix $\mathbf{K}$, by contrast, is *not* determined by $\mathbf{S}$: writing $m_k := \dim E_k$, a matrix $\mathbf{K}' \in O(p)$ satisfies (2.4) if and only if $\mathbf{K}' = \mathbf{K} \text{diag}(\mathbf{R}_1, \dots, \mathbf{R}_r)$ for some $\mathbf{R}_k \in O(m_k)$.

2. (Variational characterization.) For every $k \in \{1, \dots, p\}$,

$$\lambda_k = \max_{\substack{V \subseteq \mathbb{R}^p \\ \dim V = k}} \min_{\substack{\mathbf{v} \in V \\ \mathbf{v}^T \mathbf{v} = 1}} \mathbf{v}^T \mathbf{S} \mathbf{v} = \min_{\substack{V \subseteq \mathbb{R}^p \\ \dim V = p-k+1}} \max_{\substack{\mathbf{v} \in V \\ \mathbf{v}^T \mathbf{v} = 1}} \mathbf{v}^T \mathbf{S} \mathbf{v}; \quad (2.6)$$

in particular $\lambda_1 = \max_{\mathbf{v}^T \mathbf{v} = 1} \mathbf{v}^T \mathbf{S} \mathbf{v}$ is a convex function of $\mathbf{S}$ and $\lambda_p = \min_{\mathbf{v}^T \mathbf{v} = 1} \mathbf{v}^T \mathbf{S} \mathbf{v}$ is a concave function of $\mathbf{S}$.

3. (Ky Fan.) Let $1 \leq q < p$. For every $\mathbf{U} \in \mathbb{R}^{p \times q}$ with $\mathbf{U}^T \mathbf{U} = \mathbf{I}_q$,

$$\text{tr}(\mathbf{U}^T \mathbf{S} \mathbf{U}) \leq \sum_{j=1}^q \lambda_j. \quad (2.7)$$

If moreover $\lambda_q > \lambda_{q+1}$, then equality holds in (2.7) if and only if the column space of $\mathbf{U}$ equals $\text{span}(\mathbf{k}_1, \dots, \mathbf{k}_q)$, a subspace which by 1 does not depend on the choice of $\mathbf{K}$ in (2.4).

4. (Weyl.) For every $\mathbf{T} \in \text{Sym}(p)$,

$$\max_{1 \leq k \leq p} |\lambda_k(\mathbf{S}) - \lambda_k(\mathbf{T})| \leq \|\mathbf{S} - \mathbf{T}\|. \quad (2.8)$$

In particular each $\lambda_k : \text{Sym}(p) \rightarrow \mathbb{R}$ is 1-Lipschitz, hence continuous.

Proof. 1. The existence of (2.4) is the spectral theorem. Formula (2.5) is checked on eigenvectors: if $\mathbf{v} \in E_m$ then the right hand side maps $\mathbf{v}$ to $\left(\prod_{l \neq k} \frac{\mu_m - \mu_l}{\mu_k - \mu_l}\right) \mathbf{v}$, which is $\mathbf{v}$ when $m = k$ and 0 when $m \neq k$, the factor indexed by $l = m$ vanishing; since $\mathbb{R}^p = E_1 \oplus \dots \oplus E_r$ orthogonally, this determines both sides. As for $\mathbf{K}$, the columns of a matrix satisfying (2.4) are exactly orthonormal eigenvectors listed so that the associated eigenvalues are non-increasing, that is, consecutive blocks of $m_1, \dots, m_r$ columns forming orthonormal bases of $E_1, \dots, E_r$; two orthonormal bases of the same m_k -dimensional space differ precisely by right multiplication by an element of $O(m_k)$.

2. This is the Courant–Fischer min-max theorem ([Horn and Johnson(2013)], Section 4.2). The convexity of λ_1 and the concavity of λ_p follow because $\mathbf{S} \mapsto \mathbf{v}^T \mathbf{S} \mathbf{v}$ is linear for each fixed $\mathbf{v}$, and a pointwise supremum, respectively infimum, of linear functions is convex, respectively concave.

3. Put $\mathbf{M} := \mathbf{K}^T \mathbf{U}$, which again satisfies $\mathbf{M}^T \mathbf{M} = \mathbf{I}_q$, and let $c_j := \sum_{k=1}^q M_{jk}^2$ be the squared norm of its j -th row, that is the j -th diagonal entry of the rank q orthogonal projector $\mathbf{M} \mathbf{M}^T$; hence $c_j \in [0, 1]$ and $\sum_{j=1}^p c_j = \text{tr}(\mathbf{M} \mathbf{M}^T) = q$. By (2.4), $\text{tr}(\mathbf{U}^T \mathbf{S} \mathbf{U}) = \text{tr}(\mathbf{M}^T \text{diag}(\lambda_1, \dots, \lambda_p) \mathbf{M}) = \sum_{j=1}^p \lambda_j c_j$. Using $c_j - 1 \leq 0$ and $\lambda_j \geq \lambda_q$ for $j \leq q$, and $c_j \geq 0$ and $\lambda_j \leq \lambda_{q+1}$ for $j > q$, we get

$$\sum_{j=1}^p \lambda_j c_j - \sum_{j=1}^q \lambda_j = \sum_{j \leq q} \lambda_j (c_j - 1) + \sum_{j > q} \lambda_j c_j \leq \lambda_q \sum_{j \leq q} (c_j - 1) + \lambda_{q+1} \sum_{j > q} c_j = -(\lambda_q - \lambda_{q+1}) \sum_{j > q} c_j \leq 0,$$

where the last equality used $\sum_{j \leq q} (c_j - 1) = -\sum_{j > q} c_j$. This is (2.7). If $\lambda_q > \lambda_{q+1}$, the displayed chain shows that equality forces $c_j = 0$ for every $j > q$, that is, the last $p - q$ rows of $\mathbf{M}$ vanish, that is, the column space of $\mathbf{U} = \mathbf{K} \mathbf{M}$ is contained in $\text{span}(\mathbf{k}_1, \dots, \mathbf{k}_q)$, hence equal to it by equality of dimensions. Conversely, if the two spaces coincide then $\mathbf{U} = (\mathbf{k}_1, \dots, \mathbf{k}_q) \mathbf{R}$ for some $\mathbf{R} \in O(q)$ and $\text{tr}(\mathbf{U}^T \mathbf{S} \mathbf{U}) = \text{tr}(\mathbf{R}^T \text{diag}(\lambda_1, \dots, \lambda_q) \mathbf{R}) = \sum_{j \leq q} \lambda_j$.

4. Write $\mathbf{E} := \mathbf{T} - \mathbf{S}$. For every unit vector $\mathbf{v}$ we have $\mathbf{v}^T \mathbf{T} \mathbf{v} \leq \mathbf{v}^T \mathbf{S} \mathbf{v} + \lambda_1(\mathbf{E})$, so taking minima over a k -dimensional subspace and then the maximum over such subspaces in (2.6) gives $\lambda_k(\mathbf{T}) \leq \lambda_k(\mathbf{S}) + \lambda_1(\mathbf{E})$; exchanging the roles of $\mathbf{S}$ and $\mathbf{T}$ gives $\lambda_k(\mathbf{S}) \leq \lambda_k(\mathbf{T}) - \lambda_p(\mathbf{E})$. Hence $|\lambda_k(\mathbf{T}) - \lambda_k(\mathbf{S})| \leq \max(|\lambda_1(\mathbf{E})|, |\lambda_p(\mathbf{E})|) = \|\mathbf{E}\|$, the spectral norm of a symmetric matrix being the largest modulus of its eigenvalues. $\square$

Remark 10. Part 3 of Lemma 9 is the reason the word *principal* appears in the name of the model. For any $\mathbf{U} \in \mathbb{R}^{p \times q}$ with $\mathbf{U}^T \mathbf{U} = \mathbf{I}_q$ the matrix $\mathbf{U} \mathbf{U}^T$ is the orthogonal projector onto the column space of $\mathbf{U}$, so that

$$\frac{1}{n} \sum_{i=1}^n \|\mathbf{X}_i - \mathbf{U} \mathbf{U}^T \mathbf{X}_i\|^2 = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i^T (\mathbf{I}_p - \mathbf{U} \mathbf{U}^T) \mathbf{X}_i = \text{tr}(\mathbf{S}_n) - \text{tr}(\mathbf{U}^T \mathbf{S}_n \mathbf{U}) : \quad (2.9)$$

minimizing the mean squared reconstruction error over q -dimensional subspaces is the same as maximizing the retained empirical variance $\text{tr}(\mathbf{U}^T \mathbf{S}_n \mathbf{U})$, and by (2.7) the optimum is $\sum_{j \leq q} \lambda_j(\mathbf{S}_n)$, attained exactly at the span of the q leading eigenvectors of $\mathbf{S}_n$ as soon as $\lambda_q(\mathbf{S}_n) > \lambda_{q+1}(\mathbf{S}_n)$. Note also that part 1 already contains, in embryo, the non-identifiability that forced the quotient construction of Section 1.1: what the data determine is the leading eigenspace, whereas an orthonormal basis $\mathbf{K}_q$ of it is determined only up to right multiplication by an element of $O(q)$. This is exactly the ambiguity $\hat{\mathbf{W}} = \mathbf{K}_q (\mathbf{\Lambda}_q - \hat{\sigma}^2 \mathbf{I}_q)^{\frac{1}{2}} \mathbf{R}$ recorded in Proposition 15, and it disappears on $\mathbf{M}$. Finally, when a group of eigenvalues is separated from the rest, the projector onto the corresponding invariant subspace is not merely well defined but real analytic in $\mathbf{S}$; this is taken up in Lemma 21 on the set $\mathcal{O}$ of (2.17), which by part 4 above is open.

We now turn to the properties of $\mathbf{S}_n$ as a random matrix under the sampling scheme fixed at the beginning of this section.

Lemma 11. Let $\mathbf{X}_1, \mathbf{X}_2, \dots$ be independent and identically distributed $N_p(\mathbf{0}, \Sigma_0)$ random vectors and let $\mathbf{S}_n$ be as in (2.1).

1. $\mathbf{S}_n \in \text{Sym}(p)$ and $\mathbf{S}_n \succeq \mathbf{0}$, with $\text{rank}(\mathbf{S}_n) \leq \min(n, p)$; in particular $\mathbf{S}_n \succ \mathbf{0}$ is impossible when $n < p$.
2. $n\mathbf{S}_n$ follows the Wishart law $W_p(n, \Sigma_0)$, and for $n \geq p$ its distribution has a density with respect to the Lebesgue measure of $\text{Sym}(p)$ ([Gupta and Nagar(2000)], Chapter 3).
3. $\mathbf{S}_n$ is an unbiased estimator of Σ_0 , and for all $\mathbf{B}, \mathbf{C} \in \text{Sym}(p)$ and all $i, j, k, l \in \{1, \dots, p\}$,

$$\begin{aligned} \mathbb{E}[\mathbf{S}_n] &= \Sigma_0, \quad \text{Cov}(\text{tr}(\mathbf{B}\mathbf{S}_n), \text{tr}(\mathbf{C}\mathbf{S}_n)) = \frac{2}{n} \text{tr}(\mathbf{B}\Sigma_0\mathbf{C}\Sigma_0), \\ \text{Cov}((\mathbf{S}_n)_{ij}, (\mathbf{S}_n)_{kl}) &= \frac{1}{n} ((\Sigma_0)_{ik}(\Sigma_0)_{jl} + (\Sigma_0)_{il}(\Sigma_0)_{jk}). \end{aligned} \quad (2.10)$$

4. $\mathbf{S}_n$ is a sufficient statistic for $\mathcal{P}$: the joint density of $(\mathbf{X}_1, \dots, \mathbf{X}_n)$ at $[\mathbf{W}, \sigma^2] \in M$ is

$$\begin{aligned} \prod_{i=1}^n p(\mathbf{x}_i; [\mathbf{W}, \sigma^2]) &= (2\pi)^{-\frac{np}{2}} (\det \Sigma([\mathbf{W}, \sigma^2]))^{-\frac{n}{2}} \exp\left(-\frac{n}{2} \text{tr}\left((\Sigma([\mathbf{W}, \sigma^2]))^{-1} \mathbf{s}_n\right)\right), \\ \mathbf{s}_n &:= \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^T. \end{aligned} \quad (2.11)$$

5. $\mathbf{S}_n \rightarrow \Sigma_0$ almost surely, and

$$\sqrt{n}(\mathbf{S}_n - \Sigma_0) \rightsquigarrow \mathbf{G} \quad \text{in } \text{Sym}(p), \quad (2.12)$$

where $\mathbf{G}$ is the centred Gaussian random element of $\text{Sym}(p)$ with $\text{Cov}(\langle \mathbf{B}, \mathbf{G} \rangle, \langle \mathbf{C}, \mathbf{G} \rangle) = 2 \text{tr}(\mathbf{B}\Sigma_0\mathbf{C}\Sigma_0)$ for $\mathbf{B}, \mathbf{C} \in \text{Sym}(p)$. In particular $\|\mathbf{S}_n - \Sigma_0\| = O_{\mathbb{P}}(n^{-\frac{1}{2}})$.

6. Consequently $\max_{1 \leq k \leq p} |\lambda_k(\mathbf{S}_n) - \lambda_k(\Sigma_0)| \rightarrow 0$ almost surely, and this quantity is $O_{\mathbb{P}}(n^{-\frac{1}{2}})$. The convergence of the eigenvalues is not monotone in any useful sense, but it is biased outwards in the mean: $\mathbb{E}[\lambda_1(\mathbf{S}_n)] \geq \lambda_1(\Sigma_0)$ and $\mathbb{E}[\lambda_p(\mathbf{S}_n)] \leq \lambda_p(\Sigma_0)$.

Proof. 1. Symmetry is clear and $\mathbf{v}^T \mathbf{S}_n \mathbf{v} = \frac{1}{n} \sum_{i=1}^n (\mathbf{X}_i^T \mathbf{v})^2 \geq 0$; the column space of $\mathbf{S}_n$ is contained in the span of $\mathbf{X}_1, \dots, \mathbf{X}_n$, whence the bound on the rank.

2. By definition, a sum of n independent $\mathbf{X}_i \mathbf{X}_i^T$ with $\mathbf{X}_i \sim N_p(\mathbf{0}, \Sigma_0)$ is $W_p(n, \Sigma_0)$ distributed, and this law is non-singular, with the usual density on the cone of positive definite matrices, as soon as $n \geq p$ and $\Sigma_0 \succ \mathbf{0}$; the latter holds by Lemma 12 below. The absolute continuity of the law of $\mathbf{S}_n$ is re-derived by a more elementary route, avoiding any Wishart computation, in the proof of Lemma 23.

3. Since $\mathbb{E}[\mathbf{X}_1] = \mathbf{0}$ we have $\mathbb{E}[\mathbf{X}_1 \mathbf{X}_1^T] = \Sigma_0$, and averaging gives the first identity of (2.10). For the second, note that $\text{tr}(\mathbf{B}\mathbf{S}_n) = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i^T \mathbf{B} \mathbf{X}_i$ is an average of independent quadratic forms, so that, by Theorem 7.7.2 of [Gupta and Nagar(2000)] already used in the proof of Theorem 7,

$$\text{Cov}(\text{tr}(\mathbf{B}\mathbf{S}_n), \text{tr}(\mathbf{C}\mathbf{S}_n)) = \frac{1}{n} \text{Cov}(\mathbf{X}_1^T \mathbf{B} \mathbf{X}_1, \mathbf{X}_1^T \mathbf{C} \mathbf{X}_1) = \frac{2}{n} \text{tr}(\mathbf{B}\Sigma_0\mathbf{C}\Sigma_0).$$

The entrywise formula is the special case $\mathbf{B} = \frac{1}{2}(\mathbf{e}_i \mathbf{e}_j^T + \mathbf{e}_j \mathbf{e}_i^T)$ and $\mathbf{C} = \frac{1}{2}(\mathbf{e}_k \mathbf{e}_l^T + \mathbf{e}_l \mathbf{e}_k^T)$, for which $\text{tr}(\mathbf{B}\mathbf{S}_n) = (\mathbf{S}_n)_{ij}$, $\text{tr}(\mathbf{C}\mathbf{S}_n) = (\mathbf{S}_n)_{kl}$ and, expanding the four resulting terms and using the symmetry of Σ_0 , $\text{tr}(\mathbf{B}\Sigma_0\mathbf{C}\Sigma_0) = \frac{1}{2}((\Sigma_0)_{ik}(\Sigma_0)_{jl} + (\Sigma_0)_{il}(\Sigma_0)_{jk})$.

4. Immediate from (1.17) and the identity $\mathbf{x}^T \mathbf{B} \mathbf{x} = \text{tr}(\mathbf{B} \mathbf{x} \mathbf{x}^T)$, which turns $\sum_{i=1}^n \mathbf{x}_i^T (\boldsymbol{\Sigma}([\mathbf{W}, \sigma^2]))^{-1} \mathbf{x}_i$ into $n \text{tr}((\boldsymbol{\Sigma}([\mathbf{W}, \sigma^2]))^{-1} \mathbf{s}_n)$; sufficiency then follows from the Fisher–Neyman factorization criterion, the right hand side of (2.11) depending on the sample only through $\mathbf{s}_n$.

5. The entries of $\mathbf{X}_1 \mathbf{X}_1^T$ are products of two jointly Gaussian variables, hence integrable with finite second moments, so the strong law of large numbers applied to each of the finitely many entries gives $\mathbf{S}_n \rightarrow \boldsymbol{\Sigma}_0$ almost surely, and the multivariate central limit theorem ([van der Vaart(1998)], Chapter 2) applied to the independent and identically distributed random elements $\mathbf{X}_i \mathbf{X}_i^T$ of the $\frac{p(p+1)}{2}$ -dimensional Euclidean space $\text{Sym}(p)$ gives (2.12), the covariance of the limit being that of $\mathbf{X}_1 \mathbf{X}_1^T$, computed in 3. The last assertion follows because all norms on $\text{Sym}(p)$ are equivalent and a weakly convergent sequence is uniformly tight.

6. The first two claims follow from (2.8) and 5. For the last two, λ_1 is convex and λ_p is concave on $\text{Sym}(p)$ by part 2 of Lemma 9, so Jensen's inequality and $\mathbb{E}[\mathbf{S}_n] = \boldsymbol{\Sigma}_0$ give $\mathbb{E}[\lambda_1(\mathbf{S}_n)] \geq \lambda_1(\mathbb{E}[\mathbf{S}_n]) = \lambda_1(\boldsymbol{\Sigma}_0)$ and the reverse inequality for λ_p ; the expectations are finite because $|\lambda_k(\mathbf{S}_n)| \leq \text{tr}(\mathbf{S}_n)$, which is integrable. $\square$

It remains to record the spectrum of the matrix that $\mathbf{S}_n$ approximates. The point of the next lemma is that $\boldsymbol{\Sigma}_0$ has exactly the *spiked* shape — q eigenvalues above a flat floor of height σ_0^2 — which the model postulates, and that the corresponding spectral gap is strictly positive; by Lemma 11 the sample covariance matrix inherits both features for n large.

Lemma 12. Let $\mathbf{W}_0 = \mathbf{U}_0 \mathbf{D}_0 \mathbf{V}_0^T$ be a singular value decomposition of $\mathbf{W}_0$, with $\mathbf{U}_0 \in \mathbb{R}^{p \times q}$ having orthonormal columns, $\mathbf{D}_0 = \text{diag}(\sigma_1(\mathbf{W}_0), \dots, \sigma_q(\mathbf{W}_0))$ and $\mathbf{V}_0 \in O(q)$, and let $\mathbf{P}_0 := \mathbf{U}_0 \mathbf{U}_0^T$ be the orthogonal projector onto the column space of $\mathbf{W}_0$. Then

$$\boldsymbol{\Sigma}_0 = \mathbf{U}_0 (\mathbf{D}_0^2 + \sigma_0^2 \mathbf{I}_q) \mathbf{U}_0^T + \sigma_0^2 (\mathbf{I}_p - \mathbf{P}_0), \quad \boldsymbol{\Sigma}_0^{-1} = \mathbf{U}_0 (\mathbf{D}_0^2 + \sigma_0^2 \mathbf{I}_q)^{-1} \mathbf{U}_0^T + \frac{1}{\sigma_0^2} (\mathbf{I}_p - \mathbf{P}_0), \quad (2.13)$$

so that the eigenvalues of $\boldsymbol{\Sigma}_0$ are $\lambda_j(\boldsymbol{\Sigma}_0) = \sigma_j(\mathbf{W}_0)^2 + \sigma_0^2$ for $j = 1, \dots, q$ and $\lambda_j(\boldsymbol{\Sigma}_0) = \sigma_0^2$ for $j = q+1, \dots, p$. Consequently $\boldsymbol{\Sigma}_0 \succeq \sigma_0^2 \mathbf{I}_p \succ \mathbf{0}$, $\|\boldsymbol{\Sigma}_0^{-1}\| = \sigma_0^{-2}$, $\det \boldsymbol{\Sigma}_0 = \sigma_0^{2(p-q)} \prod_{j=1}^q (\sigma_j(\mathbf{W}_0)^2 + \sigma_0^2)$, the eigenspace of the q largest eigenvalues is the column space of $\mathbf{W}_0$, and

$$\lambda_q(\boldsymbol{\Sigma}_0) - \lambda_{q+1}(\boldsymbol{\Sigma}_0) = \sigma_{\min}(\mathbf{W}_0)^2 =: \delta_0 > 0. \quad (2.14)$$

Moreover, almost surely there is a (random) N such that for all $n \geq N$,

$$\mathbf{S}_n \succeq \frac{\sigma_0^2}{2} \mathbf{I}_p \succ \mathbf{0} \quad \text{and} \quad \lambda_q(\mathbf{S}_n) - \lambda_{q+1}(\mathbf{S}_n) \geq \frac{\delta_0}{2}. \quad (2.15)$$

Proof. Since $\mathbf{W}_0 \mathbf{W}_0^T = \mathbf{U}_0 \mathbf{D}_0^2 \mathbf{U}_0^T$ and $\mathbf{I}_p = \mathbf{P}_0 + (\mathbf{I}_p - \mathbf{P}_0)$, the first identity of (2.13) follows from $\boldsymbol{\Sigma}_0 = \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 \mathbf{I}_p$; completing the columns of $\mathbf{U}_0$ to a basis of $\mathbb{R}^p$ by an orthonormal basis of the orthogonal complement of the column space of $\mathbf{W}_0$ exhibits (2.13) as a spectral decomposition (2.4), whence the list of eigenvalues, the identification of the eigenspace of the q largest ones with the column space of $\mathbf{W}_0$, and the formula for the determinant. Since $\text{rank}(\mathbf{W}_0) = q$ we have $\sigma_j(\mathbf{W}_0) > 0$ for all $j \leq q$, so all q leading eigenvalues are strictly larger than σ_0^2 , which is therefore the smallest eigenvalue; this gives $\boldsymbol{\Sigma}_0 \succeq \sigma_0^2 \mathbf{I}_p$, $\|\boldsymbol{\Sigma}_0^{-1}\| = \lambda_p(\boldsymbol{\Sigma}_0)^{-1} = \sigma_0^{-2}$ and (2.14). The second identity of (2.13) is checked by multiplying the two expressions and using $\mathbf{U}_0^T \mathbf{U}_0 = \mathbf{I}_q$ together with $\mathbf{U}_0^T (\mathbf{I}_p - \mathbf{P}_0) = \mathbf{0}$.

For (2.15), let $\epsilon := \frac{1}{4} \min(\sigma_0^2, \delta_0) > 0$. By part 5 of Lemma 11 we have almost surely $\|\mathbf{S}_n - \boldsymbol{\Sigma}_0\| \leq \epsilon$ for all large n , and then (2.8) gives $\lambda_p(\mathbf{S}_n) \geq \lambda_p(\boldsymbol{\Sigma}_0) - \epsilon = \sigma_0^2 - \epsilon \geq \frac{\sigma_0^2}{2}$ as well as $\lambda_q(\mathbf{S}_n) - \lambda_{q+1}(\mathbf{S}_n) \geq \lambda_q(\boldsymbol{\Sigma}_0) - \lambda_{q+1}(\boldsymbol{\Sigma}_0) - 2\epsilon \geq \frac{\delta_0}{2}$. $\square$

The maximum likelihood estimator

The log-likelihood of the model (1.6) is the function on M given by

$$\mathbb{L}_n([\mathbf{W}, \sigma^2]) := \sum_{i=1}^n \log p(\mathbf{X}_i; [\mathbf{W}, \sigma^2]) = -\frac{np}{2} \log(2\pi) - \frac{n}{2} \log \det \Sigma([\mathbf{W}, \sigma^2]) - \frac{n}{2} \operatorname{tr} \left((\Sigma([\mathbf{W}, \sigma^2]))^{-1} \mathbf{S}_n \right); \quad (2.16)$$

it depends on the sample only through $\mathbf{S}_n$ and it is genuinely a function on the quotient M , since the right hand side depends on $(\mathbf{W}, \sigma^2)$ only through Σ . This is precisely the reason why the model had to be quotiented in the first place: $\mathbb{L}_n$ is exactly constant along the fibers of π , so no maximizer of $\mathbb{L}_n$ on Θ can ever be unique, whereas on M it can. The following classical computation identifies the maximizer. We write $\lambda_1(\mathbf{S}) \geq \dots \geq \lambda_p(\mathbf{S})$ for the eigenvalues of $\mathbf{S} \in \operatorname{Sym}(p)$, the space of real symmetric $p \times p$ matrices, repeated according to multiplicity, and we let

$$\mathcal{O} := \{\mathbf{S} \in \operatorname{Sym}(p) : \lambda_q(\mathbf{S}) > \lambda_{q+1}(\mathbf{S})\}, \quad (2.17)$$

which is an open subset of $\operatorname{Sym}(p)$ because the eigenvalues depend continuously on $\mathbf{S}$. For $\mathbf{S} \in \mathcal{O}$ we write $\mathbf{P}_q(\mathbf{S})$ for the orthogonal projector of $\mathbb{R}^p$ onto the sum of the eigenspaces of $\lambda_1(\mathbf{S}), \dots, \lambda_q(\mathbf{S})$; this is well defined precisely because of the spectral gap in (2.17).

The next lemma collects the identities that let one pass between the projector form and the eigenvector form of the maximizer below. They are less innocent than they look: the second one uses that the columns of $\mathbf{K}_q$ are eigenvectors and not merely an orthonormal basis of the range of $\mathbf{P}_q$, as Remark 14 shows.

Lemma 13. Let $\mathbf{S} \in \mathcal{O}$, put $\lambda_j := \lambda_j(\mathbf{S})$, $\mathbf{P}_q := \mathbf{P}_q(\mathbf{S})$ and $\Lambda_q := \operatorname{diag}(\lambda_1, \dots, \lambda_q)$, and let $\mathbf{K}_q \in \mathbb{R}^{p \times q}$ be a matrix whose columns are orthonormal eigenvectors of $\mathbf{S}$ associated with $\lambda_1, \dots, \lambda_q$ in this order, that is

$$\mathbf{K}_q^T \mathbf{K}_q = \mathbf{I}_q \quad \text{and} \quad \mathbf{S} \mathbf{K}_q = \mathbf{K}_q \Lambda_q. \quad (2.18)$$

Then the column space of $\mathbf{K}_q$ is the range of $\mathbf{P}_q$, the projector $\mathbf{P}_q$ commutes with $\mathbf{S}$, and

$$\mathbf{P}_q = \mathbf{K}_q \mathbf{K}_q^T, \quad \mathbf{S} \mathbf{P}_q = \mathbf{P}_q \mathbf{S} = \mathbf{P}_q \mathbf{S} \mathbf{P}_q = \mathbf{K}_q \Lambda_q \mathbf{K}_q^T, \quad \mathbf{P}_q \mathbf{S} \mathbf{P}_q - c \mathbf{P}_q = \mathbf{K}_q (\Lambda_q - c \mathbf{I}_q) \mathbf{K}_q^T \quad (2.19)$$

for every $c \in \mathbb{R}$. Finally, a matrix $\mathbf{K}'_q$ satisfies (2.18) if and only if $\mathbf{K}'_q = \mathbf{K}_q \mathbf{R}$ for some $\mathbf{R} \in O(q)$ commuting with Λ_q , that is block diagonal along the blocks of equal values in $(\lambda_1, \dots, \lambda_q)$; in particular none of the matrices in (2.19) depends on the choice of $\mathbf{K}_q$.

Proof. Let E denote the sum of the eigenspaces of $\lambda_1, \dots, \lambda_q$, so that $\mathbf{P}_q$ is by definition the orthogonal projector onto E . Because $\lambda_q > \lambda_{q+1}$, no eigenvalue λ_j with $j > q$ is equal to any of $\lambda_1, \dots, \lambda_q$, so the total multiplicity of the latter is exactly q and $\dim E = q$. The columns of $\mathbf{K}_q$ are eigenvectors associated with those eigenvalues, hence lie in E , and they are orthonormal, hence span a q -dimensional subspace of E ; therefore the column space of $\mathbf{K}_q$ is E . Since $\mathbf{K}_q^T \mathbf{K}_q = \mathbf{I}_q$, the matrix $\mathbf{K}_q \mathbf{K}_q^T$ is symmetric and idempotent with range E , that is $\mathbf{P}_q = \mathbf{K}_q \mathbf{K}_q^T$, which is the first identity.

For the second, $\mathbf{S} \mathbf{P}_q = \mathbf{S} \mathbf{K}_q \mathbf{K}_q^T = \mathbf{K}_q \Lambda_q \mathbf{K}_q^T$ by (2.18); this matrix is symmetric, so $\mathbf{P}_q \mathbf{S} = (\mathbf{S} \mathbf{P}_q)^T = \mathbf{S} \mathbf{P}_q$, which is the asserted commutation, and $\mathbf{P}_q \mathbf{S} \mathbf{P}_q = \mathbf{K}_q \mathbf{K}_q^T \mathbf{K}_q \Lambda_q \mathbf{K}_q^T = \mathbf{K}_q \Lambda_q \mathbf{K}_q^T$ again by $\mathbf{K}_q^T \mathbf{K}_q = \mathbf{I}_q$. Subtracting $c \mathbf{P}_q = c \mathbf{K}_q \mathbf{K}_q^T$ gives the third identity.

Suppose now that $\mathbf{K}'_q$ also satisfies (2.18). By the first part both matrices have column space E , so $\mathbf{K}'_q = \mathbf{K}_q \mathbf{R}$ with $\mathbf{R} := \mathbf{K}_q^T \mathbf{K}'_q$, and $\mathbf{R}^T \mathbf{R} = \mathbf{K}_q'^T \mathbf{K}_q \mathbf{K}_q^T \mathbf{K}'_q = \mathbf{K}_q'^T \mathbf{P}_q \mathbf{K}'_q = \mathbf{I}_q$ because the columns of $\mathbf{K}'_q$ lie in E ; thus $\mathbf{R} \in O(q)$. Applying $\mathbf{S}$ gives $\mathbf{K}_q \Lambda_q \mathbf{R} = \mathbf{S} \mathbf{K}_q \mathbf{R} = \mathbf{S} \mathbf{K}'_q = \mathbf{K}'_q \Lambda_q = \mathbf{K}_q \mathbf{R} \Lambda_q$, and multiplying on the left by $\mathbf{K}_q^T$ yields $\Lambda_q \mathbf{R} = \mathbf{R} \Lambda_q$; comparing entries, $(\lambda_i - \lambda_j) R_{ij} = 0$, so $R_{ij} = 0$ whenever $\lambda_i \neq \lambda_j$, which is the asserted block structure. Conversely such an $\mathbf{R}$ clearly returns a matrix satisfying (2.18). Independence of the choice is then immediate, since by (2.19) every matrix occurring there is expressed through $\mathbf{S}$ and $\mathbf{P}_q$ alone. $\square$

Remark 14. The hypothesis that the columns of $\mathbf{K}_q$ be eigenvectors, and not merely an orthonormal basis of the range of $\mathbf{P}_q$, cannot be dropped from the second identity of (2.19), even though it is irrelevant for the first. If $\mathbf{U} = \mathbf{K}_q \mathbf{R}$ with $\mathbf{R} \in O(q)$ arbitrary then $\mathbf{U}\mathbf{U}^T = \mathbf{P}_q$ still, whereas $\mathbf{U}\mathbf{\Lambda}_q\mathbf{U}^T = \mathbf{K}_q \mathbf{R}\mathbf{\Lambda}_q \mathbf{R}^T \mathbf{K}_q^T$ equals $\mathbf{K}_q \mathbf{\Lambda}_q \mathbf{K}_q^T$ only when $\mathbf{R}$ commutes with $\mathbf{\Lambda}_q$. For instance, for $p = 3$, $q = 2$ and $\mathbf{S} = \text{diag}(2, 1, \frac{1}{2}) \in \mathcal{O}$, the vectors $\frac{1}{\sqrt{2}}(1, 1, 0)^T$ and $\frac{1}{\sqrt{2}}(1, -1, 0)^T$ form an orthonormal basis of the range of $\mathbf{P}_2 = \text{diag}(1, 1, 0)$, but the corresponding matrix $\mathbf{U}\mathbf{\Lambda}_2\mathbf{U}^T$ has upper left 2×2 block $\begin{pmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{pmatrix}$, whereas $\mathbf{P}_2 \mathbf{S} \mathbf{P}_2$ has upper left block $\text{diag}(2, 1)$. This is also the reason why the ordering requirement in (2.18) matters: the eigenvalues must be attached to the eigenvectors that carry them.

Proposition 15 (Tipping–Bishop). Let $\mathbf{S} \in \mathcal{O}$ be positive definite and put $\lambda_j := \lambda_j(\mathbf{S})$ and $\mathbf{P}_q := \mathbf{P}_q(\mathbf{S})$. Then the function

$$[\mathbf{W}, \sigma^2] \mapsto -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma([\mathbf{W}, \sigma^2]) - \frac{1}{2} \text{tr} \left((\Sigma([\mathbf{W}, \sigma^2]))^{-1} \mathbf{S} \right)$$

attains its maximum over M at one and only one point $[\hat{\mathbf{W}}, \hat{\sigma}^2] \in M$, determined by

$$\hat{\sigma}^2 = \frac{1}{p-q} \sum_{j=q+1}^p \lambda_j, \quad \hat{\mathbf{W}} \hat{\mathbf{W}}^T = \mathbf{P}_q \mathbf{S} \mathbf{P}_q - \hat{\sigma}^2 \mathbf{P}_q. \quad (2.20)$$

Concretely, $\hat{\mathbf{W}} = \mathbf{K}_q (\mathbf{\Lambda}_q - \hat{\sigma}^2 \mathbf{I}_q)^{\frac{1}{2}} \mathbf{R}$ for an arbitrary $\mathbf{R} \in O(q)$, where the columns of $\mathbf{K}_q \in \mathbb{R}^{p \times q}$ are orthonormal eigenvectors of $\mathbf{S}$ associated with $\lambda_1, \dots, \lambda_q$ and $\mathbf{\Lambda}_q = \text{diag}(\lambda_1, \dots, \lambda_q)$.

Proof. This is Section 3.2 together with Appendix A of [Tipping and Bishop(1999)]: equations (7) and (8) there give the stationary points and equations (17)–(20) of their Appendix A.2 show that the global maximum is attained exactly when the q largest eigenvalues of $\mathbf{S}$ are retained. Since $\lambda_q > \lambda_{q+1}$ the set of retained eigenvalues, and hence the projector $\mathbf{P}_q$, is unambiguous, so the maximizer is a single point of M . The two descriptions agree by Lemma 13: the last identity of (2.19) with $c = \hat{\sigma}^2$ gives

$$\mathbf{P}_q \mathbf{S} \mathbf{P}_q - \hat{\sigma}^2 \mathbf{P}_q = \mathbf{K}_q (\mathbf{\Lambda}_q - \hat{\sigma}^2 \mathbf{I}_q) \mathbf{K}_q^T = \left(\mathbf{K}_q (\mathbf{\Lambda}_q - \hat{\sigma}^2 \mathbf{I}_q)^{\frac{1}{2}} \mathbf{R} \right) \left(\mathbf{K}_q (\mathbf{\Lambda}_q - \hat{\sigma}^2 \mathbf{I}_q)^{\frac{1}{2}} \mathbf{R} \right)^T = \hat{\mathbf{W}} \hat{\mathbf{W}}^T$$

for every $\mathbf{R} \in O(q)$. Finally $[\hat{\mathbf{W}}, \hat{\sigma}^2]$ is really a point of M : $\hat{\sigma}^2 > 0$ because $\mathbf{S}$ is positive definite, and $\text{rank}(\hat{\mathbf{W}}) = q$ because $\hat{\sigma}^2 \leq \lambda_{q+1} < \lambda_q$ forces every diagonal entry of $\mathbf{\Lambda}_q - \hat{\sigma}^2 \mathbf{I}_q$ to be strictly positive. $\square$

Definition 16. Whenever $\mathbf{S}_n$ is positive definite and lies in $\mathcal{O}$, we let $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \in M$ denote the maximum likelihood estimator, that is the unique maximizer of (2.16) over M supplied by Proposition 15 applied to $\mathbf{S} = \mathbf{S}_n$. On the complementary event we set $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] := [\mathbf{W}_0, \sigma_0^2]$; Lemma 23 below shows that this convention is invoked only on a null set as soon as $n \geq p$.

The fitted covariance matrix and the exact fit equation

It is worth recording what the maximum likelihood estimator does to the covariance matrix rather than to the parameter, if only because (2.16) immediately suggests a competitor: instead of maximizing the likelihood, why not simply ask the model to reproduce the sample covariance matrix, that is, why not solve

$$\mathbf{S} = \mathbf{W} \mathbf{W}^T + \sigma^2 \mathbf{I}_p, \quad [\mathbf{W}, \sigma^2] \in M? \quad (2.21)$$

The next two propositions answer this: the maximum likelihood fit is the sample covariance matrix with its trailing eigenvalues flattened, (2.21) is almost surely unsolvable, and its least squares relaxation returns the maximum likelihood estimator again.

Proposition 17. Let $S \in \mathcal{O}$ be positive definite, let $[\hat{W}, \hat{\sigma}^2]$ be the maximizer supplied by Proposition 15 and let

$$\hat{\Sigma} := \Sigma([\hat{W}, \hat{\sigma}^2]) = \hat{W}\hat{W}^T + \hat{\sigma}^2 I_p \quad (2.22)$$

be the *fitted covariance matrix*. Write $\lambda_j := \lambda_j(S)$, let $\mathbf{k}_1, \dots, \mathbf{k}_p$ be orthonormal eigenvectors of S ordered as in (2.4) and put $P_q := P_q(S)$. Then

1. $\hat{\Sigma}$ has the same eigenvectors as S , keeps its q largest eigenvalues and replaces the remaining $p - q$ by their average $\hat{\sigma}^2$:

$$\hat{\Sigma} = \sum_{j=1}^q \lambda_j \mathbf{k}_j \mathbf{k}_j^T + \hat{\sigma}^2 \sum_{j=q+1}^p \mathbf{k}_j \mathbf{k}_j^T, \quad \hat{\Sigma} - S = \sum_{j=q+1}^p (\hat{\sigma}^2 - \lambda_j) \mathbf{k}_j \mathbf{k}_j^T. \quad (2.23)$$

2. $\text{tr } \hat{\Sigma} = \text{tr } S$ and $P_q \hat{\Sigma} P_q = P_q S P_q$; moreover $\hat{\Sigma}$ is positive definite, lies in $\mathcal{O}$, satisfies $P_q(\hat{\Sigma}) = P_q(S)$, and the map $S \mapsto \hat{\Sigma}$ is idempotent.

3. The fit is exact if and only if the trailing spectrum of S is flat:

$$\hat{\Sigma} = S \iff \lambda_{q+1} = \lambda_{q+2} = \dots = \lambda_p \iff S \in \Sigma(M), \quad (2.24)$$

that is, if and only if (2.21) is solvable, in which case $[\hat{W}, \hat{\sigma}^2]$ is its unique solution in M .

Proof. 1. Lemma 13 applied to $K_q := (\mathbf{k}_1, \dots, \mathbf{k}_q)$, which satisfies (2.18) by (2.4), gives $P_q = \sum_{j \leq q} \mathbf{k}_j \mathbf{k}_j^T$ and $P_q S P_q = \sum_{j \leq q} \lambda_j \mathbf{k}_j \mathbf{k}_j^T$, so (2.20) gives

$$\hat{\Sigma} = (P_q S P_q - \hat{\sigma}^2 P_q) + \hat{\sigma}^2 (P_q + (I_p - P_q)) = \sum_{j \leq q} \lambda_j \mathbf{k}_j \mathbf{k}_j^T + \hat{\sigma}^2 \sum_{j > q} \mathbf{k}_j \mathbf{k}_j^T,$$

and subtracting $S = \sum_{j=1}^p \lambda_j \mathbf{k}_j \mathbf{k}_j^T$ gives the second identity of (2.23).

2. Taking traces in (2.23) and using $\hat{\sigma}^2 = \frac{1}{p-q} \sum_{j>q} \lambda_j$ gives $\text{tr } \hat{\Sigma} = \sum_{j \leq q} \lambda_j + (p - q) \hat{\sigma}^2 = \text{tr } S$, and $P_q \hat{\Sigma} P_q = \sum_{j \leq q} \lambda_j \mathbf{k}_j \mathbf{k}_j^T = P_q S P_q$ because the $\mathbf{k}_j$ are orthonormal. The eigenvalues of $\hat{\Sigma}$ are $\lambda_1, \dots, \lambda_q$ and $\hat{\sigma}^2$ with multiplicity $p - q$; they are all strictly positive, and $\hat{\sigma}^2 \leq \lambda_{q+1} < \lambda_q$ shows both that $\hat{\Sigma} \in \mathcal{O}$ and that its leading q -dimensional eigenspace is $\text{span}(\mathbf{k}_1, \dots, \mathbf{k}_q)$, that is $P_q(\hat{\Sigma}) = P_q(S)$. Idempotence follows: the trailing eigenvalues of $\hat{\Sigma}$ are already all equal to $\hat{\sigma}^2$, so applying (2.23) to $\hat{\Sigma}$ reproduces it.

3. By the second identity of (2.23) the difference $\hat{\Sigma} - S$ is symmetric with eigenvalues $\hat{\sigma}^2 - \lambda_j$, $j > q$, in the directions $\mathbf{k}_{q+1}, \dots, \mathbf{k}_p$, so it vanishes if and only if $\lambda_j = \hat{\sigma}^2$ for every $j > q$, that is if and only if the trailing eigenvalues are all equal, in which case their common value is their average $\hat{\sigma}^2$. If this holds then $S = \hat{\Sigma} \in \Sigma(M)$; conversely, if $S = \Sigma([\mathbf{W}, \sigma^2])$ for some $[\mathbf{W}, \sigma^2] \in M$, then Lemma 12 applied to $(\mathbf{W}, \sigma^2)$ in place of $(\mathbf{W}_0, \sigma_0^2)$ shows that the $p - q$ smallest eigenvalues of S all equal σ^2 . Uniqueness of the solution is uniqueness of the maximizer in Proposition 15, since any solution of (2.21) satisfies $\Sigma([\mathbf{W}, \sigma^2]) = S = \hat{\Sigma}$ and Σ is injective on M by construction. $\square$

Remark 18. Two consequences of (2.24) deserve to be separated. If $q = p - 1$ the trailing spectrum consists of the single eigenvalue λ_p , the flatness condition is vacuous, and the fit is always exact: $\hat{\Sigma} = S$, and (2.21) is always solvable. This is not an accident, since $\dim M = p(p - 1) + 1 - \frac{(p-1)(p-2)}{2} = \frac{p(p+1)}{2} = \dim \text{Sym}(p)$ when $q = p - 1$: the model is then saturated and imposes no constraint on Σ beyond positive definiteness and a simple smallest eigenvalue. If $q < p - 1$, by contrast, an exact fit forces S to have a repeated eigenvalue. Matrices with a repeated eigenvalue form a Lebesgue null subset of $\text{Sym}(p)$, being the zero set of the discriminant of the characteristic polynomial as in the proof of Lemma 21, and the law of S_n is absolutely continuous for $n \geq p$ by Lemma 23; hence almost surely $\hat{\Sigma}_n := \Sigma([\hat{W}_n, \hat{\sigma}_n^2]) \neq S_n$ and (2.21) has no solution at all. The two matrices are nevertheless close: by (2.23), (2.8) and part 5 of Lemma 11,

$$\|\hat{\Sigma}_n - S_n\| = \max_{j>q} |\hat{\sigma}_n^2 - \lambda_j(S_n)| \leq 2 \max_{j>q} |\lambda_j(S_n) - \sigma_0^2| \leq 2 \|S_n - \Sigma_0\| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right),$$

the middle inequality holding because $\hat{\sigma}_n^2$ is an average of the $\lambda_j(\mathbf{S}_n)$, $j > q$. The discrepancy disappears in the limit precisely because the population matrix does have a flat trailing spectrum, $\lambda_{q+1}(\boldsymbol{\Sigma}_0) = \dots = \lambda_p(\boldsymbol{\Sigma}_0) = \sigma_0^2$, by Lemma 12.

Since (2.21) is almost surely unsolvable, the natural way to rescue the estimator it was meant to define is to require it only approximately, that is to replace it by the minimum distance, or least squares, estimator

$$[\tilde{\mathbf{W}}_n, \tilde{\sigma}_n^2] := \arg \min_{[\mathbf{W}, \sigma^2] \in M} \|\mathbf{S}_n - \mathbf{W}\mathbf{W}^T - \sigma^2 \mathbf{I}_p\|_F^2. \quad (2.25)$$

This turns out to produce nothing new.

Proposition 19. Let $\mathbf{S} \in \mathcal{O}$ be positive definite and keep the notation of Proposition 17. Then

$$\min_{[\mathbf{W}, \sigma^2] \in M} \|\mathbf{S} - \mathbf{W}\mathbf{W}^T - \sigma^2 \mathbf{I}_p\|_F^2 = \sum_{j=q+1}^p (\lambda_j - \hat{\sigma}^2)^2, \quad (2.26)$$

the minimum being attained at $[\hat{\mathbf{W}}, \hat{\sigma}^2]$, and every minimizer has $\sigma^2 = \hat{\sigma}^2$. In particular the least squares estimator (2.25) may be taken to be the maximum likelihood estimator of Definition 16.

Proof. Write $\mathbf{B} := \mathbf{W}\mathbf{W}^T$, which ranges exactly over the positive semi-definite matrices of rank q as $[\mathbf{W}, \sigma^2]$ ranges over M with $\sigma^2 > 0$ fixed, put $\mathbf{C} := \mathbf{S} - \sigma^2 \mathbf{I}_p$ and $\mathbf{R} := \mathbf{C} - \mathbf{B}$, and note $\lambda_j(\mathbf{C}) = \lambda_j - \sigma^2$. We first bound $\|\mathbf{R}\|_F^2$ from below for fixed σ^2 . Since $\mathbf{B} \succeq \mathbf{0}$ we have $\mathbf{v}^T \mathbf{R} \mathbf{v} \leq \mathbf{v}^T \mathbf{C} \mathbf{v}$ for every $\mathbf{v}$, so (2.6) gives

$$\lambda_j(\mathbf{R}) \leq \lambda_j(\mathbf{C}) \quad \text{for } j = 1, \dots, p. \quad (2.27)$$

Moreover $K := \ker \mathbf{B}$ has dimension at least $p - q$, and for a subspace V with $\dim V = j + q$ the intersection $V \cap K$ has dimension at least j ; choosing $W \subseteq V \cap K$ with $\dim W = j$ and using $\mathbf{v}^T \mathbf{B} \mathbf{v} = 0$ on K ,

$$\min_{\mathbf{v} \in V, \mathbf{v}^T \mathbf{v} = 1} \mathbf{v}^T \mathbf{C} \mathbf{v} \leq \min_{\mathbf{v} \in W, \mathbf{v}^T \mathbf{v} = 1} \mathbf{v}^T \mathbf{C} \mathbf{v} = \min_{\mathbf{v} \in W, \mathbf{v}^T \mathbf{v} = 1} \mathbf{v}^T \mathbf{R} \mathbf{v} \leq \lambda_j(\mathbf{R}),$$

and taking the maximum over such V in (2.6),

$$\lambda_{j+q}(\mathbf{C}) \leq \lambda_j(\mathbf{R}) \quad \text{for } j = 1, \dots, p - q. \quad (2.28)$$

Let now $n_\sigma := \#\{j : \lambda_j \geq \sigma^2\}$. For $1 \leq j \leq n_\sigma - q$ we have $\lambda_{j+q}(\mathbf{C}) \geq 0$, so (2.28) gives $\lambda_j(\mathbf{R})^2 \geq (\lambda_{j+q} - \sigma^2)^2$, while for $j > n_\sigma$ we have $\lambda_j(\mathbf{C}) < 0$, so (2.27) gives $\lambda_j(\mathbf{R})^2 \geq (\sigma^2 - \lambda_j)^2$. These two ranges of indices are disjoint, whence

$$\|\mathbf{R}\|_F^2 = \sum_{j=1}^p \lambda_j(\mathbf{R})^2 \geq \sum_{j=q+1}^{n_\sigma} (\lambda_j - \sigma^2)^2 + \sum_{j>n_\sigma} (\sigma^2 - \lambda_j)^2 = \sum_{j \leq q} \left[(\sigma^2 - \lambda_j)_+ \right]^2 + \sum_{j>q} (\lambda_j - \sigma^2)^2 =: L(\sigma^2),$$

where $(x)_+ := \max(x, 0)$, the first sum being empty when $n_\sigma \leq q$; the last equality is checked separately in the cases $n_\sigma \geq q$ and $n_\sigma < q$. The function L is strictly convex and continuously differentiable on $\mathbb{R}$, with $L'(x) = 2 \sum_{j \leq q} (x - \lambda_j)_+ - 2 \sum_{j>q} (\lambda_j - x)$. Since $\hat{\sigma}^2 \leq \lambda_{q+1} \leq \lambda_j$ for $j \leq q$ the first sum vanishes at $x = \hat{\sigma}^2$, and the second vanishes there because $\hat{\sigma}^2$ is the average of $\lambda_{q+1}, \dots, \lambda_p$; hence $\hat{\sigma}^2$ is the unique minimizer of L , with $L(\hat{\sigma}^2) = \sum_{j>q} (\lambda_j - \hat{\sigma}^2)^2$. Therefore every $[\mathbf{W}, \sigma^2] \in M$ satisfies $\|\mathbf{S} - \mathbf{W}\mathbf{W}^T - \sigma^2 \mathbf{I}_p\|_F^2 \geq L(\sigma^2) \geq L(\hat{\sigma}^2)$, with the second inequality strict unless $\sigma^2 = \hat{\sigma}^2$. Finally the bound is attained at $[\hat{\mathbf{W}}, \hat{\sigma}^2]$: by (2.23), $\mathbf{S} - \hat{\mathbf{W}}\hat{\mathbf{W}}^T - \hat{\sigma}^2 \mathbf{I}_p = \mathbf{S} - \hat{\boldsymbol{\Sigma}} = \sum_{j>q} (\lambda_j - \hat{\sigma}^2) \mathbf{k}_j \mathbf{k}_j^T$, whose squared Frobenius norm is exactly $\sum_{j>q} (\lambda_j - \hat{\sigma}^2)^2$. $\square$

Remark 20. The coincidence in Proposition 19 is a small piece of good luck rather than a general principle. Maximizing (2.16) is, up to constants not depending on the parameter, the same as minimizing over M the Stein loss $\text{tr}(\Sigma^{-1}S_n) - \log \det(\Sigma^{-1}S_n) - p$ with $\Sigma = \Sigma([W, \sigma^2])$, which is a very different discrepancy from the squared Frobenius distance minimized in (2.25); that the two are minimized at the same point here reflects the special structure of (2.23), namely that the optimal fit acts only on the eigenvalues of S_n and leaves its eigenvectors alone. The practical conclusion is that the exact fit equation (2.21) defines no new estimator: it is either unsolvable, which is the generic case, or solved by the maximum likelihood estimator, and its least squares relaxation is the maximum likelihood estimator as well.

Genericity of the sample covariance matrix

Both the well-posedness of the estimation error and the consistency of the maximum likelihood estimator rest on the behaviour of the spectral projector P_q , which we now record.

Lemma 21.

1. $\mathcal{O}$ is open and connected in $\text{Sym}(p)$.
2. $P_q : \mathcal{O} \rightarrow \text{Sym}(p)$ is real analytic.
3. Let $U_0 \in \mathbb{R}^{p \times q}$ have orthonormal columns and define $\phi : \mathcal{O} \rightarrow \mathbb{R}$ by $\phi(S) := \det(U_0^T P_q(S) U_0)$. Then ϕ is real analytic and $\phi^{-1}(\{0\})$ is a Lebesgue null subset of $\text{Sym}(p) \cong \mathbb{R}^{\frac{p(p+1)}{2}}$.
4. Let $S_* \in \mathcal{O}$ and put

$$\delta_* := \lambda_q(S_*) - \lambda_{q+1}(S_*) > 0, \quad \rho_* := \lambda_1(S_*) - \lambda_{q+1}(S_*) \geq \delta_*. \quad (2.29)$$

Then every $S \in \text{Sym}(p)$ with $\|S - S_*\| \leq \frac{\delta_*}{8}$ belongs to $\mathcal{O}$ and satisfies

$$\|P_q(S) - P_q(S_*)\| \leq \frac{16\rho_*}{\delta_*^2} \|S - S_*\|. \quad (2.30)$$

In particular P_q is Lipschitz on a neighbourhood of every point of $\mathcal{O}$, with a constant controlled by the spectral gap at that point alone.

5. P_q is continuous on $\mathcal{O}$; concretely, if $S_* \in \mathcal{O}$ and $S_n \rightarrow S_*$ in $\text{Sym}(p)$, then $S_n \in \mathcal{O}$ for all large n and $P_q(S_n) \rightarrow P_q(S_*)$.

Proof. 1. Openness was already observed. For connectedness, fix $S \in \mathcal{O}$ and diagonalize $S = K \text{diag}(\lambda) K^T$ with $K \in O(p)$ and $\lambda = (\lambda_1(S), \dots, \lambda_p(S))$ non-increasing; replacing one column of K by its negative if necessary, which does not change S , we may assume $\det(K) = 1$. Let $\mu := (2, \dots, 2, 1, \dots, 1)$ with the value 2 repeated q times, and consider first the path $t \mapsto K \text{diag}((1-t)\lambda + t\mu) K^T$ for $t \in [0, 1]$. The vector $(1-t)\lambda + t\mu$ is non-increasing, being a convex combination of non-increasing vectors, so it lists the ordered eigenvalues, and the gap at position q equals $(1-t)(\lambda_q(S) - \lambda_{q+1}(S)) + t > 0$; the path therefore stays inside $\mathcal{O}$ and joins S to $K \text{diag}(\mu) K^T$. Next, $SO(p)$ is path connected, so there is a path $u \mapsto K(u)$ in $SO(p)$ from K to I_p ; the induced path $u \mapsto K(u) \text{diag}(\mu) K(u)^T$ has constant spectrum μ , hence constant gap $1 > 0$ at position q , stays in $\mathcal{O}$, and ends at $\text{diag}(\mu)$. Every point of $\mathcal{O}$ is thus joined to the fixed point $\text{diag}(\mu)$ by a path inside $\mathcal{O}$.

2. Fix $S_* \in \mathcal{O}$, choose $c \in \mathbb{R}$ with $\lambda_{q+1}(S_*) < c < \lambda_q(S_*)$ and let $\Gamma \subset \mathbb{C}$ be the positively oriented circle of centre $\frac{1}{2}(\lambda_1(S_*) + c)$ and radius $\frac{1}{2}(\lambda_1(S_*) - \lambda_{q+1}(S_*))$, which meets the real axis in the interval with endpoints $\frac{1}{2}(c + \lambda_{q+1}(S_*)) \in (\lambda_{q+1}(S_*), c)$ and $\lambda_1(S_*) + \frac{1}{2}(c - \lambda_{q+1}(S_*)) > \lambda_1(S_*)$, so that Γ encircles $\lambda_1(S_*), \dots, \lambda_q(S_*)$ and no other eigenvalue of S_* . Put $\delta := \min_{z \in \Gamma} \text{dist}(z, \text{spec}(S_*)) > 0$. If $S \in \text{Sym}(p)$ satisfies $\|S - S_*\|_2 < \delta$ then $zI_p - S$ is invertible for every $z \in \Gamma$, Weyl's inequality keeps exactly q eigenvalues of S inside Γ , and diagonalizing S shows, by the residue theorem applied to each eigenvalue separately, that

$$P_q(\mathbf{S}) = \frac{1}{2\pi i} \oint_{\Gamma} (z\mathbf{I}_p - \mathbf{S})^{-1} dz. \quad (2.31)$$

Expanding the resolvent in a Neumann series,

$$(z\mathbf{I}_p - \mathbf{S})^{-1} = \sum_{k=0}^{\infty} [(z\mathbf{I}_p - \mathbf{S}_*)^{-1}(\mathbf{S} - \mathbf{S}_*)]^k (z\mathbf{I}_p - \mathbf{S}_*)^{-1},$$

a series which converges absolutely and uniformly in $z \in \Gamma$ whenever $\|\mathbf{S} - \mathbf{S}_*\|_2 < \delta$, since $\|(z\mathbf{I}_p - \mathbf{S}_*)^{-1}\|_2 \leq \delta^{-1}$ there. Each summand is a homogeneous polynomial of degree k in the entries of $\mathbf{S} - \mathbf{S}_*$ whose coefficients are continuous in z , so integrating term by term over the compact contour Γ exhibits P_q as a convergent power series in the entries of $\mathbf{S} - \mathbf{S}_*$ on that ball.

The argument just given is the standard Riesz projection argument of analytic perturbation theory and is not new: it is carried out in [Kato(1995)], Chapter II, §1.4, where the projection associated with the eigenvalues enclosed by Γ is defined by the contour integral (2.31) and expanded, by term by term integration of the Neumann series of §1.3 there, into a convergent power series in the perturbation, and in §2.1 of the same chapter for a group of eigenvalues, which is the case needed here; §6.1 specializes the discussion to symmetric operators. The use of the contour integral in perturbation theory goes back to Sz.-Nagy and to Kato himself, replacing the earlier method of Rellich, as the footnote to §1.4 of [Kato(1995)] records. We have reproduced the proof because in the real symmetric finite-dimensional case it is short, and because part 4 below needs the resulting bound with an explicit constant.

3. Real analyticity of ϕ is immediate from 2, the determinant being a polynomial. Moreover ϕ does not vanish identically: taking $\mathbf{S}_* := \mathbf{I}_p + \mathbf{U}_0 \mathbf{U}_0^T$, whose eigenvalues are 2 with multiplicity q , with eigenspace the column space of $\mathbf{U}_0$, and 1 with multiplicity $p - q$, we get $\mathbf{S}_* \in \mathcal{O}$, $P_q(\mathbf{S}_*) = \mathbf{U}_0 \mathbf{U}_0^T$ and $\phi(\mathbf{S}_*) = \det(\mathbf{I}_q) = 1$. By 1 the domain $\mathcal{O}$ is open and connected, so the zero set of the non-trivial real analytic function ϕ is Lebesgue null in $\text{Sym}(p)$ by [Mityagin(2020)]. Finally $\text{Sym}(p) \setminus \mathcal{O}$ is itself Lebesgue null: it is contained in the zero set of the discriminant of the characteristic polynomial of $\mathbf{S}$, which is a polynomial in the entries of $\mathbf{S}$ and does not vanish at $\text{diag}(1, 2, \dots, p)$.

4. Continuity is already contained in 2; what has to be produced here is the explicit constant, and for that we specialize the contour of 2 to the midpoint $c := \frac{1}{2}(\lambda_q(\mathbf{S}_*) + \lambda_{q+1}(\mathbf{S}_*))$ of the spectral gap. Then Γ is the circle of centre $a := \frac{1}{2}(\lambda_1(\mathbf{S}_*) + c)$ and radius $r := \frac{\rho_*}{2}$, which meets the real axis at $a - r = \lambda_{q+1}(\mathbf{S}_*) + \frac{\delta_*}{4}$ and $a + r = \lambda_1(\mathbf{S}_*) + \frac{\delta_*}{4}$. A real number λ is at distance $|r - |\lambda - a||$ from Γ , and $\lambda \mapsto r - |\lambda - a|$ is concave, so on the interval $[\lambda_q(\mathbf{S}_*), \lambda_1(\mathbf{S}_*)]$, which contains $\lambda_1(\mathbf{S}_*), \dots, \lambda_q(\mathbf{S}_*)$, it is minimized at an endpoint, where it takes the values $a + r - \lambda_1(\mathbf{S}_*) = \frac{\delta_*}{4}$ and $\lambda_q(\mathbf{S}_*) - (a - r) = \frac{3\delta_*}{4}$; and every eigenvalue $\lambda_j(\mathbf{S}_*)$ with $j > q$ lies to the left of $a - r$ at distance at least $(a - r) - \lambda_{q+1}(\mathbf{S}_*) = \frac{\delta_*}{4}$. Hence the quantity δ of part 2 satisfies $\delta \geq \frac{\delta_*}{4}$.

Let now $\mathbf{E} := \mathbf{S} - \mathbf{S}_*$ with $\|\mathbf{E}\| \leq \frac{\delta_*}{8} \leq \frac{\delta}{2}$. By (2.8), $\lambda_q(\mathbf{S}) - \lambda_{q+1}(\mathbf{S}) \geq \delta_* - 2\|\mathbf{E}\| \geq \frac{3\delta_*}{4} > 0$, so $\mathbf{S} \in \mathcal{O}$, and every eigenvalue of $\mathbf{S}$ is within $\|\mathbf{E}\|$ of an eigenvalue of $\mathbf{S}_*$, so that $\text{dist}(z, \text{spec}(\mathbf{S})) \geq \delta - \|\mathbf{E}\| \geq \frac{\delta}{2}$ for $z \in \Gamma$. Both resolvents are therefore defined on Γ and (2.31) holds for $\mathbf{S}$ as well as for $\mathbf{S}_*$. Diagonalizing shows that a real symmetric matrix has $\|(z\mathbf{I}_p - \mathbf{S})^{-1}\| = \text{dist}(z, \text{spec}(\mathbf{S}))^{-1}$ for every $z \notin \text{spec}(\mathbf{S})$, so subtracting the two copies of (2.31), inserting the resolvent identity

$$(z\mathbf{I}_p - \mathbf{S})^{-1} - (z\mathbf{I}_p - \mathbf{S}_*)^{-1} = (z\mathbf{I}_p - \mathbf{S})^{-1} \mathbf{E} (z\mathbf{I}_p - \mathbf{S}_*)^{-1},$$

and estimating the integrand along the contour, of length $2\pi r$, gives

$$\|P_q(\mathbf{S}) - P_q(\mathbf{S}_*)\| \leq \frac{1}{2\pi} (2\pi r) \frac{2}{\delta} \cdot \frac{1}{\delta} \|\mathbf{E}\| = \frac{2r}{\delta^2} \|\mathbf{E}\| \leq \frac{16\rho_*}{\delta_*^2} \|\mathbf{E}\|,$$

using $r = \frac{\rho_*}{2}$ and $\delta \geq \frac{\delta_*}{4}$.

5. This is contained in 2, and also in 4, but it needs neither and may be quoted directly: it is Theorem 5.1 of [Kato(1995)], Chapter II, §5.1, a section entitled “Continuity of the eigenvalues and the total projection”, which

asserts that when the operator depends continuously on its argument its eigenvalues are continuous and so is the total projection attached to the group of eigenvalues encircled by a fixed contour; the discussion immediately preceding that theorem states the conclusion in the form used here, for a sequence of operators converging to a limit. One applies it to each of the distinct eigenvalues of $\mathbf{S}_*$ among $\lambda_1(\mathbf{S}_*), \dots, \lambda_q(\mathbf{S}_*)$, finitely many, whose total projections sum to $\mathbf{P}_q$; that $\mathbf{S}_n$ eventually lies in $\mathcal{O}$, so that $\mathbf{P}_q$ may be evaluated at it at all, is (2.8) once more. $\square$

Remark 22. The factor ρ_* in (2.30) is an artefact of estimating a circular contour that must enclose the whole leading part of the spectrum; the sharp form of the estimate, the Davis–Kahan $\sin \Theta$ theorem, replaces $\frac{16\rho_*}{\delta_*^2}$ by $\frac{2}{\delta_*}$. We do not need the sharp constant anywhere: only the fact that the Lipschitz constant is finite and depends on $\mathbf{S}_*$ solely through its spectrum.

Lemma 23. Suppose $n \geq p$. Then, almost surely, $\mathbf{S}_n$ is positive definite, $\mathbf{S}_n \in \mathcal{O}$, and the $q \times q$ matrix $\mathbf{W}_0^T \hat{\mathbf{W}}_n$ is invertible. In particular the orthogonal matrix

$$\mathbf{Q}_n := \mathbf{Q}(\mathbf{W}_0, \hat{\mathbf{W}}_n) = \left(\hat{\mathbf{W}}_n^T \mathbf{W}_0 \mathbf{W}_0^T \hat{\mathbf{W}}_n \right)^{-\frac{1}{2}} \hat{\mathbf{W}}_n^T \mathbf{W}_0 \in O(q) \quad (2.32)$$

of Proposition 5 is almost surely well defined.

Proof. We first check that the law of $\mathbf{S}_n$ is absolutely continuous with respect to the Lebesgue measure of $\text{Sym}(p)$. Let $\mathbf{Y}_1 \in \mathbb{R}^{p \times p}$ be the matrix whose rows are $\mathbf{X}_1^T, \dots, \mathbf{X}_p^T$ and let $\mathbf{Y}_2$ collect the remaining $n - p$ observations, so that $\mathbf{S}_n = \frac{1}{n} (\mathbf{Y}_1^T \mathbf{Y}_1 + \mathbf{Y}_2^T \mathbf{Y}_2)$ and $\mathbf{Y}_1$ is independent of $\mathbf{Y}_2$ with a strictly positive Lebesgue density on $\mathbb{R}^{p \times p}$. Condition on $\mathbf{Y}_2$ and consider $T(\mathbf{Y}) := \frac{1}{n} (\mathbf{Y}^T \mathbf{Y} + \mathbf{Y}_2^T \mathbf{Y}_2)$ on the open set of invertible $\mathbf{Y} \in \mathbb{R}^{p \times p}$, whose complement is Lebesgue null. Its differential at $\mathbf{Y}$ is $\mathbf{H} \mapsto \frac{1}{n} (\mathbf{H}^T \mathbf{Y} + \mathbf{Y}^T \mathbf{H})$, which is onto $\text{Sym}(p)$: given $\mathbf{B} \in \text{Sym}(p)$, the choice $\mathbf{H} = \frac{n}{2} \mathbf{Y}^{-T} \mathbf{B}$ returns $\mathbf{B}$. Thus T is a smooth submersion, and by the rank theorem ([Lee(2013)], Theorem 4.12) every point has a neighbourhood on which T is, up to diffeomorphisms, a linear projection $\mathbb{R}^{p^2} \rightarrow \mathbb{R}^{\frac{p(p+1)}{2}}$. Since preimages of Lebesgue null sets under such projections are Lebesgue null by Fubini's theorem, and diffeomorphisms preserve Lebesgue null sets, $T^{-1}(N)$ is Lebesgue null for every Lebesgue null $N \subseteq \text{Sym}(p)$, after covering the domain by countably many such neighbourhoods. Hence the conditional law of $\mathbf{S}_n$ given $\mathbf{Y}_2$ charges no Lebesgue null set, and the same holds for its mixture over $\mathbf{Y}_2$.

By Lemma 21, the set of $\mathbf{S} \in \text{Sym}(p)$ that either lie outside $\mathcal{O}$ or satisfy $\phi(\mathbf{S}) = 0$ is Lebesgue null; here we take $\mathbf{U}_0 \in \mathbb{R}^{p \times q}$ to be any matrix with orthonormal columns spanning the column space of $\mathbf{W}_0$, which has dimension q because $\text{rank}(\mathbf{W}_0) = q$. It follows that almost surely $\mathbf{S}_n \in \mathcal{O}$ and $\phi(\mathbf{S}_n) \neq 0$. Positive definiteness, by contrast, is not a generic property in $\text{Sym}(p)$ and must come from the structure of $\mathbf{S}_n$ instead: the singular matrices form a Lebesgue null subset of $\mathbb{R}^{p \times p}$, so $\mathbf{Y}_1$ is almost surely invertible and

$$\mathbf{S}_n = \frac{1}{n} \mathbf{Y}_1^T \mathbf{Y}_1 + \frac{1}{n} \mathbf{Y}_2^T \mathbf{Y}_2 \succeq \frac{1}{n} \mathbf{Y}_1^T \mathbf{Y}_1 \succ \mathbf{0}$$

almost surely.

It remains to see that $\phi(\mathbf{S}_n) \neq 0$ is exactly the invertibility of $\mathbf{W}_0^T \hat{\mathbf{W}}_n$. Write $\mathbf{W}_0 = \mathbf{U}_0 \mathbf{T}$ with $\mathbf{T} \in \mathbb{R}^{q \times q}$ invertible, which is possible since the columns of $\mathbf{U}_0$ form a basis of the column space of $\mathbf{W}_0$, and use the representative $\hat{\mathbf{W}}_n = \mathbf{K}_q (\Lambda_q - \hat{\sigma}_n^2 \mathbf{I}_q)^{\frac{1}{2}}$ of Proposition 15; note that invertibility of $\mathbf{W}_0^T \hat{\mathbf{W}}_n$ is unchanged if $\hat{\mathbf{W}}_n$ is replaced by $\hat{\mathbf{W}}_n \mathbf{R}$ with $\mathbf{R} \in O(q)$, so it is a property of the class $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$. Since $\Lambda_q - \hat{\sigma}_n^2 \mathbf{I}_q$ is invertible, we get

$$\mathbf{W}_0^T \hat{\mathbf{W}}_n \text{ is invertible} \iff \mathbf{U}_0^T \mathbf{K}_q \text{ is invertible} \iff \det(\mathbf{U}_0^T \mathbf{K}_q \mathbf{K}_q^T \mathbf{U}_0) \neq 0 \iff \phi(\mathbf{S}_n) \neq 0,$$

where we used $\det(\mathbf{U}_0^T \mathbf{K}_q)^2 = \det(\mathbf{U}_0^T \mathbf{K}_q \mathbf{K}_q^T \mathbf{U}_0)$ and $\mathbf{K}_q \mathbf{K}_q^T = \mathbf{P}_q(\mathbf{S}_n)$. $\square$

Consistency and the aligned estimation error

Proposition 24. Almost surely, $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \rightarrow [\mathbf{W}_0, \sigma_0^2]$ in M .

Proof. Write P_0 for the orthogonal projector onto the column space of W_0 , so that $P_0 W_0 = W_0$, and abbreviate

$$P_n := P_q(S_n), \quad (2.33)$$

the orthogonal projector of $\mathbb{R}^p$ onto the sum of the eigenspaces of the q largest eigenvalues of S_n , which is defined as soon as $S_n \in \mathcal{O}$. By Lemma 12 the eigenvalues of Σ_0 are $\sigma_j(W_0)^2 + \sigma_0^2$ for $j = 1, \dots, q$ and σ_0^2 with multiplicity $p - q$, the eigenspace of the q largest ones is the column space of W_0 , and the gap of Σ_0 at position q is $\delta_0 = \sigma_{\min}(W_0)^2 > 0$ by (2.14); its spread, in the sense of (2.29), is

$$\rho_0 := \lambda_1(\Sigma_0) - \lambda_{q+1}(\Sigma_0) = \sigma_1(W_0)^2. \quad (2.34)$$

In particular $\delta_0 > 0$ says that $\Sigma_0 \in \mathcal{O}$, so that $P_q(\Sigma_0)$ is defined, and we claim that

$$P_q(\Sigma_0) = P_0. \quad (2.35)$$

Indeed, let $W_0 = U_0 D_0 V_0^T$ be the singular value decomposition used in Lemma 12, so that the columns of U_0 are orthonormal and span the column space of W_0 , whence $P_0 = U_0 U_0^T$, and $W_0 W_0^T = U_0 D_0^2 U_0^T$. Then

$$\Sigma_0 U_0 = W_0 W_0^T U_0 + \sigma_0^2 U_0 = U_0 (D_0^2 + \sigma_0^2 I_q) = U_0 \Lambda_q, \quad \Lambda_q := \text{diag}(\lambda_1(\Sigma_0), \dots, \lambda_q(\Sigma_0)),$$

the identification of the diagonal entries being that of Lemma 12, and they are listed in non-increasing order because the singular values $\sigma_j(W_0)$ are. Thus U_0 satisfies (2.18) for $S = \Sigma_0$, and the first identity of (2.19) in Lemma 13 gives $P_q(\Sigma_0) = U_0 U_0^T = P_0$, which is (2.35). The inclusion of the column space of W_0 in the range of $P_q(\Sigma_0)$ is the easy half of this; what the gap $\delta_0 > 0$ contributes, through Lemma 13, is that the range is not larger, no eigenvalue of Σ_0 below the cut being equal to one above it. Equivalently, every v orthogonal to the column space of W_0 satisfies $W_0^T v = 0$ and hence $\Sigma_0 v = W_0 (W_0^T v) + \sigma_0^2 v = \sigma_0^2 v$, so the orthogonal complement of that column space is exactly the eigenspace of the smallest eigenvalue σ_0^2 .

By part 5 of Lemma 11, $S_n \rightarrow \Sigma_0$ almost surely; we fix from now on a realization lying in that almost sure event, so that the whole argument below is a deterministic statement about a convergent sequence of symmetric matrices.

We first make precise in what sense $P_n \rightarrow P_0$. Individual eigenvectors of S_n need not converge to anything, as Remark 10 recalls: they are determined only up to sign, and up to a rotation inside an eigenspace when eigenvalues collide. What converges is the projector, and it does so because it is a locally Lipschitz function of the matrix near Σ_0 , the Lipschitz constant being governed by the spectral gap δ_0 . Precisely, choose N with $\|S_n - \Sigma_0\| \leq \frac{\delta_0}{8}$ for all $n \geq N$, which is possible since $S_n \rightarrow \Sigma_0$ and $\delta_0 > 0$. Part 4 of Lemma 21, applied with $S_* = \Sigma_0$ and $S = S_n$, then shows that $S_n \in \mathcal{O}$ for every $n \geq N$, so that P_n is well defined, and that

$$\|P_n - P_0\| = \|P_q(S_n) - P_q(\Sigma_0)\| \leq \frac{16\rho_0}{\delta_0^2} \|S_n - \Sigma_0\| = \frac{16\sigma_1(W_0)^2}{\sigma_{\min}(W_0)^4} \|S_n - \Sigma_0\| \rightarrow 0. \quad (2.36)$$

The constant depends only on the true parameter, through the singular values of W_0 ; combined with part 5 of Lemma 11 the same inequality gives the rate $\|P_n - P_0\| = O_{\mathbb{P}}(n^{-\frac{1}{2}})$, which we do not need here but which is the quantitative form of the statement. For the convergence (2.36) alone, without the constant, no argument is needed at all: it is part 5 of Lemma 21, quoted there from Theorem 5.1 of [Kato(1995)].

By (2.15) the matrix S_n is also eventually positive definite, so Proposition 15 applies to it for all large n . Since $\text{tr}((I_p - P_n)S_n) = \text{tr}(S_n) - \text{tr}(P_n S_n) = \sum_{j>q} \lambda_j(S_n)$ by Lemma 13, formula (2.20) reads $\hat{\sigma}_n^2 = \frac{1}{p-q} \text{tr}((I_p - P_n)S_n)$, and the trace being a continuous bilinear function of its two arguments we get, using $P_n \rightarrow P_0$ and $S_n \rightarrow \Sigma_0$,

$$\hat{\sigma}_n^2 = \frac{1}{p-q} \text{tr}((I_p - P_n)S_n) \rightarrow \frac{1}{p-q} \text{tr}((I_p - P_0)\Sigma_0) = \frac{\sigma_0^2}{p-q} \text{tr}(I_p - P_0) = \sigma_0^2,$$

where the third equality holds because $(\mathbf{I}_p - \mathbf{P}_0)\Sigma_0 = \sigma_0^2(\mathbf{I}_p - \mathbf{P}_0)$, itself a consequence of $\Sigma_0 = \mathbf{W}_0\mathbf{W}_0^T + \sigma_0^2\mathbf{I}_p$ and $(\mathbf{I}_p - \mathbf{P}_0)\mathbf{W}_0 = \mathbf{0}$, and the fourth because $\mathbf{P}_0$ is an orthogonal projector of rank q , so that $\text{tr}(\mathbf{I}_p - \mathbf{P}_0) = p - \text{tr}(\mathbf{P}_0) = p - q$. In the same way

$$\hat{\mathbf{W}}_n \hat{\mathbf{W}}_n^T = \mathbf{P}_n \mathbf{S}_n \mathbf{P}_n - \hat{\sigma}_n^2 \mathbf{P}_n \longrightarrow \mathbf{P}_0 \Sigma_0 \mathbf{P}_0 - \sigma_0^2 \mathbf{P}_0 = \mathbf{W}_0 \mathbf{W}_0^T,$$

using $\mathbf{P}_0 \Sigma_0 \mathbf{P}_0 = \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 \mathbf{P}_0$.

It remains to convert this into convergence in M . We claim that if $\mathbf{B}_n \in \mathbb{R}^{p \times q}$ satisfy $\mathbf{B}_n \mathbf{B}_n^T \rightarrow \mathbf{W}_0 \mathbf{W}_0^T$, then $\min_{\mathbf{Q} \in O(q)} \|\mathbf{B}_n \mathbf{Q} - \mathbf{W}_0\|_F \rightarrow 0$. Indeed $\|\mathbf{B}_n\|_F^2 = \text{tr}(\mathbf{B}_n \mathbf{B}_n^T)$ is convergent, hence bounded. If the claim failed there would be $\epsilon > 0$ and a subsequence along which $\min_{\mathbf{Q}} \|\mathbf{B}_n \mathbf{Q} - \mathbf{W}_0\|_F \geq \epsilon$; by boundedness we may extract a further subsequence with $\mathbf{B}_n \rightarrow \mathbf{B}$, and then $\mathbf{B} \mathbf{B}^T = \mathbf{W}_0 \mathbf{W}_0^T$. This matrix has rank q , so $\text{rank}(\mathbf{B}) = q$ and $(\mathbf{B}, \sigma_0^2) \sim (\mathbf{W}_0, \sigma_0^2)$; by the description of the equivalence classes recalled in Section 1.1 there is $\mathbf{Q}_0 \in O(q)$ with $\mathbf{B} = \mathbf{W}_0 \mathbf{Q}_0$. Consequently $\min_{\mathbf{Q}} \|\mathbf{B}_n \mathbf{Q} - \mathbf{W}_0\|_F \leq \|\mathbf{B}_n \mathbf{Q}_0^T - \mathbf{B} \mathbf{Q}_0^T\|_F = \|\mathbf{B}_n - \mathbf{B}\|_F \rightarrow 0$ along that subsequence, a contradiction.

Applying the claim to $\mathbf{B}_n = \hat{\mathbf{W}}_n$ we obtain, almost surely, matrices $\mathbf{Q}'_n \in O(q)$ with $\hat{\mathbf{W}}_n \mathbf{Q}'_n \rightarrow \mathbf{W}_0$; combined with $\hat{\sigma}_n^2 \rightarrow \sigma_0^2$ this says $(\hat{\mathbf{W}}_n \mathbf{Q}'_n, \hat{\sigma}_n^2) \rightarrow (\mathbf{W}_0, \sigma_0^2)$ in Θ , and the continuity of π gives

$$[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] = \pi\left((\hat{\mathbf{W}}_n \mathbf{Q}'_n, \hat{\sigma}_n^2)\right) \longrightarrow \pi\left((\mathbf{W}_0, \sigma_0^2)\right) = [\mathbf{W}_0, \sigma_0^2]$$

in M , as required. $\square$

The proof just given passes through the spectral projector, and therefore through the perturbation theory of Lemma 21. It is worth recording that this is avoidable: once Proposition 17 is available, the whole spectral input reduces to Weyl's inequality, which we quote from Chapter 2 of [Chen et al.(2021)Chen, Chi, Fan, and Ma], a chapter devoted to exactly the situation at hand — there, in the notation of its equation (2.1), $\mathbf{M}$ is “an observed or estimated data matrix such as the sample covariance matrix” and $\mathbf{M}^*$ is “the target matrix such as the population covariance matrix”.

Alternative proof of Proposition 24. Fix a realization in the almost sure event $\{\mathbf{S}_n \rightarrow \Sigma_0\}$ of part 5 of Lemma 11 and set $\mathbf{E}_n := \mathbf{S}_n - \Sigma_0$, so that $\|\mathbf{E}_n\| \rightarrow 0$; this is (2.1) read as the perturbation $\mathbf{S}_n = \Sigma_0 + \mathbf{E}_n$ of [Chen et al.(2021)Chen, Chi, Fan, and Ma], equation (2.1). Write $\lambda_j := \lambda_j(\mathbf{S}_n)$.

Step 1: the eigenvalues. Lemma 2.2 of [Chen et al.(2021)Chen, Chi, Fan, and Ma], Weyl's inequality for eigenvalues, states that $|\lambda_j(\mathbf{A}) - \lambda_j(\mathbf{A} + \mathbf{E})| \leq \|\mathbf{E}\|$ for real symmetric $\mathbf{A}$ and $\mathbf{E}$ and every j ; with $\mathbf{A} = \Sigma_0$ and $\mathbf{E} = \mathbf{E}_n$ this gives

$$\max_{1 \leq j \leq p} |\lambda_j - \lambda_j(\Sigma_0)| \leq \|\mathbf{E}_n\| \longrightarrow 0. \quad (2.37)$$

By Lemma 12 we have $\lambda_j(\Sigma_0) = \sigma_0^2$ for every $j > q$, $\lambda_p(\Sigma_0) = \sigma_0^2 > 0$ and $\lambda_q(\Sigma_0) - \lambda_{q+1}(\Sigma_0) = \delta_0 > 0$, so (2.37) shows that for all large n the matrix $\mathbf{S}_n$ is positive definite and lies in $\mathcal{O}$, whence Proposition 15 applies to it, and that

$$\hat{\sigma}_n^2 = \frac{1}{p-q} \sum_{j>q} \lambda_j \longrightarrow \frac{1}{p-q} (p-q) \sigma_0^2 = \sigma_0^2, \quad \max_{j>q} |\lambda_j - \hat{\sigma}_n^2| \leq 2 \max_{j>q} |\lambda_j - \sigma_0^2| \leq 2 \|\mathbf{E}_n\|, \quad (2.38)$$

the middle inequality because $\hat{\sigma}_n^2$ is an average of the λ_j with $j > q$, each of which is within $\|\mathbf{E}_n\|$ of σ_0^2 .

Step 2: the loading matrix. Let $\hat{\Sigma}_n := \Sigma([\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]) = \hat{\mathbf{W}}_n \hat{\mathbf{W}}_n^T + \hat{\sigma}_n^2 \mathbf{I}_p$ be the fitted covariance matrix (2.22). By the second identity of (2.23) the difference $\hat{\Sigma}_n - \mathbf{S}_n$ is symmetric with eigenvalues $\hat{\sigma}_n^2 - \lambda_j$, $j > q$, so $\|\hat{\Sigma}_n - \mathbf{S}_n\| = \max_{j>q} |\hat{\sigma}_n^2 - \lambda_j| \leq 2 \|\mathbf{E}_n\| \rightarrow 0$ by (2.38). Consequently $\hat{\Sigma}_n \rightarrow \Sigma_0$, and therefore

$$\hat{\mathbf{W}}_n \hat{\mathbf{W}}_n^T = \hat{\Sigma}_n - \hat{\sigma}_n^2 \mathbf{I}_p \longrightarrow \Sigma_0 - \sigma_0^2 \mathbf{I}_p = \mathbf{W}_0 \mathbf{W}_0^T.$$

No spectral projector, and hence no continuity or analyticity statement about P_q , has been used.

Step 3: convergence in M . From $\hat{\sigma}_n^2 \rightarrow \sigma_0^2$ and $\hat{\mathbf{W}}_n \hat{\mathbf{W}}_n^T \rightarrow \mathbf{W}_0 \mathbf{W}_0^T$ the conclusion follows exactly as in the last part of the proof above, which is a statement about the quotient and involves no spectral theory. $\square$

Remark 25. If the spectral projector itself is wanted, and not merely the products in which it appears, the same chapter of [Chen et al.(2021)Chen, Chi, Fan, and Ma] supplies it in quantitative form: the Davis–Kahan $\sin \Theta$ theorem, Theorem 2.7 there, and its Corollary 2.8, bound the distance between the eigenspaces of Σ_0 and of $\mathbf{S}_n$ spanned by the q leading eigenvectors by a multiple of $\|\mathbf{E}_n\|$ divided by the eigengap, under a condition on $\|\mathbf{E}_n\|$ relative to that gap. Since Σ_0 and, eventually, $\mathbf{S}_n$ are positive definite, ordering their eigenvalues by magnitude is the same as ordering them by value, so Corollary 2.8 applies to $\mathbf{P}_n$ and $\mathbf{P}_0$ as they are defined here and is an alternative to parts 4 and 5 of Lemma 21.

Remark 26. Proposition 24 recovers, for the independent and identically distributed Gaussian sampling scheme, the strong consistency result of [Datta and Chakrabarty(2023)] and [Datta(2026)], Corollary 1, which is proved there by a Wald-type argument valid under considerably weaker assumptions. The proof above is included because it is short once Lemma 21 is available, and because it makes the section self-contained.

We can now fix once and for all the normal chart in which the whole analysis will be carried out.

Lemma 27. There exists $\rho \in (0, \min(\sigma_{\min}(\mathbf{W}_0), \sigma_0^2))$ such that, writing $B_\rho := \{\mathbf{u} \in T_{[\mathbf{W}_0, \sigma_0^2]}M : |\mathbf{u}| < \rho\} \subset \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$, the map $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}$ restricted to B_ρ is a diffeomorphism onto an open subset $\mathcal{U}_0 := \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(B_\rho) \subseteq M$, and such that $\mathcal{U}_0 \subseteq U$, where U is the neighbourhood supplied by Proposition 5 at $[\mathbf{W}_0, \sigma_0^2]$.

Proof. By Proposition 2 the exponential map is defined on $\mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$, and it is a local diffeomorphism at $\mathbf{0}$, so there is an open neighbourhood $\mathcal{N}$ of $\mathbf{0}$ inside $\mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$ that is mapped diffeomorphically onto an open subset of M . Since $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}$ is continuous and $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{0}) = [\mathbf{W}_0, \sigma_0^2] \in U$, we may pick $\rho > 0$ so small that $B_\rho \subseteq \mathcal{N}$ and $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(B_\rho) \subseteq U$. A restriction of a diffeomorphism to an open subset is again a diffeomorphism onto its (open) image. $\square$

We fix such a ρ for the rest of the section, and define the *aligned estimation error*

$$e_n := \left(\hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0, \hat{\sigma}_n^2 - \sigma_0^2 \right) \in \mathbb{R}^{p \times q} \times \mathbb{R}, \quad \hat{\mathbf{u}}_n := \begin{cases} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left([\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \right) & \text{on } \mathcal{A}_n, \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad (2.39)$$

where $\mathbf{Q}_n$ is as in (2.32) and $\mathcal{A}_n := \{[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \in \mathcal{U}_0\}$.

Proposition 28. For $n \geq p$ we have $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$; on $\mathcal{A}_n$ the vector $\hat{\mathbf{u}}_n$ belongs to B_ρ and its horizontal lift at $(\mathbf{W}_0, \sigma_0^2)$ is exactly e_n , that is $\iota_{(\mathbf{W}_0, \sigma_0^2)}(\hat{\mathbf{u}}_n) = e_n$; and $\hat{\mathbf{u}}_n \rightarrow \mathbf{0}$ almost surely.

Proof. By Lemma 23 the matrix $\mathbf{Q}_n$ is almost surely well defined for $n \geq p$, so e_n is. By Proposition 24 and the fact that $\mathcal{U}_0$ is an open neighbourhood of $[\mathbf{W}_0, \sigma_0^2]$, almost surely $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \in \mathcal{U}_0$ for all large n ; in particular $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$. On $\mathcal{A}_n$ we have $\hat{\mathbf{u}}_n \in B_\rho$ by Lemma 27, and since $\mathcal{U}_0 \subseteq U$ the formula of Proposition 5 applies and reads

$$\hat{\mathbf{u}}_n = d\pi_{(\mathbf{W}_0, \sigma_0^2)} \left(\left(\hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0, \hat{\sigma}_n^2 - \sigma_0^2 \right) \right) = d\pi_{(\mathbf{W}_0, \sigma_0^2)}(e_n).$$

As observed in the proof of Proposition 5, e_n is a horizontal vector at $(\mathbf{W}_0, \sigma_0^2)$, so $\iota_{(\mathbf{W}_0, \sigma_0^2)}(\hat{\mathbf{u}}_n) = e_n$ by (2.2). Finally let $\epsilon \in (0, \rho)$. Then $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(B_\epsilon)$ is an open neighbourhood of $[\mathbf{W}_0, \sigma_0^2]$, so almost surely $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$ lies in it for large n , and injectivity of $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}$ on B_ρ then forces $|\hat{\mathbf{u}}_n| < \epsilon$. $\square$

Remark 29. Assumption III.2 of [Sun et al.(2026)Sun, Lin, and Liu] postulates that the estimator almost surely avoids the cut locus of the true parameter, so that the logarithmic map may be applied to it, and Proposition 28 is what plays that role here. We cannot verify that assumption literally, because M is incomplete and the discussion following Proposition 5 shows that $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1}$ genuinely fails to be globally defined. What Proposition 28 provides instead is the only thing the asymptotic theory ever uses: an event of probability tending to one on which the intrinsic error is well defined, together with the explicit Procrustes description (2.39) of it. Since modifying a sequence of random vectors on a sequence of events of vanishing probability affects neither its limit in probability nor its limit in distribution, all statements below are insensitive to the convention adopted off $\mathcal{A}_n$.

Asymptotic linearity

We can now state the definition, which is Definition 3.10 of the companion draft and, in the language of [Sun et al.(2026)Sun, Lin, and Liu], the hypothesis of their Lemma III.3.

Definition 30. Let $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$ be a sequence of estimators for which the intrinsic estimation error $\hat{\mathbf{u}}_n$ of (2.39) is defined on events of probability tending to one. The sequence is said to be *asymptotically linear at $[\mathbf{W}_0, \sigma_0^2]$ with influence operator IF* if $\text{IF} : \mathbb{R}^p \rightarrow T_{[\mathbf{W}_0, \sigma_0^2]}M$ is measurable, centred and square integrable, that is

$$\mathbb{E} [\text{IF}(\mathbf{X}_1)] = \mathbf{0} \quad \text{and} \quad \mathbb{E} [|\text{IF}(\mathbf{X}_1)|^2] < \infty, \quad (2.40)$$

and if

$$\sqrt{n} \hat{\mathbf{u}}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \text{IF}(\mathbf{X}_i) + o_{\mathbb{P}}(1), \quad (2.41)$$

the remainder being $o_{\mathbb{P}}(1)$ with respect to the norm $|\cdot|$ of $T_{[\mathbf{W}_0, \sigma_0^2]}M$.

Remark 31. Both sides of (2.41) live in the single fixed vector space $T_{[\mathbf{W}_0, \sigma_0^2]}M$, so the expression is meaningful even though M is curved; this is exactly the device by which [Sun et al.(2026)Sun, Lin, and Liu] transport the classical theory to manifolds. Note also that IF is required to take values in the *tangent* space, whereas the score (1.20) is a linear functional, that is an element of the *cotangent* space; the inverse Fisher operator (2.3) is precisely what converts one into the other.

The following theorem is the main result of this section. It is the analogue of Theorem 5.3 of the companion draft, and it instantiates Corollary V.1 of [Sun et al.(2026)Sun, Lin, and Liu] for the probabilistic principal component analysis model.

Theorem 32. Let $1 \leq q < p$ and let $[\mathbf{W}_0, \sigma_0^2] \in M$. Then the maximum likelihood estimator $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$ of Definition 16 is asymptotically linear at $[\mathbf{W}_0, \sigma_0^2]$ in the sense of Definition 30, with influence operator

$$\text{IF}(\mathbf{x}) := G_{[\mathbf{W}_0, \sigma_0^2]}^{-1} (s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])) \in T_{[\mathbf{W}_0, \sigma_0^2]}M. \quad (2.42)$$

That is,

$$\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} ([\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]) = \frac{1}{\sqrt{n}} \sum_{i=1}^n G_{[\mathbf{W}_0, \sigma_0^2]}^{-1} (s(\mathbf{X}_i; [\mathbf{W}_0, \sigma_0^2])) + o_{\mathbb{P}}(1), \quad (2.43)$$

and consequently

$$\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} ([\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]) \rightsquigarrow N(\mathbf{0}, G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}) \quad \text{on } T_{[\mathbf{W}_0, \sigma_0^2]}M, \quad (2.44)$$

the Gaussian measure on $T_{[\mathbf{W}_0, \sigma_0^2]}M$ whose covariance operator is the inverse Fisher information operator. Equivalently, in terms of the Procrustes aligned error (2.39),

$$\sqrt{n} (\hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0, \hat{\sigma}_n^2 - \sigma_0^2) \rightsquigarrow \iota_{(\mathbf{W}_0, \sigma_0^2)} \left(N(\mathbf{0}, G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}) \right) \quad (2.45)$$

as random elements of $\mathbb{R}^{p \times q} \times \mathbb{R}$; the limit is a centred Gaussian law carried by the d -dimensional subspace $H_{(\mathbf{W}_0, \sigma_0^2)}$.

Proof. Write $G := G_{[\mathbf{W}_0, \sigma_0^2]}$, $\iota := \iota_{(\mathbf{W}_0, \sigma_0^2)}$ and $s(\mathbf{x}) := s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])$ throughout. The strategy is the one of Step 2 of the companion draft: in the normal chart supplied by Lemma 27 the family $\mathcal{P}$ becomes an ordinary parametric family indexed by an open subset of a Euclidean space, and $\hat{\mathbf{u}}_n$ becomes an ordinary M -estimator, so that Theorem 5.23 of [van der Vaart(1998)] applies once its four hypotheses have been verified. The essential structural point, which makes every verification a finite computation, is that by Proposition 2 the exponential map is *linear* in the horizontal lift, so no approximation of Exp is ever needed.

Step 1: the model in normal coordinates. Fix a $g_{[\mathbf{W}_0, \sigma_0^2]}$ -orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$ of $T_{[\mathbf{W}_0, \sigma_0^2]}M$ and identify $T_{[\mathbf{W}_0, \sigma_0^2]}M$ with $\mathbb{R}^d$ through it; under this identification B_ρ is the open Euclidean ball of radius ρ and $|\cdot|$ is the Euclidean norm. For $\mathbf{u} \in B_\rho$ put $(\mathbf{U}_u, b_u) := \iota(\mathbf{u})$ and

$$\Sigma_u := \Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{u}) \right) = (\mathbf{W}_0 + \mathbf{U}_u)(\mathbf{W}_0 + \mathbf{U}_u)^T + (\sigma_0^2 + b_u)\mathbf{I}_p, \quad (2.46)$$

using Proposition 2 and (1.5). Because ι is linear, expanding (2.46) gives the *exact* identity

$$\Sigma_u = \Sigma_0 + (\mathbf{U}_u \mathbf{W}_0^T + \mathbf{W}_0 \mathbf{U}_u^T + b_u \mathbf{I}_p) + \mathbf{U}_u \mathbf{U}_u^T = \Sigma_0 + D_u \Sigma([\mathbf{W}_0, \sigma_0^2]) + \mathbf{U}_u \mathbf{U}_u^T, \quad (2.47)$$

by (1.21): as a function of $\mathbf{u}$, the covariance matrix is a quadratic polynomial with no remainder whatsoever. Set, for $\mathbf{u} \in B_\rho$ and $\mathbf{x} \in \mathbb{R}^p$,

$$m(\mathbf{u}; \mathbf{x}) := \log p \left(\mathbf{x}; \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{u}) \right) = -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma_u - \frac{1}{2} \mathbf{x}^T \Sigma_u^{-1} \mathbf{x}, \quad (2.48)$$

which in the notation (1.18) is $l_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{u})(\mathbf{x})$, and write $\mathbb{M}_n(\mathbf{u}) := \frac{1}{n} \sum_{i=1}^n m(\mathbf{u}; \mathbf{X}_i)$ and $M(\mathbf{u}) := \mathbb{E}[m(\mathbf{u}; \mathbf{X}_1)]$. By (2.16) and Lemma 27, on the event $\mathcal{A}_n$ the vector $\hat{\mathbf{u}}_n$ almost surely maximizes $\mathbb{M}_n$ over B_ρ , since by Lemma 23 and Definition 16 the point $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$ almost surely maximizes $\mathbb{L}_n$ over all of M , and there it lies in the chart. On the remaining null event $\hat{\mathbf{u}}_n = \mathbf{0}$, because $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] = [\mathbf{W}_0, \sigma_0^2]$ by Definition 16.

Step 2: uniform spectral bounds. Let $\mathbf{u} \in B_\rho$. Since ι is an isometry, $\|\mathbf{U}_u\|_F \leq \rho$ and $|b_u| \leq \rho$; as $\rho < \sigma_0^2$, formula (2.46) gives

$$c_1 \mathbf{I}_p \preceq \Sigma_u \preceq c_2 \mathbf{I}_p, \quad c_1 := \sigma_0^2 - \rho > 0, \quad c_2 := (\sigma_{\max}(\mathbf{W}_0) + \rho)^2 + \sigma_0^2 + \rho, \quad (2.49)$$

the lower bound because $(\mathbf{W}_0 + \mathbf{U}_u)(\mathbf{W}_0 + \mathbf{U}_u)^T \succeq \mathbf{0}$, and the upper bound because $\sigma_{\max}(\mathbf{W}_0 + \mathbf{U}_u) \leq \sigma_{\max}(\mathbf{W}_0) + \|\mathbf{U}_u\|_F$. In particular $\|\Sigma_u^{-1}\|_2 \leq c_1^{-1}$ and $\|\Sigma_u\|_F \leq \sqrt{p} c_2$ uniformly on B_ρ .

Step 3: smoothness and the derivative of the criterion. On the open cone of positive definite matrices the maps $\Sigma \mapsto \Sigma^{-1}$ and $\Sigma \mapsto \log \det \Sigma$ are real analytic, and $\mathbf{u} \mapsto \Sigma_u$ is a polynomial by (2.47); hence for every fixed $\mathbf{x}$ the function $\mathbf{u} \mapsto m(\mathbf{u}; \mathbf{x})$ is C^∞ on B_ρ . Its differential is computed exactly as in (1.24): for $\mathbf{h} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$,

$$\nabla m(\mathbf{u}; \mathbf{x})[\mathbf{h}] = \frac{1}{2} \text{tr} \left\{ \Sigma_u^{-1} [\mathbf{x} \mathbf{x}^T - \Sigma_u] \Sigma_u^{-1} \left(D_{\mathbf{h}} \Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{u}) \right) \right) \right\}, \quad (2.50)$$

where, by (1.23),

$$D_{\mathbf{h}} \Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{u}) \right) = \mathbf{U}_{\mathbf{h}} (\mathbf{W}_0 + \mathbf{U}_u)^T + (\mathbf{W}_0 + \mathbf{U}_u) \mathbf{U}_{\mathbf{h}}^T + b_{\mathbf{h}} \mathbf{I}_p, \quad (2.51)$$

which is linear in $\mathbf{h}$ and affine in $\mathbf{u}$. Taking $\mathbf{u} = \mathbf{0}$ in (2.50) and comparing with (1.20) shows

$$\nabla m(\mathbf{0}; \mathbf{x})[\mathbf{h}] = s(\mathbf{x})(\mathbf{h}), \quad (2.52)$$

so that the gradient of the criterion at the truth is the score.

Step 4: a square integrable Lipschitz envelope. Let $\mathbf{u} \in B_\rho$ and let $\mathbf{h}$ be a unit vector. From (2.51) and $\|\mathbf{U}_{\mathbf{h}} C^T\|_F \leq \|\mathbf{U}_{\mathbf{h}}\|_F \|C\|_2$,

$$\left\| D_{\mathbf{h}} \Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{u}) \right) \right\|_F \leq 2(\sigma_{\max}(\mathbf{W}_0) + \rho) + \sqrt{p} =: c_3.$$

Combining this with $|\text{tr}(\mathbf{Z}\mathbf{Y})| \leq \|\mathbf{Z}\|_F \|\mathbf{Y}\|_F$ and the bounds of Step 2, (2.50) gives

$$\sup_{|\mathbf{h}|=1} |\nabla m(\mathbf{u}; \mathbf{x})[\mathbf{h}]| \leq \frac{c_3}{2c_1^2} (\|\mathbf{x}\|^2 + \sqrt{p}c_2) =: \dot{m}(\mathbf{x}) \quad \text{for all } \mathbf{u} \in B_\rho. \quad (2.53)$$

Since $\mathbf{X}_1$ is Gaussian, $\mathbb{E}[\|\mathbf{X}_1\|^4] < \infty$ and therefore $\mathbb{E}[\dot{m}(\mathbf{X}_1)^2] < \infty$. As B_ρ is convex and $m(\cdot; \mathbf{x})$ is C^1 there, the mean value theorem yields

$$|m(\mathbf{u}_1; \mathbf{x}) - m(\mathbf{u}_2; \mathbf{x})| \leq \dot{m}(\mathbf{x}) \|\mathbf{u}_1 - \mathbf{u}_2\| \quad \text{for all } \mathbf{u}_1, \mathbf{u}_2 \in B_\rho. \quad (2.54)$$

Step 5: the population criterion. Since $\mathbb{E}[\mathbf{X}_1 \mathbf{X}_1^T] = \Sigma_0$, (2.48) gives the closed form

$$M(\mathbf{u}) = -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma_{\mathbf{u}} - \frac{1}{2} \text{tr}(\Sigma_{\mathbf{u}}^{-1} \Sigma_0), \quad (2.55)$$

which, by the argument of Step 3, is C^∞ on B_ρ ; in particular no differentiation under the integral sign is needed anywhere below. Subtracting $M(\mathbf{0})$,

$$M(\mathbf{0}) - M(\mathbf{u}) = \frac{1}{2} \left[\text{tr}(\Sigma_{\mathbf{u}}^{-1} \Sigma_0) - p + \log \frac{\det \Sigma_{\mathbf{u}}}{\det \Sigma_0} \right] = \frac{1}{2} \sum_{j=1}^p (\mu_j - 1 - \log \mu_j), \quad (2.56)$$

where $\mu_1, \dots, \mu_p > 0$ are the eigenvalues of the positive definite matrix $\Sigma_{\mathbf{u}}^{-\frac{1}{2}} \Sigma_0 \Sigma_{\mathbf{u}}^{-\frac{1}{2}}$, which is similar to $\Sigma_{\mathbf{u}}^{-1} \Sigma_0$. The elementary inequality $\mu - 1 - \log \mu \geq 0$, with equality only at $\mu = 1$, shows that $M(\mathbf{0}) - M(\mathbf{u}) \geq 0$ with equality if and only if all $\mu_j = 1$, that is if and only if $\Sigma_{\mathbf{u}} = \Sigma_0$. By the injectivity of Σ established in Proposition 1 this happens exactly when $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{u}) = [\mathbf{W}_0, \sigma_0^2]$, and by Lemma 27 exactly when $\mathbf{u} = \mathbf{0}$. Hence $\mathbf{0}$ is the unique maximizer of M over B_ρ .

We now compute the second order Taylor expansion of M at $\mathbf{0}$. Fix $\mathbf{h} \in T_{[\mathbf{W}_0, \sigma_0^2]} M$, write $(\mathbf{U}, b) := \iota(\mathbf{h})$, and let $\Sigma(t) := \Sigma_{t\mathbf{h}}$. By (2.47),

$$\Sigma(t) = \Sigma_0 + t\Delta + t^2 \mathbf{U} \mathbf{U}^T, \quad \Delta := D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2]) = \mathbf{U} \mathbf{W}_0^T + \mathbf{W}_0 \mathbf{U}^T + b \mathbf{I}_p, \quad (2.57)$$

so $\dot{\Sigma}(0) = \Delta$ and $\ddot{\Sigma}(0) = 2\mathbf{U} \mathbf{U}^T$ exactly. Using $\frac{d}{dt} \log \det \Sigma = \text{tr}(\Sigma^{-1} \dot{\Sigma})$ and $\frac{d}{dt} \Sigma^{-1} = -\Sigma^{-1} \dot{\Sigma} \Sigma^{-1}$, the two terms of (2.55) contribute

$$\left. \frac{d}{dt} \right|_{t=0} \left[-\frac{1}{2} \log \det \Sigma(t) \right] = -\frac{1}{2} \text{tr}(\Sigma_0^{-1} \Delta), \quad \left. \frac{d}{dt} \right|_{t=0} \left[-\frac{1}{2} \text{tr}(\Sigma(t)^{-1} \Sigma_0) \right] = \frac{1}{2} \text{tr}(\Sigma_0^{-1} \Delta),$$

whose sum vanishes, so $\nabla M(\mathbf{0}) = 0$, as it must since $\mathbf{0}$ is an interior maximizer. Differentiating once more,

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \left[-\frac{1}{2} \log \det \Sigma(t) \right] = -\frac{1}{2} \text{tr}(\Sigma_0^{-1} \ddot{\Sigma}(0)) + \frac{1}{2} \text{tr}((\Sigma_0^{-1} \Delta)^2)$$

and, since $\frac{d}{dt}(\Sigma^{-1} \dot{\Sigma} \Sigma^{-1}) = \Sigma^{-1} \ddot{\Sigma} \Sigma^{-1} - 2\Sigma^{-1} \dot{\Sigma} \Sigma^{-1} \dot{\Sigma} \Sigma^{-1}$,

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \left[-\frac{1}{2} \text{tr}(\Sigma(t)^{-1} \Sigma_0) \right] = \frac{1}{2} \text{tr}(\Sigma_0^{-1} \ddot{\Sigma}(0)) - \text{tr}((\Sigma_0^{-1} \Delta)^2).$$

The terms in $\ddot{\Sigma}(0)$ cancel identically and we are left with

$$\left. \frac{d^2}{dt^2} \right|_{t=0} M(t\mathbf{h}) = -\frac{1}{2} \text{tr}((\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2]))^2) = -G(\mathbf{h}, \mathbf{h}) \quad (2.58)$$

by (1.22). As M is twice continuously differentiable on B_ρ , its Hessian at $\mathbf{0}$ is the symmetric bilinear form determined by the quadratic form (2.58), namely $-G$, and Taylor's theorem gives

$$M(\mathbf{u}) = M(\mathbf{0}) - \frac{1}{2}G(\mathbf{u}, \mathbf{u}) + o(|\mathbf{u}|^2) \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}. \quad (2.59)$$

By Theorem 7 the form G is positive definite, so the second derivative matrix of M at $\mathbf{0}$ is symmetric and nonsingular. Incidentally, (2.58) is the manifold version of the classical identity between the Fisher information and the expected negative Hessian of the log-likelihood, recorded as Remark III.4 of [Sun et al.(2026)Sun, Lin, and Liu]; here it comes out of (2.47) by a finite computation precisely because the exponential map is linear in the horizontal lift.

Step 6: application of the M -estimation theorem. We check the four hypotheses of Theorem 5.23 of [van der Vaart(1998)] for the criterion functions $\{m(\mathbf{u}; \cdot)\}_{\mathbf{u} \in B_\rho}$ indexed by the open subset B_ρ of $\mathbb{R}^d$, at the point $\mathbf{u} = \mathbf{0}$.

1. For each $\mathbf{u}$ the map $\mathbf{x} \mapsto m(\mathbf{u}; \mathbf{x})$ is measurable, being continuous, and for each $\mathbf{x}$ the map $\mathbf{u} \mapsto m(\mathbf{u}; \mathbf{x})$ is differentiable at $\mathbf{0}$ with derivative $s(\mathbf{x}) \in L^2$, by Step 3 and (2.52); square integrability of the score is (1.28).
2. The Lipschitz condition with square integrable envelope is (2.54) and (2.53).
3. The map $\mathbf{u} \mapsto M(\mathbf{u})$ admits the second order Taylor expansion (2.59) at the maximizer $\mathbf{0}$, with nonsingular symmetric second derivative matrix $-G$, by Step 5.
4. By Proposition 28, $\hat{\mathbf{u}}_n \rightarrow \mathbf{0}$ in probability, and $\mathbb{M}_n(\hat{\mathbf{u}}_n) \geq \mathbb{M}_n(\mathbf{0})$ holds exactly: on $\mathcal{A}_n$ because $\hat{\mathbf{u}}_n$ maximizes $\mathbb{M}_n$ over $B_\rho \ni \mathbf{0}$ by Step 1, and off $\mathcal{A}_n$ because there $\hat{\mathbf{u}}_n = \mathbf{0}$. In particular $\mathbb{M}_n(\hat{\mathbf{u}}_n) \geq \mathbb{M}_n(\mathbf{0}) - o_{\mathbb{P}}(n^{-1})$.

The cited theorem therefore yields

$$\sqrt{n} \hat{\mathbf{u}}_n = \mathbf{G}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{s}(\mathbf{X}_i) + o_{\mathbb{P}}(1), \quad (2.60)$$

where, in the orthonormal coordinates of Step 1, $\mathbf{G}$ is the $d \times d$ matrix with entries $\mathbf{G}_{jk} = G(\mathbf{e}_j, \mathbf{e}_k)$ and $\mathbf{s}(\mathbf{x}) \in \mathbb{R}^d$ is the vector with entries $\mathbf{s}(\mathbf{x})_j = s(\mathbf{x})(\mathbf{e}_j)$. Now if $\text{IF}(\mathbf{x}) = \sum_j c_j(\mathbf{x}) \mathbf{e}_j$ is defined by (2.42), then (2.3) reads $\sum_k \mathbf{G}_{jk} c_k(\mathbf{x}) = s(\mathbf{x})(\mathbf{e}_j)$ for every j , that is $\mathbf{c}(\mathbf{x}) = \mathbf{G}^{-1} \mathbf{s}(\mathbf{x})$; so (2.60) is precisely (2.43) and (2.41).

The requirements (2.40) hold: the score has mean zero and covariance G by Theorem 7, that is $\mathbb{E}[\mathbf{s}(\mathbf{X}_1)] = \mathbf{0}$ and $\mathbb{E}[\mathbf{s}(\mathbf{X}_1) \mathbf{s}(\mathbf{X}_1)^T] = \mathbf{G}$, whence $\mathbb{E}[\mathbf{c}(\mathbf{X}_1)] = \mathbf{0}$ and $\mathbb{E}[\|\mathbf{c}(\mathbf{X}_1)\|^2] = \text{tr}(\mathbf{G}^{-1}) < \infty$. The vectors $\mathbf{s}(\mathbf{X}_1), \mathbf{s}(\mathbf{X}_2), \dots$ being independent, identically distributed, centred and square integrable, the multivariate central limit theorem gives $n^{-\frac{1}{2}} \sum_{i=1}^n \mathbf{s}(\mathbf{X}_i) \rightsquigarrow N(\mathbf{0}, \mathbf{G})$, so that by (2.60), Slutsky's lemma and the continuous mapping theorem,

$$\sqrt{n} \hat{\mathbf{u}}_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1} \mathbf{G} \mathbf{G}^{-1}) = N(\mathbf{0}, \mathbf{G}^{-1}),$$

which is (2.44): in the $g_{[\mathbf{W}_0, \sigma_0^2]}$ -orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$ the covariance operator $G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}$ is represented by the matrix $\mathbf{G}^{-1}$.

Step 7: the Procrustes form. By Proposition 28, $\mathbf{e}_n = \iota(\hat{\mathbf{u}}_n)$ on $\mathcal{A}_n$, and $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$. If $\boldsymbol{\xi}_n$ and $\boldsymbol{\eta}_n$ agree on events of probability tending to one and $\boldsymbol{\xi}_n \rightsquigarrow \boldsymbol{\xi}$, then $\boldsymbol{\eta}_n \rightsquigarrow \boldsymbol{\xi}$ as well, since for every bounded Lipschitz test function ψ one has $|\mathbb{E}\psi(\boldsymbol{\xi}_n) - \mathbb{E}\psi(\boldsymbol{\eta}_n)| \leq 2\|\psi\|_\infty \mathbb{P}(\mathcal{A}_n^c) \rightarrow 0$ by the portmanteau theorem. Applying this and the continuous mapping theorem to the linear isometry ι turns (2.44) into (2.45); the limit is carried by $\iota(T_{[\mathbf{W}_0, \sigma_0^2]} M) = H_{(\mathbf{W}_0, \sigma_0^2)}$, which by Proposition 1 has dimension d . $\square$

The proof just given is a proof by M -estimation. It verifies the hypotheses of a general theorem about criterion functions, and it therefore uses Theorem 7 only as a repository of formulas — the expression (1.20) for the score, its first two moments (1.28), and the positive definiteness of G — while Proposition 8 is not used at all. The substance of Theorem 7, namely that the square root likelihood is differentiable in $L^2(\mathbb{R}^p)$, never enters, and that is precisely why Step 5 must compute the Hessian of the population criterion by hand: (2.58) is the information identity, and a theorem about arbitrary criterion functions cannot be expected to know it.

The second proof below reverses that emphasis. It works throughout with the local experiments indexed by the tangent vectors $\mathbf{h} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$, that is with the points $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{h}/\sqrt{n})$ of M , and with their log-likelihood ratio process; its analytic engine is Proposition 8, that is Proposition III.1 of [Sun et al.(2026)Sun, Lin, and Liu], whose sole input is the differentiability in quadratic mean supplied by Theorem 7; and it quotes no asymptotic theorem of [van der Vaart(1998)]. What replaces the M -estimation machinery is the geometry of (M, g) . Two features of it are used, and they are used in different places. The first is that the exponential map is affine in the horizontal lift, Proposition 2, so that (2.47) is an exact polynomial identity: every expansion below terminates, and none of the curvature corrections that Lemma A.1 and Lemma A.6 of [Sun et al.(2026)Sun, Lin, and Liu] would supply on a general manifold is needed. The second is that the vertical and horizontal spaces meet trivially, Proposition 1, which is what makes G positive definite in Theorem 7 and hence what produces the quadratic separation (2.75) of the Kullback–Leibler divergence; that separation, and not a maximal inequality, is what delivers the $\sqrt{n}$ rate.

Throughout the proof G , ι and $s(\mathbf{x})$ keep the meaning fixed at the beginning of the proof of Theorem 32; the orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$, the identification $T_{[\mathbf{W}_0, \sigma_0^2]}M \cong \mathbb{R}^d$, the ball B_ρ , the matrix $\mathbf{G}$ and the coordinate vector $\mathbf{s}(\mathbf{x})$ are those of its Steps 1 and 6; the functions m , $\mathbb{M}_n$ and M are those of (2.48) and (2.55); and c_1 , c_2 , c_3 are the constants of its Steps 2 and 4. We write $\lambda_{\min}(\mathbf{G}) > 0$ for the smallest eigenvalue of $\mathbf{G}$, positive by Theorem 7, and we use freely that all norms on $\text{Sym}(p)$ are equivalent.

Alternative proof of Theorem 32. Step 1: the local experiment, and the score as a linear functional of S_n .
Abbreviate

$$\mathbf{Z}_n := \sqrt{n}(\mathbf{S}_n - \Sigma_0) \in \text{Sym}(p), \quad \Delta_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{s}(\mathbf{X}_i) \in \mathbb{R}^d, \quad (2.61)$$

and, for $\mathbf{h} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$ with $|\mathbf{h}| < \rho\sqrt{n}$, let

$$\Lambda_n(\mathbf{h}) := \sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]} \left(\frac{\mathbf{h}}{\sqrt{n}} \right)\right)}{p(\mathbf{X}_i; [\mathbf{W}_0, \sigma_0^2])} = n \left[\mathbb{M}_n \left(\frac{\mathbf{h}}{\sqrt{n}} \right) - \mathbb{M}_n(\mathbf{0}) \right] \quad (2.62)$$

be the log-likelihood ratio process of the local experiments, which is the object appearing in Proposition 8. Since $\mathbf{x}\mathbf{x}^T$ enters (1.20) linearly and $\mathbf{Z}_n = n^{-\frac{1}{2}} \sum_{i=1}^n (\mathbf{X}_i \mathbf{X}_i^T - \Sigma_0)$, formula (1.20) gives the exact identity

$$\Delta_n^T \mathbf{h} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{s}(\mathbf{X}_i)(\mathbf{h}) = \frac{1}{2} \text{tr} \left\{ \Sigma_0^{-1} \mathbf{Z}_n \Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2]) \right\} = \frac{1}{2} \text{tr}(\mathbf{B}_{\mathbf{h}} \mathbf{Z}_n), \quad (2.63)$$

$$\mathbf{B}_{\mathbf{h}} := \Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2]) \Sigma_0^{-1} \in \text{Sym}(p).$$

The score is thus an exact linear image of the single random matrix $\mathbf{Z}_n$; this is the sufficiency of $\mathbf{S}_n$, part 4 of Lemma 11, made quantitative, and it is what allows the entire stochastic side of the argument to rest on the one weak limit (2.12). Two consequences are recorded at once. First, by (2.10) applied with $\mathbf{B} = \mathbf{B}_{\mathbf{h}_1}$ and $\mathbf{C} = \mathbf{B}_{\mathbf{h}_2}$,

$$\mathbb{E}[\Delta_n] = \mathbf{0}, \quad \text{Cov}(\Delta_n^T \mathbf{h}_1, \Delta_n^T \mathbf{h}_2) = \frac{n}{4} \cdot \frac{2}{n} \text{tr}(\mathbf{B}_{\mathbf{h}_1} \Sigma_0 \mathbf{B}_{\mathbf{h}_2} \Sigma_0) = G(\mathbf{h}_1, \mathbf{h}_2) \quad (2.64)$$

for every n , the last equality being (1.22). Second, applying the continuous mapping theorem to the linear map $\mathbf{Z} \mapsto \frac{1}{2} (\text{tr}(\mathbf{B}_{\mathbf{e}_j} \mathbf{Z}))_{j=1}^d$ and to (2.12),

$$\Delta_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}), \quad \text{and in particular} \quad \Delta_n = O_{\mathbb{P}}(1), \quad \|\mathbf{Z}_n\|_F = O_{\mathbb{P}}(1). \quad (2.65)$$

No separate central limit theorem for the score is needed.

Step 2: the criterion is exactly affine in S_n . By (2.48) and $\mathbf{x}^T \Sigma_u^{-1} \mathbf{x} = \text{tr}(\Sigma_u^{-1} \mathbf{x} \mathbf{x}^T)$,

$$\mathbb{M}_n(\mathbf{u}) = -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma_u - \frac{1}{2} \text{tr}(\Sigma_u^{-1} S_n), \quad \mathbf{u} \in B_\rho,$$

which differs from (2.55) only in that S_n replaces Σ_0 ; hence $\mathbb{M}_n(\mathbf{u}) - M(\mathbf{u}) = -\frac{1}{2} \text{tr}(\Sigma_u^{-1}(S_n - \Sigma_0))$ and, writing $K(\mathbf{u}) := M(\mathbf{0}) - M(\mathbf{u})$ for the Kullback–Leibler divergence evaluated in (2.56),

$$\mathbb{M}_n(\mathbf{0}) - \mathbb{M}_n(\mathbf{u}) = K(\mathbf{u}) + \frac{1}{2} \text{tr}[(\Sigma_u^{-1} - \Sigma_0^{-1})(S_n - \Sigma_0)] \quad \text{for every } \mathbf{u} \in B_\rho \text{ and every } n. \quad (2.66)$$

This identity is exact: no remainder, no expectation, no limit. Equivalently, in terms of (2.62),

$$\Lambda_n(\mathbf{h}) = -nK\left(\frac{\mathbf{h}}{\sqrt{n}}\right) - \frac{\sqrt{n}}{2} \text{tr}\left[\left(\Sigma_{\frac{\mathbf{h}}{\sqrt{n}}}^{-1} - \Sigma_0^{-1}\right) \mathbf{Z}_n\right], \quad |\mathbf{h}| < \rho\sqrt{n}. \quad (2.67)$$

The log-likelihood ratio process is therefore a deterministic smooth function of $\mathbf{h}$ plus a linear functional of $\mathbf{Z}_n$ whose coefficient is again deterministic and smooth in $\mathbf{h}$. This is the structural reason why no empirical process, no envelope function and no maximal inequality will be required below.

Step 3: elementary bounds on the chart. Let $\mathbf{u} \in B_\rho$. Since ι is an isometry, $\|U_{\mathbf{u}}\|_F \leq |\mathbf{u}|$, and $\|D_{\mathbf{u}}\Sigma([\mathbf{W}_0, \sigma_0^2])\|_F \leq c_3|\mathbf{u}|$ by the bound of Step 4 of the previous proof read at $\mathbf{u} = \mathbf{0}$ together with linearity in the direction, so that (2.47) gives

$$\|\Sigma_{\mathbf{u}} - \Sigma_0\|_F \leq c_3|\mathbf{u}| + |\mathbf{u}|^2 \leq (c_3 + \rho)|\mathbf{u}|. \quad (2.68)$$

Since $\Sigma_u^{-1} - \Sigma_0^{-1} = \Sigma_u^{-1}(\Sigma_0 - \Sigma_u)\Sigma_0^{-1}$ and $\|\Sigma_u^{-1}\|_2 \leq c_1^{-1}$, $\|\Sigma_0^{-1}\|_2 \leq c_1^{-1}$ by (2.49),

$$\|\Sigma_u^{-1} - \Sigma_0^{-1}\|_F \leq c_1^{-2} \|\Sigma_u - \Sigma_0\|_F \leq c_4|\mathbf{u}|, \quad c_4 := \frac{c_3 + \rho}{c_1^2}. \quad (2.69)$$

Consequently, for every $K > 0$, every $\mathbf{h}$ with $|\mathbf{h}| \leq K$ and every $n > K^2/\rho^2$,

$$\left\| \sqrt{n} \left(\Sigma_{\frac{\mathbf{h}}{\sqrt{n}}}^{-1} - \Sigma_0^{-1} \right) + \mathbf{B}_{\mathbf{h}} \right\|_F \leq \frac{c_5(K)}{\sqrt{n}}, \quad c_5(K) := (c_4 c_3 c_1^{-1} + c_1^{-2}) K^2. \quad (2.70)$$

Indeed, taking $\mathbf{u} = \mathbf{h}/\sqrt{n}$ in the factorization above and using (2.47),

$$\sqrt{n} \left(\Sigma_{\frac{\mathbf{h}}{\sqrt{n}}}^{-1} - \Sigma_0^{-1} \right) + \mathbf{B}_{\mathbf{h}} = - \left(\Sigma_{\frac{\mathbf{h}}{\sqrt{n}}}^{-1} - \Sigma_0^{-1} \right) D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2]) \Sigma_0^{-1} - \frac{1}{\sqrt{n}} \Sigma_{\frac{\mathbf{h}}{\sqrt{n}}}^{-1} U_{\mathbf{h}} U_{\mathbf{h}}^T \Sigma_0^{-1},$$

and (2.69) bounds the first term in Frobenius norm by $c_4 c_3 c_1^{-1} K^2 n^{-\frac{1}{2}}$ and the second by $c_1^{-2} K^2 n^{-\frac{1}{2}}$. Combining (2.70) with (2.63) and $|\text{tr}(\mathbf{Y} \mathbf{Z})| \leq \|\mathbf{Y}\|_F \|\mathbf{Z}\|_F$,

$$\sup_{|\mathbf{h}| \leq K} \left| -\frac{\sqrt{n}}{2} \text{tr} \left[\left(\Sigma_{\frac{\mathbf{h}}{\sqrt{n}}}^{-1} - \Sigma_0^{-1} \right) \mathbf{Z}_n \right] - \Delta_n^T \mathbf{h} \right| \leq \frac{c_5(K)}{2\sqrt{n}} \|\mathbf{Z}_n\|_F = O_{\mathbb{P}} \left(n^{-\frac{1}{2}} \right). \quad (2.71)$$

Step 4: the Hessian of the Kullback–Leibler divergence, read off from local asymptotic normality. By (2.47) the map $\mathbf{u} \mapsto \Sigma_{\mathbf{u}}$ is polynomial and, by (2.49), takes its values in the open cone of positive definite matrices, on which log det and inversion are real analytic; hence K is C^∞ on B_ρ . By (2.56) we have $K \geq 0$ and $K(\mathbf{0}) = 0$, so $\mathbf{0}$ is an interior minimizer of K and therefore

$$K(\mathbf{0}) = 0, \quad \nabla K(\mathbf{0}) = \mathbf{0}; \quad (2.72)$$

this requires no computation, only the nonnegativity of the Kullback–Leibler divergence. Now fix $\mathbf{h} \in T_{[\mathbf{W}_0, \sigma_0^2]} M$. Proposition 8, applied at the true parameter $[\mathbf{W}_0, \sigma_0^2]$ and written in the coordinates of Step 1, states that $\Lambda_n(\mathbf{h}) = \Delta_n^T \mathbf{h} - \frac{1}{2} G(\mathbf{h}, \mathbf{h}) + o_{\mathbb{P}}(1)$. Substituting this and (2.71) into (2.67),

$$nK\left(\frac{\mathbf{h}}{\sqrt{n}}\right) = -\Lambda_n(\mathbf{h}) + \Delta_n^T \mathbf{h} + o_{\mathbb{P}}(1) = \frac{1}{2}G(\mathbf{h}, \mathbf{h}) + o_{\mathbb{P}}(1).$$

The left hand side is a deterministic sequence of real numbers, and a deterministic sequence which equals a constant up to $o_{\mathbb{P}}(1)$ converges to that constant; hence $nK(\mathbf{h}/\sqrt{n}) \rightarrow \frac{1}{2}G(\mathbf{h}, \mathbf{h})$ for every $\mathbf{h}$. On the other hand (2.72) and Taylor's theorem give $nK(\mathbf{h}/\sqrt{n}) = \frac{1}{2}\mathbf{h}^T \nabla^2 K(\xi_{n,\mathbf{h}})\mathbf{h}$ for some $\xi_{n,\mathbf{h}}$ on the segment joining $\mathbf{0}$ to $\mathbf{h}/\sqrt{n}$, and $\nabla^2 K$ is continuous, so $nK(\mathbf{h}/\sqrt{n}) \rightarrow \frac{1}{2}\mathbf{h}^T \nabla^2 K(\mathbf{0})\mathbf{h}$. Comparing the two limits and polarizing,

$$\nabla^2 K(\mathbf{0}) = \mathbf{G}, \quad \text{equivalently} \quad -\nabla^2 M(\mathbf{0}) = \mathbf{G}. \quad (2.73)$$

This is (2.58) again, the manifold form of the identity between the Fisher information and the Hessian of the Kullback–Leibler divergence recorded as Remark III.4 of [Sun et al.(2026)Sun, Lin, and Liu]. It has not been computed here: it has been read off the local asymptotic normality expansion, which is itself only a restatement of the differentiability in quadratic mean of Theorem 7. Two corollaries follow. First, since $\nabla^2 K$ is continuous, the Taylor argument above is uniform on compacta,

$$\sup_{|\mathbf{h}| \leq K} \left| nK\left(\frac{\mathbf{h}}{\sqrt{n}}\right) - \frac{1}{2}\mathbf{h}^T \mathbf{G}\mathbf{h} \right| \leq \frac{K^2}{2} \sup_{|\xi| \leq \frac{K}{\sqrt{n}}} \|\nabla^2 K(\xi) - \nabla^2 K(\mathbf{0})\| \rightarrow 0,$$

so that (2.67) and (2.71) upgrade Proposition 8 to its locally uniform form: for every $K > 0$,

$$\sup_{|\mathbf{h}| \leq K} \left| \Lambda_n(\mathbf{h}) - \Delta_n^T \mathbf{h} + \frac{1}{2}\mathbf{h}^T \mathbf{G}\mathbf{h} \right| \rightarrow 0 \quad \text{in probability.} \quad (2.74)$$

The upgrade is free precisely because, by (2.67), all the dependence of Λ_n on $\mathbf{h}$ is deterministic and smooth, the only randomness being the single matrix $\mathbf{Z}_n$. Second, (2.72), (2.73) and Taylor's theorem give $K(\mathbf{u}) = \frac{1}{2}\mathbf{u}^T \mathbf{G}\mathbf{u} + o(|\mathbf{u}|^2)$, so there is $\rho_1 \in (0, \rho]$ with

$$K(\mathbf{u}) \geq \frac{\lambda_{\min}(\mathbf{G})}{4} |\mathbf{u}|^2 \quad \text{whenever } |\mathbf{u}| \leq \rho_1. \quad (2.75)$$

This quadratic separation is where the positive definiteness of the Fisher operator, that is, by the proof of Theorem 7, the transversality $V_{(\mathbf{w}_0, \sigma_0^2)} \cap H_{(\mathbf{w}_0, \sigma_0^2)} = \{\mathbf{0}\}$ of Proposition 1, is used quantitatively.

Step 5: the $\sqrt{n}$ rate, by the basic inequality. On $\mathcal{A}_n$ the vector $\hat{\mathbf{u}}_n$ maximizes $\mathbb{M}_n$ over B_ρ , by Step 1 of the previous proof, and $\mathbf{0} \in B_\rho$; hence $\mathbb{M}_n(\mathbf{0}) - \mathbb{M}_n(\hat{\mathbf{u}}_n) \leq 0$, and (2.66) together with (2.69) gives

$$K(\hat{\mathbf{u}}_n) \leq \frac{1}{2} |\text{tr}[(\Sigma_{\hat{\mathbf{u}}_n}^{-1} - \Sigma_0^{-1})(\mathbf{S}_n - \Sigma_0)]| \leq \frac{c_4}{2} |\hat{\mathbf{u}}_n| \|\mathbf{S}_n - \Sigma_0\|_F \quad \text{on } \mathcal{A}_n. \quad (2.76)$$

By Proposition 28, $\hat{\mathbf{u}}_n \rightarrow \mathbf{0}$ almost surely, so for every $\varepsilon > 0$ there is n_ε with $\mathbb{P}(|\hat{\mathbf{u}}_n| \leq \rho_1 \text{ for all } n \geq n_\varepsilon) \geq 1 - \varepsilon$. On the intersection of that event with $\mathcal{A}_n$ the separation (2.75) applies to $\hat{\mathbf{u}}_n$ and, combined with (2.76), yields $\frac{\lambda_{\min}(\mathbf{G})}{4} |\hat{\mathbf{u}}_n|^2 \leq \frac{c_4}{2} |\hat{\mathbf{u}}_n| \|\mathbf{S}_n - \Sigma_0\|_F$, that is

$$|\hat{\mathbf{u}}_n| \leq \frac{2c_4}{\lambda_{\min}(\mathbf{G})} \|\mathbf{S}_n - \Sigma_0\|_F.$$

Since $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$, since ε was arbitrary, and since $\|\mathbf{S}_n - \Sigma_0\|_F = O_{\mathbb{P}}(n^{-\frac{1}{2}})$ by part 5 of Lemma 11, we conclude that

$$\hat{\mathbf{h}}_n := \sqrt{n} \hat{\mathbf{u}}_n = O_{\mathbb{P}}(1). \quad (2.77)$$

This is the classical basic inequality argument in its simplest possible form. It is available here because (2.66) is exact and affine in $\mathbf{S}_n$, so that the stochastic part of the criterion is controlled by the single Lipschitz bound (2.69), and because its deterministic part is quadratically separated by (2.75). Beyond Proposition 24, only the geometry of the chart has been used.

Step 6: identification of the local maximizer. By (2.62) and Step 1 of the previous proof, on $\mathcal{A}_n$ the vector $\hat{\mathbf{h}}_n$ maximizes Λ_n over $\{|\mathbf{h}| < \rho\sqrt{n}\}$. Since $\mathbf{G}$ is positive definite by Theorem 7, the quadratic

$$Q_n(\mathbf{h}) := \Delta_n^T \mathbf{h} - \frac{1}{2} \mathbf{h}^T \mathbf{G} \mathbf{h} = \frac{1}{2} \Delta_n^T \mathbf{G}^{-1} \Delta_n - \frac{1}{2} (\mathbf{h} - \mathbf{G}^{-1} \Delta_n)^T \mathbf{G} (\mathbf{h} - \mathbf{G}^{-1} \Delta_n) \quad (2.78)$$

has the unique maximizer $\mathbf{G}^{-1} \Delta_n$, which is $O_{\mathbb{P}}(1)$ by (2.65). Let $\varepsilon > 0$. By (2.77), (2.65) and $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$ there are $K > 0$ and $n_0 > K^2/\rho^2$ such that the event

$$\mathcal{E}_n := \mathcal{A}_n \cap \left\{ |\hat{\mathbf{h}}_n| \leq K \right\} \cap \left\{ |\mathbf{G}^{-1} \Delta_n| \leq K \right\}$$

has $\mathbb{P}(\mathcal{E}_n) \geq 1 - \varepsilon$ for all $n \geq n_0$; on $\mathcal{E}_n$ both $\hat{\mathbf{h}}_n$ and $\mathbf{G}^{-1} \Delta_n$ then lie in the ball $\{|\mathbf{h}| \leq K\}$, on which Λ_n is defined. Put $r_n := \sup_{|\mathbf{h}| \leq K} |\Lambda_n(\mathbf{h}) - Q_n(\mathbf{h})|$, which is $o_{\mathbb{P}}(1)$ by (2.74). On $\mathcal{E}_n$, using the maximality of $\hat{\mathbf{h}}_n$ for Λ_n in the middle step,

$$Q_n(\hat{\mathbf{h}}_n) \geq \Lambda_n(\hat{\mathbf{h}}_n) - r_n \geq \Lambda_n(\mathbf{G}^{-1} \Delta_n) - r_n \geq Q_n(\mathbf{G}^{-1} \Delta_n) - 2r_n,$$

which, by the completed square in (2.78), is exactly

$$\frac{1}{2} (\hat{\mathbf{h}}_n - \mathbf{G}^{-1} \Delta_n)^T \mathbf{G} (\hat{\mathbf{h}}_n - \mathbf{G}^{-1} \Delta_n) \leq 2r_n, \quad \text{whence} \quad \left| \hat{\mathbf{h}}_n - \mathbf{G}^{-1} \Delta_n \right|^2 \leq \frac{4r_n}{\lambda_{\min}(\mathbf{G})}.$$

Therefore $\limsup_n \mathbb{P} \left(\left| \hat{\mathbf{h}}_n - \mathbf{G}^{-1} \Delta_n \right| > \delta \right) \leq \varepsilon$ for every $\delta > 0$, and letting $\varepsilon \downarrow 0$,

$$\sqrt{n} \hat{\mathbf{u}}_n = \mathbf{G}^{-1} \Delta_n + o_{\mathbb{P}}(1), \quad (2.79)$$

which is (2.60).

Step 7: conclusion. The passage from (2.79) to (2.43), the verification of (2.40) — for which (2.64) may be quoted in place of Theorem 7 — and the Procrustes form (2.45) are word for word Steps 6 and 7 of the previous proof and are not repeated. Finally (2.79), (2.65) and Slutsky's lemma give $\sqrt{n} \hat{\mathbf{u}}_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1} \mathbf{G} \mathbf{G}^{-1}) = N(\mathbf{0}, \mathbf{G}^{-1})$, which is (2.44). $\square$

Remark 33. Theorem 32 has the shape of Corollary V.1 of [Sun et al.(2026)Sun, Lin, and Liu], but neither proof given above is an application of it. That corollary is deduced there from their Proposition V.1, whose standing hypothesis is that the loss $\mathbf{x} \mapsto -\log p(\mathbf{x}; \cdot)$ be almost surely *geodesically convex* on the parameter manifold, in the sense of their Definition V.1. That hypothesis is not available here. A function f convex along a geodesic γ with $\gamma(0) = \theta$ and $\nabla f(\theta) = \mathbf{0}$ satisfies $f(\gamma(t)) \geq f(\gamma(0))$ for all $t \in [0, 1]$, so a geodesically convex function has no stationary point that fails to minimize it among the points that a geodesic joins it to. But the stationarity equations (7) and (8) of [Tipping and Bishop(1999)], recalled in the proof of Proposition 15, exhibit a stationary point of (2.16) for every selection of q eigenvectors of $\mathbf{S}$ whose eigenvalues exceed the resulting $\hat{\sigma}^2$, and only the selection of the q largest ones gives the maximizer. As soon as $p - q \geq 2$ a second admissible selection exists for an open set of $\mathbf{S} \in \mathcal{O}$ — take $\lambda_1, \dots, \lambda_{q+1}$ nearly equal and much larger than $\lambda_{q+2}, \dots, \lambda_p$, and retain $\lambda_2, \dots, \lambda_{q+1}$ — and by part 2 of Lemma 11 the law of $\mathbf{S}_n$ charges every such open set once $n \geq p$. So $-\mathbb{L}_n$ has, with positive probability, stationary points on M that are not global minimizers, and the geodesic convexity of Definition V.1 of [Sun et al.(2026)Sun, Lin, and Liu] could hold only in the vacuous way permitted by the incompleteness of M , that is if no geodesic of (M, g) joined such a point to the maximizer. Neither proof of Theorem 32 needs it. What the second proof does use of [Sun et al.(2026)Sun, Lin, and Liu] is their Definition III.1 and Proposition III.1, verified for $\mathcal{P}$ in Theorem 7 and Proposition 8; their Remark III.4, obtained here as (2.73) rather than assumed; and their Assumption III.2, whose role is played by Proposition 28.

Corollary 34. The influence operator (2.42) satisfies the influence relation of [Sun et al.(2026)Sun, Lin, and Liu], Lemma III.3, namely

$$\mathbb{E} [\text{IF}(\mathbf{X}_1) \otimes s(\mathbf{X}_1; [\mathbf{W}_0, \sigma_0^2])] = \text{id}_{T_{[\mathbf{W}_0, \sigma_0^2]} M}. \quad (2.80)$$

Moreover it is the only influence operator with values in the span of the score for which (2.41) can hold.

Proof. In the coordinates of Step 6 of the previous proof, (2.80) reads

$$\mathbb{E} [\mathbf{c}(\mathbf{X}_1) \mathbf{s}(\mathbf{X}_1)^T] = \mathbf{G}^{-1} \mathbb{E} [\mathbf{s}(\mathbf{X}_1) \mathbf{s}(\mathbf{X}_1)^T] = \mathbf{G}^{-1} \mathbf{G} = \mathbf{I}_d.$$

For the uniqueness, an influence operator of the stated form is $\text{IF}'(\mathbf{x}) = \mathbf{C} \mathbf{s}(\mathbf{x})$ for some $d \times d$ matrix $\mathbf{C}$, and (2.80), which [Sun et al.(2026)Sun, Lin, and Liu] show to be forced for regular asymptotically linear sequences, gives $\mathbf{C} \mathbf{G} = \mathbf{I}_d$, that is $\mathbf{C} = \mathbf{G}^{-1}$. $\square$

Remark 35. Three comments on the content of Theorem 32.

First, (2.45) is the well posed replacement of the false naive statement discussed at the beginning of this section: it is not the raw error $\hat{\mathbf{W}}_n - \mathbf{W}_0$ that is asymptotically Gaussian, but the error measured after the optimal rotational alignment $\hat{\mathbf{W}}_n \mapsto \hat{\mathbf{W}}_n \mathbf{Q}_n$, and the limit lives on the horizontal space $H_{(\mathbf{W}_0, \sigma_0^2)}$ only. Viewed inside $\mathbb{R}^{p \times q} \times \mathbb{R} \cong \mathbb{R}^{pq+1}$ the limiting Gaussian law is degenerate, carrying no mass in the $\frac{q(q-1)}{2}$ vertical directions of Proposition 1. This degeneracy is a feature and not a defect: those are exactly the directions along which the likelihood (2.16) is constant, so no data can ever be informative about them.

Second, (2.44) contains as a by-product the rate of convergence, since it gives $\left| \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1}([\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]) \right| = O_{\mathbb{P}}(n^{-\frac{1}{2}})$ and hence, on $\mathcal{A}_n$, that the Riemannian distance between $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$ and $[\mathbf{W}_0, \sigma_0^2]$ is $O_{\mathbb{P}}(n^{-\frac{1}{2}})$. This is the quantitative refinement of Proposition 24.

Third, no assumption whatsoever has been made on the multiplicities of the singular values of $\mathbf{W}_0$. This is worth emphasizing, because the classical route to such a theorem, namely perturbation of the eigenvectors of $\mathbf{S}_n$ followed by the delta method, requires the leading q eigenvalues of Σ_0 to be simple. When they are not, individual principal directions are genuinely not asymptotically normal, and yet the point $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$ of M still is: the quotient level limit simply does not see the multiplicities. The reason is Theorem 7, where positive definiteness of $G_{[\mathbf{W}_0, \sigma_0^2]}$ was obtained from $V_{(\mathbf{W}_0, \sigma_0^2)} \cap H_{(\mathbf{W}_0, \sigma_0^2)} = \{\mathbf{0}\}$ alone.

Remark 36. Theorem 32 is the input required by the optimality theory of [Sun et al.(2026)Sun, Lin, and Liu]. Together with the local asymptotic normality expansion of Proposition 8 it yields, through Le Cam's third lemma, that the maximum likelihood estimator is locally regular in the sense of their Definition III.5, and then their convolution theorem, Theorem III.1, and their local asymptotic minimax theorem, Theorem III.2, identify $G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}$ as the efficiency bound over all regular estimator sequences; compare their Theorem V.1. Carrying this out requires the parallel transport of (M, g) and a comparison between the transported and the coordinate errors, which are Proposition 37 and Lemma 40 of Section 2.1; the conclusions are Theorem 45, Theorem 47 and Theorem 48 there.

2.1 Parallel Transport, Local Regularity and Optimality

Theorem 32 is a statement made entirely under the single law $\mathbb{P}_{[\mathbf{W}_0, \sigma_0^2]}$, and it concerns a single estimator. The present section adds the two things it does not contain: a statement about the behaviour of the maximum likelihood estimator under *neighbouring* parameters, which is local regularity, and a statement about *all other* estimators, which is optimality. Both require comparing tangent vectors based at two different points of M , so both require the parallel transport of (M, g) , which we compute first.

Throughout this section we abbreviate

$$\theta_0 := [\mathbf{W}_0, \sigma_0^2] \in M, \quad \hat{\theta}_n := [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \in M, \quad G := G_{\theta_0}, \quad \iota := \iota_{(\mathbf{W}_0, \sigma_0^2)}. \quad (2.81)$$

The letter θ , which in Section 1.1 denoted a point of Θ , was retired at the beginning of Section 1.4 when the coordinates on Θ were renamed $(\mathbf{W}, \sigma^2)$, so no confusion can arise: from here on θ always denotes a point of M . We keep the orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$ of $T_{\theta_0} M$ and the identification $T_{\theta_0} M \cong \mathbb{R}^d$ of Step 1 of the proof of Theorem 32, and we write $\mathbf{G}$ for the matrix of G and $\mathbf{s}(\mathbf{x}) \in \mathbb{R}^d$ for the coordinate vector of the score in that basis, exactly as in (2.60). Finally we abbreviate the normalized score

$$\Delta_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n s(\mathbf{X}_i) \in \mathbb{R}^d, \quad (2.82)$$

so that Theorem 32 reads $\sqrt{n}\hat{\mathbf{u}}_n = \mathbf{G}^{-1}\Delta_n + o_{\mathbb{P}}(1)$ and Proposition 8 reads, for $\mathbf{h} \in T_{\theta_0}M$ with coordinate vector still written $\mathbf{h}$,

$$\Lambda_n(\mathbf{h}) := \sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \text{Exp}_{\theta_0}\left(\frac{\mathbf{h}}{\sqrt{n}}\right)\right)}{p(\mathbf{X}_i; \theta_0)} = \Delta_n^T \mathbf{h} - \frac{1}{2} \mathbf{h}^T \mathbf{G} \mathbf{h} + o_{\mathbb{P}}(1), \quad (2.83)$$

because $s(\mathbf{x})(\mathbf{h}) = s(\mathbf{x})^T \mathbf{h}$ by bilinearity and $\sum_i s(\mathbf{X}_i)(\mathbf{h}/\sqrt{n}) = \Delta_n^T \mathbf{h}$.

Parallel transport along the geodesics of M

The metric g was constructed in Section 1.1 so that π is a Riemannian submersion from the *flat* manifold $\Theta \subseteq \mathbb{R}^{pq+1}$, and Proposition 2 exploited this to make the exponential map affine in the horizontal lift. The same structure makes the parallel transport an explicit linear ordinary differential equation.

Proposition 37. Let $[\mathbf{A}, s] \in M$, let $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ with horizontal lift $(\mathbf{V}, c) := \iota_{(\mathbf{A}, s)}(\mathbf{v})$, and let

$$\gamma(t) := \text{Exp}_{[\mathbf{A}, s]}(t\mathbf{v}) = [\mathbf{A}_t, s + tc], \quad \mathbf{A}_t := \mathbf{A} + t\mathbf{V}, \quad t \in [0, 1], \quad (2.84)$$

be the geodesic of Proposition 2.

1. $\text{rank}(\mathbf{A}_t) = q$ for every $t \in [0, 1]$, and for every t and every skew-symmetric $\mathbf{K} \in \mathbb{R}^{q \times q}$ the Lyapunov equation

$$(\mathbf{A}_t^T \mathbf{A}_t) \Omega + \Omega (\mathbf{A}_t^T \mathbf{A}_t) = \mathbf{K} \quad (2.85)$$

has a unique solution $\Omega =: \mathcal{L}_t(\mathbf{K})$; it is skew-symmetric, $\mathcal{L}_t$ is linear in $\mathbf{K}$, and $(t, \mathbf{K}) \mapsto \mathcal{L}_t(\mathbf{K})$ is smooth.

2. Let $\mathbf{x}(\cdot)$ be a vector field along γ , that is $\mathbf{x}(t) \in T_{\gamma(t)}M$, and let $(\mathbf{X}(t), \beta(t)) := \iota_{(\mathbf{A}_t, s+tc)}(\mathbf{x}(t)) \in H_{(\mathbf{A}_t, s+tc)}$ be its horizontal lift. Then $\mathbf{x}$ is parallel along γ if and only if

$$\dot{\mathbf{X}}(t) = \mathbf{A}_t \mathcal{L}_t(\mathbf{X}(t)^T \mathbf{V} - \mathbf{V}^T \mathbf{X}(t)), \quad \dot{\beta}(t) = 0. \quad (2.86)$$

3. Equation (2.86) is a linear ordinary differential equation on $\mathbb{R}^{p \times q} \times \mathbb{R}$ whose coefficients are smooth in $(t, \mathbf{A}, \mathbf{V}, c)$; its flow preserves the horizontal subspaces, that is $(\mathbf{X}(0), \beta(0)) \in H_{(\mathbf{A}, s)}$ implies $(\mathbf{X}(t), \beta(t)) \in H_{(\mathbf{A}_t, s+tc)}$ for all t , and it preserves the Euclidean norm. Consequently the parallel transport along γ exists, is a linear isometry, and depends smoothly on all the data.

Proof. **1.** Since $\iota_{(\mathbf{A}, s)}$ is an isometry, $\|\mathbf{V}\| \leq \|\mathbf{V}\|_F \leq |\mathbf{v}| < \min(\sigma_{\min}(\mathbf{A}), s)$, so Weyl's perturbation inequality, as quoted in the proof of Proposition 2, gives $\sigma_{\min}(\mathbf{A}_t) \geq \sigma_{\min}(\mathbf{A}) - t\|\mathbf{V}\| > 0$ for $t \in [0, 1]$; hence $\mathbf{A}_t$ has rank q and $\mathbf{P}_t := \mathbf{A}_t^T \mathbf{A}_t$ is positive definite. The linear operator $\Omega \mapsto \mathbf{P}_t \Omega + \Omega \mathbf{P}_t$ on $\mathbb{R}^{q \times q}$ is invertible: diagonalizing $\mathbf{P}_t = \mathbf{O} \text{diag}(p_1, \dots, p_q) \mathbf{O}^T$ with $\mathbf{O} \in O(q)$ and $p_i > 0$ and conjugating by $\mathbf{O}$ turns it into the entrywise multiplication by $p_i + p_j > 0$. This proves existence, uniqueness and linearity, and smoothness in $(t, \mathbf{K})$ follows since the inverse of an invertible matrix depends smoothly on it and $t \mapsto \mathbf{P}_t$ is polynomial. For the skew-symmetry, transposing (2.85) gives $\mathbf{P}_t(-\Omega^T) + (-\Omega^T)\mathbf{P}_t = -\mathbf{K}^T = \mathbf{K}$, so $-\Omega^T$ solves the same equation and uniqueness forces $\Omega^T = -\Omega$.

2. The curve $c(t) := (\mathbf{A}_t, s + tc)$ is a straight line in Θ with horizontal initial velocity, hence, as recalled in the proof of Proposition 2, a horizontal geodesic of Θ projecting to γ . Because π is a Riemannian submersion, the

horizontal lift of the covariant derivative $D_t \mathbf{x}$ along γ is the horizontal part of the covariant derivative of the lift along c ([O'Neill(1967)], Section 2; [Valencia(2021)], Section 3). The statement there is for basic vector fields, and it extends to fields along a curve because both sides of the identity are $\mathbb{R}$ -linear, satisfy the Leibniz rule in the scalar factor, and agree on fields that extend to basic fields, which are the three properties characterizing the covariant derivative along a curve. Since the metric of Θ is the Euclidean one (1.3), its covariant derivative along c is the ordinary derivative $(\dot{\mathbf{X}}, \dot{\beta})$. Therefore $\mathbf{x}$ is parallel if and only if $(\dot{\mathbf{X}}(t), \dot{\beta}(t)) \in V_{(\mathbf{A}_t, s+tc)}$, which by Proposition 1 means

$$\dot{\mathbf{X}}(t) = \mathbf{A}_t \Omega(t) \text{ for some skew-symmetric } \Omega(t), \quad \dot{\beta}(t) = 0. \quad (2.87)$$

It remains to identify $\Omega(t)$. By Proposition 1 again, horizontality of the lift at every t says that $\mathbf{S}(t) := \mathbf{A}_t^T \mathbf{X}(t) - \mathbf{X}(t)^T \mathbf{A}_t$ vanishes identically, so $\dot{\mathbf{S}}(t) = \mathbf{0}$; substituting (2.87) and $\Omega^T = -\Omega$,

$$\mathbf{0} = \dot{\mathbf{S}}(t) = \mathbf{V}^T \mathbf{X} + \mathbf{A}_t^T \mathbf{A}_t \Omega - \Omega^T \mathbf{A}_t^T \mathbf{A}_t - \mathbf{X}^T \mathbf{V} = (\mathbf{A}_t^T \mathbf{A}_t \Omega + \Omega \mathbf{A}_t^T \mathbf{A}_t) - (\mathbf{X}^T \mathbf{V} - \mathbf{V}^T \mathbf{X}), \quad (2.88)$$

which is (2.85) with $\mathbf{K} = \mathbf{X}^T \mathbf{V} - \mathbf{V}^T \mathbf{X}$, a skew-symmetric matrix. By part 1 this determines $\Omega(t)$ uniquely and gives (2.86). Conversely, (2.86) is of the form (2.87) with a skew-symmetric $\Omega(t)$, so it implies parallelism provided the lift stays horizontal, which is part 3.

3. The right hand side of (2.86) is linear in $\mathbf{X}(t)$, with coefficients smooth in t and in $(\mathbf{A}, \mathbf{V}, c)$ by part 1; solutions therefore exist on all of $[0, 1]$ and depend smoothly on the initial condition and on the parameters, by the standard theory of linear systems and smooth dependence ([Lee(2013)], Theorem D.1 and Theorem D.6). For the preservation of horizontality, let $\mathbf{X}$ solve (2.86) and put again $\mathbf{S}(t) := \mathbf{A}_t^T \mathbf{X}(t) - \mathbf{X}(t)^T \mathbf{A}_t$. The computation (2.88), read in the reverse direction, gives $\dot{\mathbf{S}}(t) = \mathbf{0}$ identically, and $\mathbf{S}(0) = \mathbf{0}$ because $(\mathbf{X}(0), \beta(0))$ is horizontal; hence $\mathbf{S} \equiv \mathbf{0}$. For the preservation of the norm, write $\Omega(t)$ for the skew-symmetric matrix appearing in (2.86) and note that $\mathbf{X}^T \mathbf{A}_t = (\mathbf{A}_t^T \mathbf{X})^T = \mathbf{A}_t^T \mathbf{X}$ is symmetric by the horizontality just proved, so

$$\frac{d}{dt} [\text{tr}(\mathbf{X} \mathbf{X}^T) + \beta^2] = 2 \text{tr}(\mathbf{X}^T \dot{\mathbf{X}}) + 0 = 2 \text{tr}(\mathbf{X}^T \mathbf{A}_t \Omega) = 2 \text{tr}((\mathbf{A}_t^T \mathbf{X}) \Omega) = 0,$$

the last equality because the trace of the product of a symmetric and a skew-symmetric matrix vanishes. Since $\iota_{(\mathbf{A}_t, s+tc)}$ is an isometry by (2.2), this is exactly the statement that $|\mathbf{x}(t)|$ is constant. $\square$

Definition 38. For $\mathbf{u} \in B_\rho$ we write

$$\Gamma_{\mathbf{u}} : T_{\text{Exp}_{\theta_0}(\mathbf{u})} M \longrightarrow T_{\theta_0} M \quad (2.89)$$

for the parallel transport from $t = 1$ to $t = 0$ along the radial geodesic $t \mapsto \text{Exp}_{\theta_0}(t\mathbf{u})$ of the normal chart of Lemma 27, that is the inverse of the linear isometry supplied by Proposition 37 applied with $[\mathbf{A}, s] = \theta_0$ and $\mathbf{v} = \mathbf{u}$. This is the transport along the minimizing geodesic joining the two points, which is the map denoted Γ in Definition III.5 of [Sun et al.(2026)Sun, Lin, and Liu]. By Proposition 37, $\Gamma_{\mathbf{u}}$ is a linear isometry, $\Gamma_0 = \text{id}$, and, after both tangent spaces are identified with subspaces of $\mathbb{R}^{p \times q} \times \mathbb{R}$ through the horizontal lifts $\iota_{(\mathbf{W}_0, \sigma_0^2)}$ and $\iota_{(\mathbf{W}_0 + \mathbf{U}_{\mathbf{u}}, \sigma_0^2 + b_{\mathbf{u}})}$, the map $(\mathbf{u}, \mathbf{z}) \mapsto \Gamma_{\mathbf{u}}(\mathbf{z})$ is smooth on $B_\rho \times \mathbb{R}^{p \times q} \times \mathbb{R}$.

The transport comparison

Definition III.5 of [Sun et al.(2026)Sun, Lin, and Liu] compares $\text{Exp}_{\theta_{n,h}}^{-1}(\hat{\theta}_n)$, transported back to $T_{\theta_0} M$, with $\text{Exp}_{\theta_0}^{-1}(\hat{\theta}_n)$. On a flat space the two would differ by exactly the vector joining the two base points; on M they differ by that vector plus a curvature correction, and the whole content of this subsection is that the correction is quadratically small. We first record that the logarithm may be taken at a moving base point.

Lemma 39. There exists $r \in (0, \frac{\rho}{2})$ such that, writing $\mathcal{V} := \text{Exp}_{\theta_0}(B_r)$, the following hold.

1. For every $x \in \mathcal{V}$ the map Exp_x^{-1} is defined and smooth on $\mathcal{V}$, and $\text{Exp}_x^{-1}(x) = \mathbf{0}$.

2. The map $(\mathbf{u}, \mathbf{w}) \mapsto \text{Exp}_{\text{Exp}_{\theta_0}(\mathbf{u})}^{-1}(\text{Exp}_{\theta_0}(\mathbf{w}))$ is smooth on $B_r \times B_r$, as a map into $\mathbb{R}^{p \times q} \times \mathbb{R}$ after the horizontal lift identification of Definition 38.

Proof. Let $\mathcal{E} \subseteq TM$ be the set of pairs $(x, \mathbf{v})$ with $\mathbf{v} \in \mathcal{E}_x$, the ball of Proposition 2. Its radius at $x = [\mathbf{A}, s]$ is $\min(\sigma_{\min}(\mathbf{A}), s)$, which is well defined on M because the singular values of $\mathbf{A}$ are unchanged by $\mathbf{A} \mapsto \mathbf{A}\mathbf{Q}$ with $\mathbf{Q} \in O(q)$, and is a continuous function of x ; hence $\mathcal{E}$ is open in TM and contains the zero section, and $\text{Exp} : \mathcal{E} \rightarrow M$ is smooth by Proposition 2, being affine in the horizontal lift. Consider

$$E : \mathcal{E} \longrightarrow M \times M, \quad E(x, \mathbf{v}) := (x, \text{Exp}_x(\mathbf{v})).$$

Its differential at $(\theta_0, \mathbf{0})$ is invertible, because in the splitting $T_{(\theta_0, \mathbf{0})}TM \cong T_{\theta_0}M \oplus T_{\theta_0}M$ it is $(\mathbf{a}, \mathbf{b}) \mapsto (\mathbf{a}, \mathbf{a} + \mathbf{b})$, using $d(\text{Exp}_{\theta_0})_{\mathbf{0}} = \text{id}$. The inverse function theorem therefore gives open sets $\mathcal{N} \ni (\theta_0, \mathbf{0})$ in $\mathcal{E}$ and $\mathcal{D} \ni (\theta_0, \theta_0)$ in $M \times M$ such that $E : \mathcal{N} \rightarrow \mathcal{D}$ is a diffeomorphism. This is the standard construction of uniformly normal neighbourhoods ([Lee(2018)], Theorem 5.14 and its proof); note that completeness of M , which fails here, is nowhere used, only the openness of the domain $\mathcal{E}$ of Exp . Writing $E^{-1}(x, y) = (x, \Xi(x, y))$, the map Ξ is smooth on $\mathcal{D}$ and satisfies $\text{Exp}_x(\Xi(x, y)) = y$ and $\Xi(x, x) = \mathbf{0}$; and $\Xi(x, \cdot)$ is a smooth inverse of Exp_x on a neighbourhood of x , so it deserves the name Exp_x^{-1} . Since Exp_{θ_0} is continuous and $(\theta_0, \theta_0) \in \mathcal{D}$, we may choose $r \in (0, \rho/2)$ so small that $\mathcal{V} \times \mathcal{V} \subseteq \mathcal{D}$ with $\mathcal{V} = \text{Exp}_{\theta_0}(B_r)$, which is part 1. Part 2 follows by composing Ξ with the smooth map $(\mathbf{u}, \mathbf{w}) \mapsto (\text{Exp}_{\theta_0}(\mathbf{u}), \text{Exp}_{\theta_0}(\mathbf{w}))$ and with the horizontal lift, itself smooth because by Proposition 1 the horizontal space at $(\mathbf{A}, s)$ is the kernel of a matrix depending polynomially on $\mathbf{A}$, of constant rank. $\square$

We may and do shrink r further so that $\text{Exp}_{\theta_0}^{-1}$ agrees on $\mathcal{V}$ with the map of Lemma 27, which is automatic since both invert Exp_{θ_0} on $B_r \subseteq B_\rho$ and the latter is injective there. Define

$$F : B_r \times B_r \longrightarrow T_{\theta_0}M, \quad F(\mathbf{u}, \mathbf{w}) := \Gamma_{\mathbf{u}} \left(\text{Exp}_{\text{Exp}_{\theta_0}(\mathbf{u})}^{-1}(\text{Exp}_{\theta_0}(\mathbf{w})) \right). \quad (2.90)$$

Lemma 40 (Transport comparison). After a further shrinking of r there is a constant $C < \infty$, depending only on $[\mathbf{W}_0, \sigma_0^2]$ and r , such that

$$|F(\mathbf{u}, \mathbf{w}) - (\mathbf{w} - \mathbf{u})| \leq C |\mathbf{u}| |\mathbf{w} - \mathbf{u}| \quad \text{for all } \mathbf{u}, \mathbf{w} \in B_r. \quad (2.91)$$

Proof. By Lemma 39 and Definition 38, F is smooth on $B_r \times B_r$; put

$$R(\mathbf{u}, \mathbf{w}) := F(\mathbf{u}, \mathbf{w}) - (\mathbf{w} - \mathbf{u}),$$

a smooth $T_{\theta_0}M$ -valued function on $B_r \times B_r$. It vanishes on two slices:

- $R(\mathbf{u}, \mathbf{u}) = \mathbf{0}$ for every $\mathbf{u} \in B_r$, because $\text{Exp}_x^{-1}(x) = \mathbf{0}$ and $\Gamma_{\mathbf{u}}$ is linear, so $F(\mathbf{u}, \mathbf{u}) = \mathbf{0} = \mathbf{u} - \mathbf{u}$;
- $R(\mathbf{0}, \mathbf{w}) = \mathbf{0}$ for every $\mathbf{w} \in B_r$, because $\Gamma_{\mathbf{0}} = \text{id}$ and $\text{Exp}_{\theta_0}^{-1}(\text{Exp}_{\theta_0}(\mathbf{w})) = \mathbf{w}$, so $F(\mathbf{0}, \mathbf{w}) = \mathbf{w} = \mathbf{w} - \mathbf{0}$.

Replace r by $r/2$, so that the closed balls $\overline{B_r} \times \overline{B_r}$ are compact subsets of the open set on which R is smooth, and set

$$C := \sup_{(\mathbf{u}, \mathbf{w}) \in \overline{B_r} \times \overline{B_r}} \|\partial_1 \partial_2 R(\mathbf{u}, \mathbf{w})\| < \infty,$$

the norm being the one induced on bilinear maps $T_{\theta_0}M \times T_{\theta_0}M \rightarrow T_{\theta_0}M$ by $|\cdot|$; the supremum is finite because $\partial_1 \partial_2 R$ is continuous on a compact set. Let $\mathbf{u}, \mathbf{w} \in B_r$. The ball B_r being convex, the segment from $\mathbf{u}$ to $\mathbf{w}$ lies in it, and the fundamental theorem of calculus in the second variable, together with $R(\mathbf{u}, \mathbf{u}) = \mathbf{0}$, gives

$$R(\mathbf{u}, \mathbf{w}) = \int_0^1 \partial_2 R(\mathbf{u}, \mathbf{u} + \tau(\mathbf{w} - \mathbf{u})) [\mathbf{w} - \mathbf{u}] \, d\tau. \quad (2.92)$$

Fix $z \in B_r$. The segment from $\mathbf{0}$ to $\mathbf{u}$ also lies in B_r , and $\partial_2 R(\mathbf{0}, z) = \mathbf{0}$ because $R(\mathbf{0}, \cdot)$ vanishes identically on B_r ; so the same argument in the first variable gives

$$\partial_2 R(\mathbf{u}, z) = \int_0^1 \partial_1 \partial_2 R(\varsigma \mathbf{u}, z)[\mathbf{u}] d\varsigma, \quad \text{whence} \quad \|\partial_2 R(\mathbf{u}, z)\| \leq C|\mathbf{u}|. \quad (2.93)$$

Substituting (2.93) into (2.92) yields (2.91). $\square$

Remark 41. Lemma 40 is the only place where the curvature of (M, g) enters the asymptotic theory, and (2.91) says exactly how much it costs. In the applications below $\mathbf{u}$ is a local perturbation of size $n^{-\frac{1}{2}}$ and $\mathbf{w} - \mathbf{u}$ is an estimation error of size $O_{\mathbb{P}}(n^{-\frac{1}{2}})$, so the right hand side is $O_{\mathbb{P}}(n^{-1})$; after the multiplication by $\sqrt{n}$ that all these statements carry, it is $O_{\mathbb{P}}(n^{-\frac{1}{2}})$ and disappears. The two vanishing slices used in the proof are the two trivial cases of the comparison — no displacement of the base point, and no error to compare — and it is their combination, not either alone, that produces the product $|\mathbf{u}||\mathbf{w} - \mathbf{u}|$.

Local regularity

We can now state the definition. For $\mathbf{h} \in T_{\theta_0}M$ and n large enough that $\frac{\mathbf{h}}{\sqrt{n}} \in B_r$ we write

$$\theta_{n,\mathbf{h}} := \text{Exp}_{\theta_0} \left(\frac{\mathbf{h}}{\sqrt{n}} \right) \in \mathcal{V}, \quad (2.94)$$

the *geodesic local alternative* in the direction $\mathbf{h}$, and $\mathbb{P}_{\theta}^n$ for the law of $(\mathbf{X}_1, \dots, \mathbf{X}_n)$ when the true parameter is θ , that is the n -fold product of $p(\cdot; \theta) d\mathbf{x}$. Given an M -valued estimator sequence T_n we put, in the notation of Lemma 39,

$$\hat{\mathbf{u}}_n[T_n] := \begin{cases} \text{Exp}_{\theta_0}^{-1}(T_n) & \text{on } \{T_n \in \mathcal{V}\}, \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] := \begin{cases} \Gamma_{\frac{\mathbf{h}}{\sqrt{n}}} \left(\text{Exp}_{\theta_{n,\mathbf{h}}}^{-1}(T_n) \right) & \text{on } \{T_n \in \mathcal{V}\}, \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad (2.95)$$

both taking values in the single vector space $T_{\theta_0}M$; the second is defined as soon as $\frac{\mathbf{h}}{\sqrt{n}} \in B_r$, by part 1 of Lemma 39. For the maximum likelihood estimator, $\hat{\mathbf{u}}_n[\hat{\theta}_n]$ coincides with the vector $\hat{\mathbf{u}}_n$ of (2.39) on the event $\{\hat{\theta}_n \in \mathcal{V}\}$, which by the last part of the proof of Proposition 28, applied with $\epsilon = r$, has probability tending to one; we therefore keep the notation $\hat{\mathbf{u}}_n$ for it, the two conventions differing only on events of vanishing probability, which by the portmanteau argument of Step 7 of the proof of Theorem 32 changes nothing below.

Definition 42. An M -valued estimator sequence T_n is *locally regular at θ_0 with limit law L* , where L is a Borel probability measure on $T_{\theta_0}M$, if for every $\mathbf{h} \in T_{\theta_0}M$

$$\sqrt{n} \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] \rightsquigarrow L \quad \text{under } \mathbb{P}_{\theta_{n,\mathbf{h}}}^n, \quad (2.96)$$

the limit L being the same for every $\mathbf{h}$. This is Definition III.5 of [Sun et al.(2026)Sun, Lin, and Liu] with $\psi = \text{id}$ and $\Psi = M$.

Lemma 43 (Contiguity). For every $\mathbf{h} \in T_{\theta_0}M$ the sequences $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$ and $\mathbb{P}_{\theta_0}^n$ are mutually contiguous.

Proof. By (1.32) of Proposition 8 the log-likelihood ratio $\Lambda_n(\mathbf{h})$ of (2.83) converges in distribution under $\mathbb{P}_{\theta_0}^n$ to $N(-\frac{1}{2}\varsigma^2, \varsigma^2)$ with $\varsigma^2 := G(\mathbf{h}, \mathbf{h})$. If $Z \sim N(-\frac{1}{2}\varsigma^2, \varsigma^2)$ then $\mathbb{E}[e^Z] = e^{-\frac{1}{2}\varsigma^2 + \frac{1}{2}\varsigma^2} = 1$, so Le Cam's first lemma in the form of Example 6.5 of [van der Vaart(1998)] gives mutual contiguity. $\square$

The next lemma is the bridge: it says that on this manifold, regularity in the intrinsic sense of Definition 42 is the same thing as regularity of the ordinary Euclidean kind read in the normal chart of Lemma 27. It is what allows the classical theory to be applied verbatim below.

Lemma 44 (Chart form of regularity). Let T_n be an M -valued estimator sequence such that $\mathbb{P}_{\theta_0}^n(T_n \in \mathcal{V}) \rightarrow 1$ and $\sqrt{n} \hat{\mathbf{u}}_n[T_n] = O_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_0}^n$. Then, for every $\mathbf{h} \in T_{\theta_0}M$,

$$\sqrt{n} \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] = \sqrt{n} \hat{\mathbf{u}}_n[T_n] - \mathbf{h} + o_{\mathbb{P}}(1) \quad \text{under } \mathbb{P}_{\theta_{n,\mathbf{h}}}^n. \quad (2.97)$$

Consequently T_n is locally regular at θ_0 with limit law L if and only if

$$\sqrt{n} \hat{\mathbf{u}}_n[T_n] - \mathbf{h} \rightsquigarrow L \quad \text{under } \mathbb{P}_{\theta_{n,\mathbf{h}}}^n \text{ for every } \mathbf{h} \in T_{\theta_0}M. \quad (2.98)$$

Moreover the two standing hypotheses are automatically satisfied by any locally regular sequence.

Proof. Fix $\mathbf{h}$ and abbreviate $\hat{\mathbf{u}}_n := \hat{\mathbf{u}}_n[T_n]$, $\mathbf{u}_n := \frac{\mathbf{h}}{\sqrt{n}}$ and $\mathcal{B}_n := \{T_n \in \mathcal{V}\}$. By hypothesis $\mathbb{P}_{\theta_0}^n(\mathcal{B}_n) \rightarrow 1$, hence $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n(\mathcal{B}_n) \rightarrow 1$ by Lemma 43; and $\sqrt{n} \hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$ as well. The latter is the only property of contiguity used beyond the transfer of $o_{\mathbb{P}}(1)$ statements, and it follows from the definition: were it false, there would be $\varepsilon > 0$, a subsequence n_k and constants $M_k \rightarrow \infty$ with $\mathbb{P}_{\theta_{n_k,\mathbf{h}}}^{n_k}(\sqrt{n_k}|\hat{\mathbf{u}}_{n_k}| > M_k) > \varepsilon$ for all k , whereas $\mathbb{P}_{\theta_0}^{n_k}(\sqrt{n_k}|\hat{\mathbf{u}}_{n_k}| > M_k) \rightarrow 0$ by boundedness in probability under $\mathbb{P}_{\theta_0}^n$, and these two are incompatible with contiguity. On $\mathcal{B}_n$, and for n large enough that $\mathbf{u}_n \in B_r$, we have $\hat{\mathbf{u}}_n \in B_r$ and, by (2.90) and (2.95),

$$\hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] = \Gamma_{\mathbf{u}_n} \left(\text{Exp}_{\text{Exp}_{\theta_0}(\mathbf{u}_n)}^{-1} \left(\text{Exp}_{\theta_0}(\hat{\mathbf{u}}_n) \right) \right) = F(\mathbf{u}_n, \hat{\mathbf{u}}_n),$$

because $T_n = \text{Exp}_{\theta_0}(\hat{\mathbf{u}}_n)$ there. Lemma 40 therefore gives, on $\mathcal{B}_n$,

$$\left| \sqrt{n} \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] - (\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h}) \right| = \sqrt{n} |F(\mathbf{u}_n, \hat{\mathbf{u}}_n) - (\hat{\mathbf{u}}_n - \mathbf{u}_n)| \leq \sqrt{n} C \frac{|\mathbf{h}|}{\sqrt{n}} \left| \hat{\mathbf{u}}_n - \frac{\mathbf{h}}{\sqrt{n}} \right| = \frac{C|\mathbf{h}|}{\sqrt{n}} |\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h}|. \quad (2.99)$$

The right hand side is $O_{\mathbb{P}}(n^{-\frac{1}{2}}) = o_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$, and $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n(\mathcal{B}_n^c) \rightarrow 0$, which proves (2.97). The equivalence of (2.96) and (2.98) is then Slutsky's lemma.

For the last assertion, suppose T_n is locally regular with limit law L and take $\mathbf{h} = \mathbf{0}$ in (2.96): since $\theta_{n,\mathbf{0}} = \theta_0$ and $\Gamma_0 = \text{id}$, this reads $\sqrt{n} \hat{\mathbf{u}}_n[T_n] \rightsquigarrow L$ under $\mathbb{P}_{\theta_0}^n$, which is boundedness in probability by Prohorov's theorem; and (2.95) is only meaningful under the convention that $T_n \in \mathcal{V}$ with probability tending to one, which we have built into the definition. $\square$

Theorem 45 (Local regularity of the maximum likelihood estimator). *Let $1 \leq q < p$ and let $\theta_0 = [\mathbf{W}_0, \sigma_0^2] \in M$. Then the maximum likelihood estimator $\hat{\theta}_n$ of Definition 16 is locally regular at θ_0 in the sense of Definition 42, with limit law*

$$L = N(\mathbf{0}, G_{\theta_0}^{-1}) \quad \text{on } T_{\theta_0}M, \quad (2.100)$$

which does not depend on $\mathbf{h}$. Explicitly, for every $\mathbf{h} \in T_{\theta_0}M$,

$$\Gamma_{\frac{\mathbf{h}}{\sqrt{n}}} \left(\sqrt{n} \text{Exp}_{\theta_{n,\mathbf{h}}}^{-1} \left(\hat{\theta}_n \right) \right) \rightsquigarrow N(\mathbf{0}, G_{\theta_0}^{-1}) \quad \text{under } \mathbb{P}_{\theta_{n,\mathbf{h}}}^n. \quad (2.101)$$

Proof. We work in the coordinates of (2.81). Fix $\mathbf{h} \in T_{\theta_0}M$.

Step 1: a single linear statistic drives everything. By Theorem 32, in the form (2.60),

$$\sqrt{n} \hat{\mathbf{u}}_n = \mathbf{G}^{-1} \Delta_n + o_{\mathbb{P}}(1) \quad \text{under } \mathbb{P}_{\theta_0}^n, \quad (2.102)$$

and by (2.83), $\Lambda_n(\mathbf{h}) = \Delta_n^T \mathbf{h} - \frac{1}{2} \mathbf{h}^T \mathbf{G} \mathbf{h} + o_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_0}^n$. Both expansions are linear in the *same* vector Δ_n of (2.82), so no separate joint central limit theorem is required: the vectors $\mathbf{s}(\mathbf{X}_1), \mathbf{s}(\mathbf{X}_2), \dots$ are independent, identically distributed, centred and square integrable with covariance matrix $\mathbf{G}$ by Theorem 7, so the multivariate central limit theorem gives $\Delta_n \rightsquigarrow N(\mathbf{0}, \mathbf{G})$ under $\mathbb{P}_{\theta_0}^n$, and the continuous mapping theorem applied to the linear map $\delta \mapsto (\mathbf{G}^{-1} \delta, \delta^T \mathbf{h} - \frac{1}{2} \mathbf{h}^T \mathbf{G} \mathbf{h})$ together with Slutsky's lemma yields

$$(\sqrt{n}\hat{\mathbf{u}}_n, \Lambda_n(\mathbf{h})) \rightsquigarrow N\left(\begin{pmatrix} \mathbf{0} \\ -\frac{1}{2}\mathbf{h}^T\mathbf{G}\mathbf{h} \end{pmatrix}, \begin{pmatrix} \mathbf{G}^{-1} & \mathbf{h} \\ \mathbf{h}^T & \mathbf{h}^T\mathbf{G}\mathbf{h} \end{pmatrix}\right) \quad \text{under } \mathbb{P}_{\theta_0}^n, \quad (2.103)$$

the off-diagonal block being $\text{Cov}(\mathbf{G}^{-1}\Delta, \Delta^T\mathbf{h}) = \mathbf{G}^{-1}\mathbf{G}\mathbf{h} = \mathbf{h}$ for $\Delta \sim N(\mathbf{0}, \mathbf{G})$.

Step 2: Le Cam's third lemma. The limit in (2.103) has the form required by Example 6.7 of [van der Vaart(1998)], the Gaussian case of Le Cam's third lemma (Theorem 6.6 there): the last coordinate is the log-likelihood ratio and is $N(-\frac{1}{2}\varsigma^2, \varsigma^2)$ with $\varsigma^2 = \mathbf{h}^T\mathbf{G}\mathbf{h} = G(\mathbf{h}, \mathbf{h})$. The cited result therefore gives

$$\sqrt{n}\hat{\mathbf{u}}_n \rightsquigarrow N(\mathbf{h}, \mathbf{G}^{-1}) \quad \text{under } \mathbb{P}_{\theta_n, \mathbf{h}}^n : \quad (2.104)$$

the limit under the contiguous alternative is the limit under the null shifted by the covariance $\mathbf{h}$ between the statistic and the log-likelihood ratio. Equivalently $\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h} \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$ under $\mathbb{P}_{\theta_n, \mathbf{h}}^n$, a law free of $\mathbf{h}$.

Step 3: from the chart to the transported error. The hypotheses of Lemma 44 hold for $T_n = \hat{\theta}_n$: the event $\{\hat{\theta}_n \in \mathcal{V}\}$ has probability tending to one, as recorded after (2.95), and $\sqrt{n}\hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_0}^n$ by (2.44). The equivalence (2.98) of that lemma turns (2.104) into (2.101), with limit law $N(\mathbf{0}, \mathbf{G}^{-1})$, which in the g_{θ_0} -orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$ represents $N(\mathbf{0}, G_{\theta_0}^{-1})$ as in Step 6 of the proof of Theorem 32. $\square$

Remark 46. It is worth being explicit about what (2.101) excludes, since Theorem 32 alone excludes nothing of the kind. Fix a point $\theta_* \in M$, known to the statistician, and let $\hat{\theta}_n^H$ be the Hodges-type modification of $\hat{\theta}_n$ that replaces it by θ_* whenever $\hat{\theta}_n \in \text{Exp}_{\theta_*}(B)$ with B the ball of radius $n^{-\frac{1}{4}}$ in $T_{\theta_*}M$, and leaves it unchanged otherwise. For any fixed true parameter $\theta_0 \neq \theta_*$ the modification occurs on an event of probability tending to zero, by Proposition 24, so $\hat{\theta}_n^H$ satisfies the exact analogue of (2.44) there. For $\theta_0 = \theta_*$ it does better: since $|\hat{\mathbf{u}}_n| = O_{\mathbb{P}}(n^{-\frac{1}{2}}) \ll n^{-\frac{1}{4}}$ by (2.44), we have $\hat{\theta}_n^H = \theta_*$ with probability tending to one, so that $\sqrt{n}\hat{\mathbf{u}}_n[\hat{\theta}_n^H] \rightsquigarrow \delta_0$, a limit strictly more concentrated than $N(\mathbf{0}, G_{\theta_0}^{-1})$; this is the superefficiency phenomenon, and it shows that no bound whatsoever can be extracted from pointwise statements of the type of (2.44) alone. And yet, at $\theta_0 = \theta_*$ and $\mathbf{h} \neq \mathbf{0}$, the same computation gives $\sqrt{n}\hat{\mathbf{u}}_n[\hat{\theta}_n^H] \rightsquigarrow \delta_0$ under $\mathbb{P}_{\theta_n, \mathbf{h}}^n$ as well — the estimator is pinned to θ_* on an event of probability tending to one, since $|\hat{\mathbf{u}}_n| = O_{\mathbb{P}}(n^{-\frac{1}{2}}) \ll n^{-\frac{1}{4}}$ under the alternative too, by Lemma 43 — so by (2.97) its transported error converges to $\delta_{-\mathbf{h}}$, which does depend on $\mathbf{h}$. Hence $\hat{\theta}_n^H$ is superefficient at θ_* and is not locally regular there. Local regularity is precisely the statement that the asymptotics of Theorem 32 do not degrade as the truth is moved at the scale $n^{-\frac{1}{2}}$, and it is a strictly stronger property.

Optimality

Theorem 45 concerns one estimator. We now turn to the statements that concern *all* estimators, and which are therefore lower bounds rather than limit theorems. The key point is Lemma 44: because on this manifold intrinsic regularity is equivalent to ordinary regularity read in the normal chart of Lemma 27, and because by Proposition 8 the local experiments in that chart are asymptotically normal in the sense of Chapter 7 of [van der Vaart(1998)], with nonsingular information matrix $\mathbf{G}$ by Theorem 7, the classical bounds apply verbatim. No curvature enters: it has already been absorbed by Lemma 40.

Throughout this subsection the *local experiments* are

$$\mathcal{E}_n := \left(\mathbb{P}_{\text{Exp}_{\theta_0}^n\left(\frac{\mathbf{h}}{\sqrt{n}}\right)}^n : \mathbf{h} \in T_{\theta_0}M \cong \mathbb{R}^d \right), \quad (2.105)$$

and (2.83) says exactly that $\mathcal{E}_n$ is asymptotically normal at $\mathbf{h} = \mathbf{0}$ with score Δ_n and information $\mathbf{G}$.

Theorem 47 (Convolution bound). *Let T_n be any estimator sequence that is locally regular at θ_0 in the sense of Definition 42, with limit law L . Then there exists a Borel probability measure Λ on $T_{\theta_0}M$ such that*

$$L = N(\mathbf{0}, G_{\theta_0}^{-1}) * \Lambda, \quad (2.106)$$

the convolution of the Gaussian law with covariance operator the inverse Fisher information and an independent factor. In particular:

1. if L has a finite second moment then so does Λ and the covariance operators satisfy $\text{cov}(L) = G_{\theta_0}^{-1} + \text{cov}(\Lambda) \succeq G_{\theta_0}^{-1}$;
2. the maximum likelihood estimator attains $\Lambda = \delta_0$, by Theorem 45, and is therefore best regular: no locally regular estimator sequence for $[\mathbf{W}, \sigma^2]$ has a limit law less dispersed than that of $\hat{\theta}_n$.

Proof. By the last assertion of Lemma 44 the sequence T_n satisfies the two standing hypotheses of that lemma, and by (2.98)

$$\sqrt{n} \hat{\mathbf{u}}_n[T_n] - \mathbf{h} \rightsquigarrow L \quad \text{under } \mathbb{P}_{\theta_{n,h}}^n \text{ for every } \mathbf{h} \in \mathbb{R}^d.$$

In the chart, $\hat{\mathbf{u}}_n[T_n]$ is an estimator of the local parameter $\mathbf{u}$, the true value is $\mathbf{u} = \mathbf{0}$, the local alternative $\theta_{n,h}$ is the parameter value $\mathbf{u} = \mathbf{h}/\sqrt{n}$, and the display reads $\sqrt{n} \left(\hat{\mathbf{u}}_n[T_n] - \frac{\mathbf{h}}{\sqrt{n}} \right) \rightsquigarrow L$ under $\mathbf{h}$; this is precisely the definition of a regular estimator sequence of Section 8.5 of [van der Vaart(1998)], with ψ the identity map. The experiments $\mathcal{E}_n$ being asymptotically normal with nonsingular $\mathbf{G}$, the convolution theorem, Theorem 8.8 of [van der Vaart(1998)], gives a probability measure Λ on $\mathbb{R}^d$ with $L = N(\mathbf{0}, \mathbf{G}^{-1}) * \Lambda$, which is (2.106) once $\mathbb{R}^d$ is identified with $T_{\theta_0}M$ through the g_{θ_0} -orthonormal basis $e_1, \dots, e_d$, in which $\mathbf{G}^{-1}$ represents $G_{\theta_0}^{-1}$.

Part 1 is the elementary fact that if Z and Y are independent with $Z \sim N(\mathbf{0}, \mathbf{G}^{-1})$ and $Z + Y \sim L$ square integrable then Y is square integrable, its law being the image of L under a measurable map determined by the deconvolution, and covariances add on independent summands; concretely, $\text{cov}(L) = \text{cov}(Z) + \text{cov}(Y) = \mathbf{G}^{-1} + \text{cov}(\Lambda)$. Part 2 is Theorem 45, which gives $L = N(\mathbf{0}, G_{\theta_0}^{-1})$, whence $\Lambda = \delta_0$ by uniqueness of the deconvolution of a Gaussian, or directly by comparing characteristic functions: $\hat{L}(\xi) = \hat{N}(\xi)\hat{\Lambda}(\xi)$ with $\hat{N}$ nowhere vanishing forces $\hat{\Lambda} \equiv 1$. $\square$

The convolution theorem restricts attention to regular estimators. The local asymptotic minimax theorem removes that restriction, at the price of replacing the limit law by a maximal risk over a shrinking neighbourhood. Recall that a function $\ell : T_{\theta_0}M \rightarrow [0, \infty)$ is called *bowl-shaped* if its sublevel sets $\{\ell \leq c\}$ are convex and symmetric about the origin for every $c \geq 0$; the squared norm $\ell(\mathbf{z}) = |\mathbf{z}|^2$ and its truncations $|\mathbf{z}|^2 \wedge m$ are the examples to keep in mind.

Theorem 48 (Local asymptotic minimax bound and its attainment). *Let ℓ be bowl-shaped.*

1. (Lower bound.) *For every sequence $\hat{\mathbf{v}}_n$ of measurable maps $(\mathbb{R}^p)^n \rightarrow T_{\theta_0}M$ — in particular for $\hat{\mathbf{v}}_n = \hat{\mathbf{u}}_n[T_n]$ arising from an arbitrary M -valued estimator sequence T_n , regular or not —*

$$\sup_I \liminf_{n \rightarrow \infty} \sup_{\mathbf{h} \in I} \mathbb{E}_{\theta_{n,h}} [\ell(\sqrt{n} \hat{\mathbf{v}}_n - \mathbf{h})] \geq \int \ell \, dN(\mathbf{0}, G_{\theta_0}^{-1}), \quad (2.107)$$

the first supremum being over all finite subsets $I \subset T_{\theta_0}M$.

2. (Attainment.) *If ℓ is in addition bounded and continuous, then the maximum likelihood estimator turns (2.107) into an equality:*

$$\sup_I \liminf_{n \rightarrow \infty} \sup_{\mathbf{h} \in I} \mathbb{E}_{\theta_{n,h}} [\ell(\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h})] = \int \ell \, dN(\mathbf{0}, G_{\theta_0}^{-1}), \quad (2.108)$$

and the same holds with $\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h}$ replaced by the transported intrinsic error $\sqrt{n} \hat{\mathbf{u}}_n^h[\hat{\theta}_n]$ of (2.95).

Proof. **1.** As in the proof of Theorem 47, in the normal chart the family $\mathcal{E}_n$ is a sequence of asymptotically normal experiments indexed by $\mathbf{h} \in \mathbb{R}^d$ with nonsingular information $\mathbf{G}$, and $\hat{\mathbf{v}}_n$ is an arbitrary estimator sequence for the local parameter. Inequality (2.107) is then Theorem 8.11 of [van der Vaart(1998)], the local asymptotic minimax theorem, whose only hypotheses are the asymptotic normality of the experiments and the bowl-shapedness of ℓ ; no property of the estimator sequence is required, which is why the bound is estimator-agnostic and closes the superefficiency loophole illustrated in Remark 46.

2. Fix a finite $I \subset T_{\theta_0}M$. By Step 2 of the proof of Theorem 45, $\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h} \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$ under $\mathbb{P}_{\theta_{n,h}}^n$ for each $\mathbf{h}$. Since ℓ is bounded and continuous, the portmanteau theorem gives $\mathbb{E}_{\theta_{n,h}}[\ell(\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h})] \rightarrow \int \ell dN(\mathbf{0}, \mathbf{G}^{-1})$ for each $\mathbf{h} \in I$, a limit not depending on $\mathbf{h}$. As I is finite, the maximum over $\mathbf{h} \in I$ of finitely many sequences with a common limit converges to that limit, so

$$\liminf_{n \rightarrow \infty} \sup_{\mathbf{h} \in I} \mathbb{E}_{\theta_{n,h}}[\ell(\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h})] = \int \ell dN(\mathbf{0}, \mathbf{G}^{-1})$$

for every finite I , and taking the supremum over I leaves the value unchanged. This is (2.108), and it matches the lower bound of part 1 exactly. For the last claim, (2.97) gives $\sqrt{n}\hat{\mathbf{u}}_n^h[\hat{\theta}_n] = \sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h} + o_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_{n,h}}^n$, so the two have the same limit in distribution and, ℓ being bounded and continuous, the same limiting risk. $\square$

Remark 49. Part 2 is stated for bounded ℓ because weak convergence alone controls the risk only for bounded losses; for $\ell(\mathbf{z}) = |\mathbf{z}|^2$ the identity (2.108) requires in addition the uniform integrability of $|\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h}|^2$ under $\mathbb{P}_{\theta_{n,h}}^n$, which we have not established and which does not follow from anything proved above. The lower bound of part 1 is unaffected: it holds for every bowl-shaped ℓ , bounded or not. Applying part 2 to the truncations $\ell \wedge m$, which are again bounded, continuous and bowl-shaped, and letting $m \rightarrow \infty$ recovers the bound but not the equality, monotone convergence acting on the wrong side of the inequality.

Consequences

The first consequence closes the small gap left in Corollary 34, whose uniqueness clause was proved by invoking the influence relation, which [Sun et al.(2026)Sun, Lin, and Liu] establish in their Lemma III.3 for sequences that are *both* asymptotically linear *and* regular. Theorem 45 supplies the missing half of that hypothesis. It is worth recording that the uniqueness in fact needs neither.

Proposition 50. Let T_n be an estimator sequence that is asymptotically linear at θ_0 in the sense of Definition 30 with influence operator IF, and suppose IF' is another influence operator for which (2.41) holds. Then $\text{IF}'(\mathbf{X}_1) = \text{IF}(\mathbf{X}_1)$ almost surely. In particular the influence operator (2.42) of the maximum likelihood estimator is the unique one, with no restriction to the span of the score, and $\hat{\theta}_n$ is a regular asymptotically linear sequence, so that the influence relation (2.80) is now available with all the hypotheses of [Sun et al.(2026)Sun, Lin, and Liu], Lemma III.3, verified.

Proof. Subtracting the two expansions (2.41) gives $n^{-\frac{1}{2}} \sum_{i=1}^n \mathbf{D}(\mathbf{X}_i) = o_{\mathbb{P}}(1)$ with $\mathbf{D} := \text{IF} - \text{IF}'$, which is centred and square integrable by (2.40). The multivariate central limit theorem gives $n^{-\frac{1}{2}} \sum_i \mathbf{D}(\mathbf{X}_i) \rightsquigarrow N(\mathbf{0}, \text{cov}(\mathbf{D}(\mathbf{X}_1)))$, and a sequence converging in distribution to a limit and also to $\mathbf{0}$ in probability forces that limit to be δ_0 , so $\text{cov}(\mathbf{D}(\mathbf{X}_1)) = \mathbf{0}$; being centred, $\mathbf{D}(\mathbf{X}_1) = \mathbf{0}$ almost surely. The remaining assertions are Theorem 32 and Theorem 45. $\square$

The second consequence is the inferential payoff, and it is the one that distinguishes local regularity from Theorem 32 in practice. We first record that the information operator may be evaluated at the estimator.

Lemma 51. For $\mathbf{u} \in B_r$ let $\mathbf{G}(\mathbf{u})$ be the $d \times d$ matrix

$$\mathbf{G}(\mathbf{u})_{jk} := G_{\text{Exp}_{\theta_0}(\mathbf{u})}(\Gamma_{\mathbf{u}}^{-1}(\mathbf{e}_j), \Gamma_{\mathbf{u}}^{-1}(\mathbf{e}_k)), \quad j, k = 1, \dots, d, \quad (2.109)$$

the matrix of the Fisher information operator at $\text{Exp}_{\theta_0}(\mathbf{u})$ in the frame obtained by transporting $\mathbf{e}_1, \dots, \mathbf{e}_d$ along the radial geodesic. Then $\mathbf{u} \mapsto \mathbf{G}(\mathbf{u})$ is smooth on B_r , with $\mathbf{G}(\mathbf{0}) = \mathbf{G}$, and it is positive definite everywhere.

Proof. Write $(\mathbf{W}_u, \sigma_u^2) := (\mathbf{W}_0 + \mathbf{U}_u, \sigma_0^2 + b_u)$ for the representative of $\text{Exp}_{\theta_0}(\mathbf{u})$ used in Definition 38, which is affine in $\mathbf{u}$, and let $(\tilde{\mathbf{U}}_j(\mathbf{u}), \tilde{b}_j(\mathbf{u})) := \iota_{(\mathbf{W}_u, \sigma_u^2)}(\Gamma_u^{-1}(\mathbf{e}_j))$ be the horizontal lift there. By (1.22) and (1.21) applied at the point $\text{Exp}_{\theta_0}(\mathbf{u})$,

$$\mathbf{G}(\mathbf{u})_{jk} = \frac{1}{2} \text{tr} \{ \Sigma_u^{-1} \Delta_j(\mathbf{u}) \Sigma_u^{-1} \Delta_k(\mathbf{u}) \}, \quad \Delta_j(\mathbf{u}) := \tilde{\mathbf{U}}_j(\mathbf{u}) \mathbf{W}_u^T + \mathbf{W}_u \tilde{\mathbf{U}}_j(\mathbf{u})^T + \tilde{b}_j(\mathbf{u}) \mathbf{I}_p.$$

Each ingredient is smooth in $\mathbf{u}$: the matrix Σ_u is a polynomial by (2.47) and is bounded below by $c_1 \mathbf{I}_p$ by (2.49), so Σ_u^{-1} is smooth; Γ_u^{-1} is smooth by Proposition 37 and Definition 38; and $\iota_{(\mathbf{W}_u, \sigma_u^2)}$ is smooth in $\mathbf{u}$, as noted at the end of the proof of Lemma 39. That $\mathbf{G}(\mathbf{0}) = \mathbf{G}$ is $\Gamma_0 = \text{id}$, and positive definiteness is Theorem 7 at the point $\text{Exp}_{\theta_0}(\mathbf{u})$, the change of frame being invertible. $\square$

Corollary 52 (Locally uniform Wald confidence sets). Let $\alpha \in (0, 1)$, let $\chi_{d,1-\alpha}^2$ be the $(1 - \alpha)$ quantile of the chi-squared law with d degrees of freedom, and define the random subset of M

$$\mathcal{C}_{n,1-\alpha} := \text{Exp}_{\hat{\theta}_n} \left(\{ \mathbf{v} \in T_{\hat{\theta}_n} M : n G_{\hat{\theta}_n}(\mathbf{v}, \mathbf{v}) \leq \chi_{d,1-\alpha}^2 \} \right), \quad (2.110)$$

which is well defined on an event of probability tending to one. Then, for every $\mathbf{h} \in T_{\theta_0} M$,

$$\mathbb{P}_{\theta_{n,\mathbf{h}}}^n (\theta_{n,\mathbf{h}} \in \mathcal{C}_{n,1-\alpha}) \longrightarrow 1 - \alpha. \quad (2.111)$$

In particular the coverage is asymptotically correct not only at θ_0 , which is the case $\mathbf{h} = \mathbf{0}$ and already follows from Theorem 32, but along every sequence of local alternatives at the estimation scale $n^{-\frac{1}{2}}$.

Proof. Fix $\mathbf{h}$ and work under $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$, which by Lemma 43 is contiguous to $\mathbb{P}_{\theta_0}^n$; all the $o_{\mathbb{P}}(1)$ and $O_{\mathbb{P}}(1)$ statements of Theorem 32 and Proposition 28 therefore remain valid. In particular $\hat{\mathbf{u}}_n \rightarrow \mathbf{0}$ in probability and $\sqrt{n} \hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$.

By Lemma 51 and the continuous mapping theorem, $\mathbf{G}(\hat{\mathbf{u}}_n) \rightarrow \mathbf{G}$ in probability; in particular $\lambda_{\min}(\mathbf{G}(\hat{\mathbf{u}}_n))$ is bounded away from zero in probability, so the ellipsoid in (2.110) is contained in a ball of radius $O_{\mathbb{P}}(n^{-\frac{1}{2}})$ and hence, with probability tending to one, in the domain $\mathcal{E}_{\hat{\theta}_n}$ of $\text{Exp}_{\hat{\theta}_n}$ supplied by Proposition 2; this is the well definedness.

Since $\text{Exp}_{\hat{\theta}_n}$ is injective on that ball for large n , membership $\theta_{n,\mathbf{h}} \in \mathcal{C}_{n,1-\alpha}$ is equivalent, on an event of probability tending to one, to

$$n G_{\hat{\theta}_n}(\mathbf{v}_n, \mathbf{v}_n) \leq \chi_{d,1-\alpha}^2, \quad \mathbf{v}_n := \text{Exp}_{\hat{\theta}_n}^{-1}(\theta_{n,\mathbf{h}}). \quad (2.112)$$

Put $\mathbf{z}_n := \Gamma_{\hat{\mathbf{u}}_n}(\mathbf{v}_n) \in T_{\theta_0} M$. Because $\Gamma_{\hat{\mathbf{u}}_n}^{-1} \mathbf{e}_1, \dots, \Gamma_{\hat{\mathbf{u}}_n}^{-1} \mathbf{e}_d$ is a basis of $T_{\hat{\theta}_n} M$ whose coordinates for $\mathbf{v}_n$ are exactly the coordinates of $\mathbf{z}_n$ in $\mathbf{e}_1, \dots, \mathbf{e}_d$, the definition (2.109) gives

$$G_{\hat{\theta}_n}(\mathbf{v}_n, \mathbf{v}_n) = \mathbf{z}_n^T \mathbf{G}(\hat{\mathbf{u}}_n) \mathbf{z}_n. \quad (2.113)$$

Now apply Lemma 40 with $\mathbf{u} = \hat{\mathbf{u}}_n$ and $\mathbf{w} = \mathbf{h}/\sqrt{n}$, both in B_r with probability tending to one, noting that $F(\hat{\mathbf{u}}_n, \mathbf{h}/\sqrt{n}) = \mathbf{z}_n$ by (2.90):

$$|\sqrt{n} \mathbf{z}_n - (\mathbf{h} - \sqrt{n} \hat{\mathbf{u}}_n)| \leq \sqrt{n} C |\hat{\mathbf{u}}_n| \left| \frac{\mathbf{h}}{\sqrt{n}} - \hat{\mathbf{u}}_n \right| = \frac{C}{\sqrt{n}} |\sqrt{n} \hat{\mathbf{u}}_n| |\mathbf{h} - \sqrt{n} \hat{\mathbf{u}}_n| = O_{\mathbb{P}} \left(n^{-\frac{1}{2}} \right),$$

so that $\sqrt{n} \mathbf{z}_n = -(\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h}) + o_{\mathbb{P}}(1)$. By Step 2 of the proof of Theorem 45, $\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h} \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$ under $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$, and the Gaussian law is symmetric, so $\sqrt{n} \mathbf{z}_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$. Combining this with $\mathbf{G}(\hat{\mathbf{u}}_n) \rightarrow \mathbf{G}$ in probability, Slutsky's lemma and the continuous mapping theorem applied to $(\mathbf{z}, \mathbf{A}) \mapsto \mathbf{z}^T \mathbf{A} \mathbf{z}$ give, through (2.113),

$$n G_{\hat{\theta}_n}(\mathbf{v}_n, \mathbf{v}_n) = (\sqrt{n} \mathbf{z}_n)^T \mathbf{G}(\hat{\mathbf{u}}_n) (\sqrt{n} \mathbf{z}_n) \rightsquigarrow \mathbf{Z}^T \mathbf{G} \mathbf{Z}, \quad \mathbf{Z} \sim N(\mathbf{0}, \mathbf{G}^{-1}),$$

and $\mathbf{Z}^T \mathbf{G} \mathbf{Z} = \|\mathbf{G}^{\frac{1}{2}} \mathbf{Z}\|^2 \sim \chi_d^2$ because $\mathbf{G}^{\frac{1}{2}} \mathbf{Z} \sim N(\mathbf{0}, \mathbf{I}_d)$. The distribution function of χ_d^2 being continuous, the probability of (2.112) converges to $\mathbb{P}(\chi_d^2 \leq \chi_{d,1-\alpha}^2) = 1 - \alpha$, which is (2.111). $\square$

Remark 53. Corollary 52 is stated along each fixed sequence of local alternatives, which is the exact strength of Definition 42 and is already enough to rule out the failure exhibited in Remark 46: for the Hodges-type estimator the analogous set is a single point with probability tending to one, and its coverage of $\theta_{n,h}$ tends to 0 for every $h \neq 0$. The genuinely uniform statement, $\sup_{|h| \leq c} \left| \mathbb{P}_{\theta_{n,h}}^n(\theta_{n,h} \in \mathcal{C}_{n,1-\alpha}) - (1 - \alpha) \right| \rightarrow 0$, requires the expansion (2.83) along arbitrary bounded sequences h_n rather than along fixed h , which Proposition 8 as stated does not provide; we do not need it anywhere.

Remark 54. It may be useful to summarize what Section 2.1 adds to Theorem 32, since the three groups of results answer three different questions.

Theorem 45 is about *other parameters*: it says that the conclusion of Theorem 32 is stable under perturbations of the truth at the scale $n^{-\frac{1}{2}}$. Nothing in (2.43)–(2.45) implies this, as Remark 46 shows by exhibiting an estimator that satisfies the analogue of (2.44) at every point of M and is not regular anywhere.

Theorems 47 and 48 are about *other estimators*: they are lower bounds over a class of procedures, and are therefore statements of a different kind from anything in Section 1.4, which concerns one estimator only. The two are complementary. The convolution bound is sharp and distributional but restricted to regular sequences; the local asymptotic minimax bound holds for every estimator sequence whatsoever but only in the weaker, maximal-risk form. It is worth noting where each hypothesis is used: the lower bound (2.107) needs nothing beyond Theorem 7 and Proposition 8, so it was already within reach at the end of Section 1.4; what needs Theorem 45 is the *attainment* (2.108), because the supremum there is taken over a neighbourhood that shrinks at the estimation scale and therefore probes exactly the local alternatives.

Finally, Proposition 37 and Lemmas 39–40 are geometric results about (M, g) in their own right, independent of any statistics; Lemma 43 is a probabilistic one. They are what the two-base-point comparison costs, and Theorem 32, which lives entirely inside the single vector space $T_{\theta_0}M$, needed none of them.

In the terminology of [Sun et al.(2026)Sun, Lin, and Liu], Theorem 45 verifies their Definition III.5 for the probabilistic principal component analysis model, Theorem 47 is the instance of their Theorem III.1, Theorem 48 of their Theorem III.2, and the two together are their Theorem V.1 for this model. We have proved them here through the chart reduction of Lemma 44 and the classical Euclidean theorems of [van der Vaart(1998)], rather than by citing the manifold versions, because Lemma 40 makes the reduction exact and because it isolates precisely where the curvature of M is felt: in the $O(\|u\|\|w - u\|)$ term of (2.91), and nowhere else.

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